On the Functional Dimension of a Family of Analytic Neural Networks

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Abstract

The functional dimension of a neural network is the number of independent directions in which its parameters change the function that it computes. Symmetries of the parameters bound it from above, and for ReLU networks the bound can fail. We study deep networks in which every layer has input weights, a fixed nonlinear map and output weights, as the layers of a transformer have. Between two such layers, the output weights of the first can be multiplied by an invertible matrix and the input weights of the second by its inverse: a $\mathrm{GL}(d_{\mathrm{model}})$ symmetry at every hidden state, and an $\mathrm{O}(d_{\mathrm{model}})$ symmetry under pre-RMSNorm. With this symmetry we remove d^2_{model} (resp. $\approx d^2_{\mathrm{model}}/2$ under pre-RMSNorm) parameters per layer without changing the network function. This is a lossless reduction of about 8.5 percent of the parameters of the skipless ViT-B of Ji et al. (2026). We then prove that for $\tanh$, SwiGLU and GeGLU layers, and for GELU layers under pre-RMSNorm, nothing more can be removed, at any depth: these symmetries, together with those inside the layers, are the only redundancy, and the networks are locally identifiable up to them.

1 Introduction

Which parameters of a neural network does the function that it computes determine? For $\tanh$ networks the answer is classical: the function determines the weights up to permutations and sign flips of the hidden units, for irreducible networks with one hidden layer (Sussmann, 1992) and, under conditions that almost all weights satisfy, at any depth (Fefferman, 1994). The layers of current networks have continuous symmetries as well, and every continuous symmetry lowers the *functional dimension* of the network, the number of independent directions in which its parameters change its function (Grigsby et al., 2025). So a count of symmetries bounds the functional dimension from above. It does not always give its value: ReLU networks lose more dimensions than their symmetries account for, on sets of parameters of positive measure (Grigsby et al., 2023).

We study the networks whose layers have the form of the layers of a transformer: input weights, a fixed nonlinear map, and output weights. We call such a layer a *block*. In a network that composes blocks, two weight matrices meet between two blocks, with a hidden state of dimension d_{model} between them. The output weights of one block can then be multiplied by an invertible matrix and the input weights of the next block by its inverse (Godfrey et al., 2022; Zhao et al., 2023), which makes d^2_{model} parameters redundant at every hidden state.

This symmetry is known, and its use to remove parameters has barely been explored. Godfrey et al. (2022) use symmetries of this kind to compare the hidden states of trained networks. Zhao et al. (2023) name model compression as a prospect of them, but do not take the step themselves. To our knowledge, Graef (2024) is the first to remove parameters in this way, and does so in the attention blocks of transformers without skip connections and without normalization. We remove them in every block, with and without normalization. Deep networks that compose such blocks directly, without skip connections, have recently been trained to varying degrees of success (He and Hofmann, 2024; He et al., 2023; Ji et al., 2025, 2026), and the reduction applies to them: in the ViT-B of Ji et al. (2026) it removes 8.5 percent of the parameters. We then ask whether anything more can be removed, that is, whether these symmetries are the only redundancy.

1.1 Results

An exact reduction. At each of the $L+1$ hidden states of a network of L blocks, of any kind, a change of basis turns a $d_{\text{model}} \times d_{\text{model}}$ submatrix of the weights into the identity. This removes $(L+1)d_{\text{model}}^2$ parameters. Under pre-RMSNorm the change of basis is orthogonal and leaves an upper triangular submatrix, which removes $(L+1)d_{\text{model}}(d_{\text{model}}-1)/2$ parameters (Proposition 3.3). The function does not change, so the reduction applies to trained weights (Section 3).

Main result. Whether more can be removed is a question on the functional dimension. Consider a network of L MLP blocks of one kind, each with at least d_{model} hidden units, without biases and with P parameters. Let $\mathcal{G}$ be the group of the changes of basis of one hidden state, $\mathrm{GL}(d_{\text{model}})$ without normalization and $\mathrm{O}(d_{\text{model}})$ under pre-RMSNorm, and let s be the dimension of the symmetries inside the blocks: zero for $\tanh$ and GELU blocks, and one rescaling per hidden unit for SwiGLU and GeGLU blocks. The symmetries give the bound $P-(L+1)\dim\mathcal{G}-s$ on the functional dimension (Theorem 4.3). We prove that at generic parameters, that is, for almost all weights, the functional dimension is

$$P-(L+1)\dim\mathcal{G}-s$$

for $\tanh$, GELU, SwiGLU and GeGLU blocks, with one case left open, GELU blocks without normalization (Theorem 6.1, Table 1). For $\tanh$ blocks without normalization, the count also follows from the theorem of Fefferman (1994) (Remark D.20). These four kinds of block are real-analytic, and they are the family of the title. So after the reduction no redundancy is left in a network of $\tanh$ or GELU blocks, and in a SwiGLU or GeGLU network only the rescalings. Equivalently, the networks are *locally identifiable* up to their symmetries: nearby parameters with the same function differ only by changes of basis at the hidden states and by rescalings inside the blocks (Corollary 6.2).

One block. For a network of one block without normalization, we prove the same count, $P-2d_{\text{model}}^2-s$, for every kind of block that we consider: the four above, attention, and ReLU, ReLU^2 and ReGLU blocks (Theorem 5.1). For a $\tanh$ block and for an attention block, the count also follows from earlier theorems (Section 5).

What is open, and where the count fails. At depth, the count is open for GELU blocks without normalization, which we state as a conjecture, and for attention blocks (Section 7). For ReLU, ReLU^2 and ReGLU blocks at depth, the count fails on an open set of weights.

Idea of the proof. A direction of the parameters along which the function does not change gives an identity in the input of the first block. By analyticity the identity holds at every input, and we follow the input to infinity along a ray, as Sussmann (1992) does for $\tanh$ networks with one hidden layer. There the hidden units of a $\tanh$ block saturate at different exponential rates, and the hidden units of a GELU block and the gates of a SwiGLU or GeGLU block differ from linear or vanishing ones by small terms of different rates. A sum of terms with different rates vanishes only if every term does. Term by term, the direction then acts on the weights of the block as a symmetry does, and what is left is the same statement for the network after this block.

1.2 Related work

Section 3 states what is known on the change of basis as a symmetry and on its use to remove parameters. Here we name the works on the two questions to which the functional dimension belongs.

Identifiability. The theorems of Sussmann (1992) and Fefferman (1994) for $\tanh$ networks are global: the function determines the weights up to a finite group of symmetries. Vlačić and Bölcskei (2022) prove the same up to permutations, at any depth and connectivity, for a family of sigmoidal nonlinearities that they construct. For ReLU networks, Stock and Gribonval (2021) and Bona-Pellissier et al. (2022) study identifiability up to permutations and rescalings of the hidden units, also locally. In these networks one weight matrix stands between two nonlinearities, and no continuous symmetry acts between two layers. In ours two matrices meet in a hidden state of dimension d_{model}, with or without a normalization between them. Without normalization, a network of $\tanh$ blocks is a $\tanh$ network of the first kind, whose weight matrices are products of two factors, and for it our counts follow from the two theorems (Remarks C.4 and D.20). For attention, Henry et al. (2025)

Proven	Value	Blocks	Result
Parameters explicitly removed without changing the function of the network, post-training			
any L	$(L+1)d_{\text{model}}^2$	any	Prop. 3.3(i)
with pre-RMSNorm	$(L+1)d_{\text{model}}(d_{\text{model}}-1)/2$	any	Prop. 3.3(ii)
Functional dimension, for P parameters and symmetries of dimension s inside the blocks			
any L	at most $P-(L+1)d_{\text{model}}^2-s$	any	Thm. 4.3
with pre-RMSNorm	at most $P-(L+1)d_{\text{model}}(d_{\text{model}}-1)/2-s$	any	Thm. 4.3
$L=1$	exactly $P-2d_{\text{model}}^2-s$	any	Thm. 5.1
any L	exactly $P-(L+1)d_{\text{model}}^2-s$	$\tanh$, gated	Thm. 6.1
with pre-RMSNorm	exactly $P-(L+1)d_{\text{model}}(d_{\text{model}}-1)/2-s$	$\tanh$, GELU, gated	Thm. 6.1

Table 1: Summary of the results, for networks of L blocks at generic parameters (scope in Section 2). "Gated" blocks are SwiGLU and GeGLU blocks. "Any" block is a $\tanh$, GELU, gated, attention, ReLU, ReGLU or ReLU^2 block. Theorem 6.1 also treats RMSNorm with $\varepsilon = 0$.

treat networks of this kind without softmax at any depth, and Henry (2026) one block with softmax. Our statements are local. They hold for the four kinds of MLP block at any depth, also with a normalization between the two matrices.

Dimension counts. Grigsby et al. (2025) bound the functional dimension of ReLU networks by their rescaling symmetries, and Grigsby et al. (2023) show that it falls below this bound on sets of positive measure. For polynomial networks it is the dimension of the functional variety (Kileel et al., 2019), and Shahverdi et al. (2026, Theorem 4.1) prove that an MLP whose activation is a generic polynomial of large enough degree has as many dimensions of functions as weights. Such an MLP is a network of our kind in which the two matrices between two layers are one, so this is the analogue of Theorem 6.1 for polynomial activations.

2 Setup

This section fixes the networks. We write every block, MLP or attention, in one form, compose blocks into networks, add normalization, and state the scope of the results.

Blocks. A hidden state is a matrix $h \in \mathbb{R}^{T\times d_{\text{model}}}$ whose rows are the T tokens of a sequence. A block reads it with a trainable matrix W_{read}, the *read*, applies a fixed map f, and writes the result with a trainable matrix W_{write}, the *write*:

$$g(h) = f(hW_{\text{read}})\, W_{\text{write}}. \tag{1}$$

In an MLP block with d_{hidden} hidden units, its width, the read is a $d_{\text{model}} \times d_{\text{hidden}}$ matrix, the write is a $d_{\text{hidden}} \times d_{\text{model}}$ matrix, and f applies an activation to every hidden unit of every token: $\tanh$, GELU, ReLU, or its square, ReLU^2. Gated blocks are MLP blocks as well. In a gated block the read is $W_{\text{read}} = [\, W_g \mid W_v \,]$, two $d_{\text{model}} \times d_{\text{hidden}}$ matrices that give each token a gate z_i and a value w_i for each hidden unit, and f maps them to $\sigma(z_i)\, w_i$, with σ the ReLU in ReGLU, SiLU in SwiGLU and GELU in GeGLU (Shazeer, 2020). In an attention block with H heads of width d_h, the read is $W_{\text{read}} = [\, W_Q \mid W_K \mid W_V \,]$, the query, key and value matrices of all heads, f is the attention map, and the write is the output projection $W_{\text{write}} = W_O$.

Networks. A network has a linear embedding $E \in \mathbb{R}^{d_{\text{in}}\times d_{\text{model}}}$, then L blocks $g_1, \dots, g_L$, then a linear readout $D \in \mathbb{R}^{d_{\text{model}}\times d_{\text{out}}}$. Block ℓ has the read W_{read}^{ℓ}, the fixed map f_ℓ and the write W_{write}^{ℓ}. Before a block or the readout reads a hidden state, it applies a map N to it: the identity, or a normalization. The network maps an input $x \in \mathbb{R}^{T\times d_{\text{in}}}$ to an output y through the hidden states $h_0, \dots, h_L$:

$$h_0 = xE, \qquad h_\ell = g_\ell\big(N(h_{\ell-1})\big) \quad (1 \le \ell \le L), \qquad y = N(h_L)\, D. \tag{2}$$

We write F_θ for the function $x \mapsto y$, where $\theta = (E, W_{\text{read}}^1, W_{\text{write}}^1, \dots, W_{\text{read}}^L, W_{\text{write}}^L, D) \in \mathbb{R}^P$ holds the parameters. A statement holds at *generic* parameters if it holds outside a set of Lebesgue

measure zero, so with probability one for weights drawn from a density. A transformer layer has two blocks, its attention and MLP sub-layers, so a transformer of 12 layers has $L = 24$.

Normalization. RMSNorm with the constant $\varepsilon \geq 0$ is the map N_ε that acts on every row $v \in \mathbb{R}^{d_{\text{model}}}$ of a hidden state h as

$$N_\varepsilon(v) \;=\; \frac{v}{\sqrt{\dfrac{\|v\|^2}{d_{\text{model}}} + \varepsilon}},$$

so that $N_\varepsilon(h)$ is the matrix of the normalized rows. The constant ε is fixed and is not a parameter. We treat three cases, and name them for later use. In the case (n) there is no normalization, and N is the identity. In the case (+), *pre-RMSNorm* with $\varepsilon > 0$, $N = N_\varepsilon$. In the case (0), pre-RMSNorm with $\varepsilon = 0$, RMSNorm keeps only the direction of a row, $N_\varepsilon(v) = \sqrt{d_{\text{model}}}\, v/\|v\|$ for $v \neq 0$, and a network is defined at the inputs for which no row of a hidden state vanishes. Below, "under pre-RMSNorm" means in the case (+), and the case (0) is named where it is meant. A learned gain $\operatorname{diag}(\gamma)$ after N_ε is absorbed in the read that follows it, $W_{\text{read}} \mapsto \operatorname{diag}(\gamma)\, W_{\text{read}}$.

Scope. Networks have no biases, and their inputs range over all of $\mathbb{R}^{T \times d_{\text{in}}}$. Every MLP block has $d_{\text{hidden}} \geq d_{\text{model}}$, and all blocks of a network are of one kind. The heads of an attention block have total width $Hd_h = d_{\text{model}}$, and attention blocks act on sequences of at least two tokens. The input and the output satisfy $d_{\text{in}}, d_{\text{out}} \geq d_{\text{model}} \geq 2$. The reduction of Section 3 needs none of this: it holds for any inputs, one-hot tokens included, for blocks of different kinds, and with biases.

3 Changes of basis, and the parameters that they remove

A symmetry is a map of the parameters that leaves the function F_θ unchanged. This section describes the one symmetry that acts between two blocks, says what is known of it, and uses it to remove parameters.

Take the first two blocks of a network without normalization. The first writes the hidden state $h_1 = f_1(h_0 W^1_{\text{read}})\, W^1_{\text{write}}$, and the second reads it as

$$h_1 W^2_{\text{read}} \;=\; f_1(h_0 W^1_{\text{read}})\, W^1_{\text{write}} W^2_{\text{read}}. \tag{3}$$

The two matrices enter the function only through their product, and for every invertible $d_{\text{model}} \times d_{\text{model}}$ matrix G we have the invariance

$$\big(W^1_{\text{write}} G\big)\big(G^{-1} W^2_{\text{read}}\big) \;=\; W^1_{\text{write}} W^2_{\text{read}}. \tag{4}$$

The hidden state h_1 becomes $h_1 G$: the same tokens, written in another basis.

Definition 3.1 (Change of basis)**.** The writes into the hidden state h_ℓ are W^ℓ_{write}, or E if $\ell = 0$, and the reads of h_ℓ are $W^{\ell+1}_{\text{read}}$, or D if $\ell = L$. For an invertible $d_{\text{model}} \times d_{\text{model}}$ matrix G, the *change of basis* G at h_ℓ maps

$$W \mapsto WG \;\text{ for the writes into } h_\ell, \qquad W \mapsto G^{-1} W \;\text{ for the reads of } h_\ell. \tag{5}$$

What is known. This section builds on results that are known, in scattered and sometimes partial form. For MLPs, Godfrey et al. (2022) show that the weights of two consecutive layers can be changed in this way by every matrix A for which the activation σ between them satisfies $\sigma \circ A = B \circ \sigma$ for some matrix B (their Proposition 3.4). With the identity as activation this is every invertible matrix, and the symmetry is the invariance (4). They use these symmetries to compare the hidden states of trained networks, by stitching two networks and by measures of similarity, and they do not remove parameters with them. Zhao et al. (2023) show the same independently (their Proposition C.1). They include activations that rescale a vector by a function of its norm, $\sigma(z) = f(\|z\|)\, z$, for which A is orthogonal. RMSNorm is such an activation, applied to every token, with $f(r) = (r^2/d_{\text{model}} + \varepsilon)^{-1/2}$. They name model compression as a prospect of such symmetries, and stop short of using them to remove parameters. Such a removal is moreover exact, a free lunch: the network keeps its function.

To our knowledge, Graef (2024) is the first to remove parameters in this way, independently and without explicitly mentioning the symmetry framework. Moreover, the work is restricted to skipless transformers without normalization, and applies the reduction only to the attention blocks.

Pre-RMSNorm. Under pre-RMSNorm a block reads $N_\varepsilon(h)$ in place of h. For an orthogonal Q,

$$N_\varepsilon(hQ) \;=\; N_\varepsilon(h)\,Q,$$

since Q preserves the norm of every row, so the change of basis Q at a hidden state is undone by its reads, $W \mapsto Q^\top W$, as in (5). For an invertible G that is not orthogonal, $N_\varepsilon(hG) = N_\varepsilon(h)M$ fails for every matrix M at some h, since $\varepsilon > 0$ (Lemma A.1). So the reads can compensate only orthogonal changes of basis. For $\varepsilon = 0$ the scale of a row is lost as well, $N_\varepsilon(h\,cQ) = N_\varepsilon(h)\,Q$ for every number $c > 0$, and the reads compensate the change of basis cQ by $W \mapsto Q^\top W$. From here on, $\mathcal{G}$ is the group of the changes of basis that the reads of one hidden state can compensate: $\mathrm{GL}(d_{\text{model}})$ without normalization, of dimension d_{model}^2; $\mathrm{O}(d_{\text{model}})$ under pre-RMSNorm, of dimension $d_{\text{model}}(d_{\text{model}} - 1)/2$; and for $\varepsilon = 0$ the group $\mathrm{CO}(d_{\text{model}})$ of the matrices cQ, of dimension $d_{\text{model}}(d_{\text{model}}-1)/2+1$.

Proposition 3.2 (Changes of basis preserve the function)**.** *Without normalization or under pre-RMSNorm, let $G_0, \ldots, G_L \in \mathcal{G}$. The parameters*

$$\Big(EG_0,\;\; \big\{G_{\ell-1}^{-1} W_{\text{read}}^{\ell},\;\; W_{\text{write}}^{\ell} G_\ell\big\}_{\ell=1}^{L},\;\; G_L^{-1} D\Big)$$

give the same function as θ, with the hidden states $h_\ell G_\ell$: one matrix of $\mathcal{G}$ per hidden state. For $\varepsilon = 0$ and $G_\ell = c_\ell Q_\ell$, the same holds with $Q_\ell^\top$ in the place of G_ℓ^{-1}.

Proof in Appendix A.

Removing parameters. A change of basis that turns a $d_{\text{model}} \times d_{\text{model}}$ submatrix of the weights into the identity removes d_{model}^2 parameters: the submatrix no longer requires storage or computation. For instance, take $1 \le \ell < L$ and write $W_{\text{read}}^{\ell+1} = [\,W_1 \mid W_2\,]$, where W_1 holds the first d_{model} columns. If W_1 is invertible, the change of basis $G = W_1$ at h_ℓ maps

$$W_{\text{read}}^{\ell+1} \;\mapsto\; W_1^{-1}\,[\,W_1 \mid W_2\,] = [\,I \mid W_1^{-1} W_2\,], \qquad W_{\text{write}}^{\ell} \;\mapsto\; W_{\text{write}}^{\ell}\, W_1.$$

Under pre-RMSNorm the change of basis must be orthogonal, and in place of the identity it leaves an upper triangular matrix, which has $d_{\text{model}}(d_{\text{model}} - 1)/2$ fixed entries, the zeros below its diagonal.

Proposition 3.3 (Exact reduction)**.** *Let d_{model} columns of the read of h_ℓ ($W_{\text{read}}^{\ell+1}$, or D if $\ell = L$) form an invertible matrix W_1.*
(i) *Without normalization, the change of basis $G = W_1$ at h_ℓ preserves the function and turns W_1 into $G^{-1}W_1 = I$.*
(ii) *Under pre-RMSNorm, let $W_1 = QR$ with Q orthogonal and R upper triangular. The change of basis Q at h_ℓ preserves the function and turns W_1 into $Q^\top W_1 = R$.*
At generic parameters every hidden state has such columns, and these changes of basis at all $L + 1$ hidden states remove $(L+1)\dim \mathcal{G}$ parameters: $(L+1)d_{\text{model}}^2$ in (i), and $(L+1)d_{\text{model}}(d_{\text{model}}-1)/2$ in (ii).

Proof in Appendix A.2.

Any d_{model} columns that form an invertible matrix serve as W_1, and rows of the matrix W_{write}^{ℓ} work too, or rows of E if $\ell = 0$: if d_{model} of them form an invertible matrix W_1, the change of basis $G = W_1^{-1}$ at h_ℓ turns them into $W_1 G = I$ (Example A.2).

In an attention block with $Hd_h = d_{\text{model}}$, the matrices W_Q and W_O are $d_{\text{model}} \times d_{\text{model}}$, and without normalization the changes of basis $G = W_Q$ at $h_{\ell-1}$ and $G = W_O^{-1}$ at h_ℓ turn both into the identity. This is the reduction of Graef (2024). Proposition 3.3 removes d_{model}^2 more, at the hidden state before the readout, and it holds for every kind of block and under pre-RMSNorm.

LayerNorm and trained models. LayerNorm is RMSNorm after a projection that removes the mean of each row, so every count under pre-RMSNorm holds with $d_{\text{model}} - 1$ in place of d_{model} (Proposition A.3). In the ViT-B (Dosovitskiy et al., 2021) that Ji et al. (2026) train without skip connections, with standard pre-LayerNorm blocks, the changes of basis at its 25 hidden states remove 7.3M parameters, 8.5 percent of the model, without changing its function (Appendix A.3).

4 Functional dimension and the upper bound

The reduction of Section 3 removes parameters without changing the function. Whether redundancy remains even after it is a question that Section 3 leaves open. The functional dimension answers it: we define it, bound it by a count of symmetries, and name the case in which nothing more can be removed.

Definition 4.1 (Functional dimension (Grigsby et al., 2025))**.** For a finite batch $X = (x_1, \dots, x_n)$ of inputs, let $F_X(\theta) = (F_\theta(x_1), \dots, F_\theta(x_n))$ be the outputs on X at the parameters $\theta \in \mathbb{R}^P$, and $\mathrm{D}F_X(\theta)$ the Jacobian of $\theta \mapsto F_X(\theta)$. The *functional dimension* and the *redundancy* at θ are

$$\mathrm{fd}(\theta) \;=\; \max_X \operatorname{rank} \mathrm{D}F_X(\theta), \qquad\qquad \mathrm{red}(\theta) \;=\; P - \mathrm{fd}(\theta),$$

where the maximum is over all finite batches X.

A continuous symmetry adds to the redundancy: if a continuous family of maps of the parameters preserves the function, the directions in which it moves θ lie in the kernel of every such Jacobian. A rank says more than a list of symmetries can. Where $\mathrm{fd}(\theta) = P - m$ near a generic θ, the functions of the network near F_θ form a family of dimension $P - m$, and no smooth reparametrization describes them with fewer parameters.

Example 4.2 (Not all reductions are maximal)**.** The linear network $x \mapsto xM_1M_2M_3$, with three $d_{\mathrm{model}} \times d_{\mathrm{model}}$ matrices, has $3d_{\mathrm{model}}^2$ parameters. Its functions are the linear maps $x \mapsto xM$ with $M = M_1M_2M_3$, so its functional dimension is d_{model}^2 at generic parameters and its redundancy is $2d_{\mathrm{model}}^2$. Merging M_2 and M_3 into $M_2' = M_2M_3$ gives $x \mapsto xM_1M_2'$, with $2d_{\mathrm{model}}^2$ parameters, the same functions and a redundancy of d_{model}^2. Merging once more, into $M = M_1M_2'$, leaves d_{model}^2 parameters and no redundancy.

Two kinds of symmetry bound the functional dimension. The first are the changes of basis in $\mathcal{G}$ at the hidden states (Proposition 3.2). The second are the *symmetries inside the blocks*, which change the read and the write of one block and leave the map of that block unchanged: the rescaling of a hidden unit of a ReLU or ReLU^2 block, the rescaling of the value of a hidden unit of a gated block, and the changes of basis inside an attention head (Elhage et al., 2021; Godfrey et al., 2022; Rolnick and Kording, 2020). A ReLU block with $d_{\mathrm{hidden}} = 4d_{\mathrm{model}}$ has $4d_{\mathrm{model}}$ rescalings, against d_{model}^2 dimensions for a change of basis. Let s be the total dimension of the symmetries inside the blocks, summed over the blocks. For $\tanh$ and GELU blocks, $s = 0$, and Appendix B writes all of them out, with their sources.

Theorem 4.3 (Upper bound)**.** *Without normalization or under pre-RMSNorm, at generic parameters, the functional dimension of a network of L blocks is at most $P - (L+1)\dim \mathcal{G} - s$.*

Proof in Appendix B.

For $\varepsilon = 0$ the same bound holds, with $\mathcal{G} = \mathrm{CO}(d_{\mathrm{model}})$, for the four kinds of block of the main result (Appendix D.1).

Definition 4.4 (Equality condition)**.** A network satisfies the *equality condition* if the bound of Theorem 4.3 is attained at generic parameters, that is, if its symmetries are its only redundancy.

The reduction of Proposition 3.3 then leaves no redundancy but that of the symmetries inside the blocks.

Definition 4.5 (Identifiability up to symmetries)**.** A network is *identifiable up to its symmetries* at θ if every θ' with $F_{\theta'} = F_\theta$ is the image of θ under a symmetry: a change of basis of Proposition 3.2, a symmetry inside a block, or a composition of these. It is *locally identifiable* at θ if this holds for every θ' in some neighborhood of θ.

The notion follows Vlačić and Bölcskei (2020, 2022), Stock and Gribonval (2021) and Bona-Pellissier et al. (2022), with the equivalence given here by the symmetries of Theorem 4.3. For the blocks of the main result, the equality condition is local identifiability.

Proposition 4.6 (The equality condition and local identifiability)**.** *Let the blocks be* tanh*, GELU, SwiGLU, GeGLU or attention blocks, without normalization or under pre-RMSNorm. A network satisfies the equality condition if and only if, at generic parameters, it is locally identifiable up to its symmetries. Every symmetry that is needed is then close to the identity.*

Proof in Appendix B.

The proposition compares only parameters that are close to each other: the equality condition does not exclude a distant parameter with the same function that no symmetry reaches.

5 One block

The bound of Theorem 4.3 counts the symmetries that we know. Whether a network has further redundancy is another question, and for some blocks it has (Section 7). In this section we answer it for networks of one block without normalization. The argument has two steps.

With one block, let $C = EW_{\text{read}}$ and $A = W_{\text{write}}D$. The network computes

$$x \mapsto f(xC)\,A. \tag{6}$$

Step 1: factorizations. The parameters enter only through C and A. A matrix of rank d_{model} has only one factorization into two factors of rank d_{model}, up to a change of basis between the factors. So the directions η of θ that leave C and A unchanged to first order are those of two changes of basis, $I + \tau X$ at h_0 and $I + \tau Y$ at h_1: their entries at E, W_{read}, W_{write} and D are

$$\eta_E = EX, \qquad \eta_{\text{read}} = -XW_{\text{read}}, \qquad \eta_{\text{write}} = W_{\text{write}}Y, \qquad \eta_D = -YD.$$

Step 2: hidden units. It remains to show that, to first order, the function determines C and A up to the symmetries inside the block. Let c_i be the columns of C and a_i the rows of A, so that a tanh or GELU network computes $x \mapsto \sum_i \sigma(xc_i)\,a_i$. Take a direction $(\psi_{c_i}, \psi_{a_i})_i$ of these columns and rows along which the function does not change:

$$\sum_i \Big[\sigma(xc_i)\,\psi_{a_i} + \sigma'(xc_i)\,(x\psi_{c_i})\,a_i\Big] \;=\; 0 \qquad \text{for all } x.$$

For tanh and GELU this forces $\psi_{c_i} = 0$ and $\psi_{a_i} = 0$ for every i. For the other kinds of MLP block the corresponding identity forces a rescaling of each hidden unit, such as $\psi_{c_i} = \lambda_i c_i$ and $\psi_{a_i} = -\lambda_i a_i$ for ReLU (Lemma C.3). The proof isolates one hidden unit at a time: by its kink for ReLU, ReLU^2 and ReGLU, and for the other activations by the rate at which it approaches its limit along a ray $x = t\,n$, $t \to \infty$. For tanh this is the argument of Sussmann (1992). For an attention block, inputs with two different tokens turn every head into a tanh hidden unit, and the same lemma applies.

Theorem 5.1 (One block)**.** *Let $L = 1$, without normalization, with a* tanh*, GELU, ReLU, ReLU^2, ReGLU, SwiGLU, GeGLU or attention block. At generic parameters the functional dimension is $P - 2d_{\text{model}}^2 - s$, so the network satisfies the equality condition.*

A special case of Theorem C.1, proved in Appendix C.

Two cases of the theorem also follow from earlier work. For a tanh block, it follows from the theorem of Sussmann (1992) on networks with one hidden layer (Remark C.4). For an attention block, it follows from Theorem 3.9 of Henry et al. (2025), for one head, and from Theorem 2.1 of Henry (2026), for several heads and sequences of at least three tokens (Remark C.5). The proof of Theorem 5.1 does not use these theorems.

6 Any depth

With several blocks, the two factorizations of Section 5 are not enough: the blocks have to be told apart inside their composition. We state the result, and then present its proof, which removes one block at a time. Section 6.1 reduces the count to one step, from the network after block $k+1$ to the network after block k, and Sections 6.2 to 6.4 make this step.

Theorem 6.1 (The bound is attained at any depth)**.** *Let $d_{\text{in}}, d_{\text{out}} \geq d_{\text{model}} \geq 2$, and consider networks without biases whose blocks are all of one kind,* $\tanh$*, GELU, SwiGLU or GeGLU, each with at least d_{model} hidden units. In the following cases the functional dimension at generic parameters is*

$$P - (L+1)\dim \mathcal{G} - s,$$

so the network satisfies the equality condition.
(i) $\tanh$ *blocks: without normalization, and under pre-RMSNorm with $\varepsilon > 0$ and with $\varepsilon = 0$.*
(ii) *GELU blocks: under pre-RMSNorm with $\varepsilon > 0$ and with $\varepsilon = 0$.*
(iii) *SwiGLU blocks, and GeGLU blocks: without normalization, and under pre-RMSNorm with $\varepsilon > 0$.*

Proof in Appendix D.

Here $s = 0$ for $\tanh$ and GELU blocks, and s is the number of hidden units of the network for SwiGLU and GeGLU blocks. For $\tanh$ blocks without normalization, this also follows from the theorem of Fefferman (1994) on $\tanh$ networks of any depth, together with the uniqueness of a factorization (Remark D.20). That theorem does not apply when a normalization stands between the two matrices. With Proposition 4.6, Theorem 6.1 gives local identifiability.

Corollary 6.2 (Local identifiability)**.** *Without normalization or under pre-RMSNorm with $\varepsilon > 0$, in the cases of Theorem 6.1, let θ be generic parameters. Every θ' near θ with the same function is obtained from θ by matrices $G_0, \dots, G_L \in \mathcal{G}$ close to the identity, one per hidden state, as in Proposition 3.2, and for SwiGLU and GeGLU blocks by a rescaling of the value of each hidden unit, close to* 1.

For $\varepsilon = 0$ we state no such corollary, since Proposition 4.6 is stated for $\varepsilon > 0$.

6.1 One claim for every hidden state

An MLP block acts on each token separately, so in this section inputs have one token: a hidden state is a row $h \in \mathbb{R}^{1\times d_{\text{model}}}$, and the input of a block is a row $n = N(h)$. Let $\mathfrak{g}$ be the Lie algebra of $\mathcal{G}$, the set of the matrices Z with $\exp(\tau Z) \in \mathcal{G}$ for all numbers τ: all $d_{\text{model}} \times d_{\text{model}}$ matrices in the case (n), the antisymmetric ones in the case (+), and the matrices $Y + \beta I$ with Y antisymmetric and β a number in the case (0).

For $k = 0, \dots, L$ and a row h, let $\mathcal{B}_{>k}(h)$ be the output of the network when its hidden state h_k is set to h:

$$\mathcal{B}_{>L}(h) = N(h)\, D, \qquad \mathcal{B}_{>k-1}(h) = \mathcal{B}_{>k}\big(g_k(N(h))\big), \qquad \text{so that} \quad F_\theta(x) = \mathcal{B}_{>0}(xE).$$

Let $\mathrm{D}\mathcal{B}_{>k}(h)\, v$ be its derivative at h along a row v, and $\mathrm{D}_\theta \mathcal{B}_{>k}(h)\, \eta$ its derivative along a direction η of the weights on which it depends, those of the blocks $k+1, \dots, L$ and of D. For $Z_k, \dots, Z_L \in \mathfrak{g}$, the changes of basis $G_\ell = \exp(\tau Z_\ell)$ of Proposition 3.2 change the hidden state h_k into $h_k \exp(\tau Z_k)$ and move the later weights along a direction η, and they do not change the output. To first order in τ, with $Z = Z_k$,

$$\mathrm{D}\mathcal{B}_{>k}(h)\,(hZ) + \mathrm{D}_\theta \mathcal{B}_{>k}(h)\,\eta \;=\; 0 \qquad \text{for all rows } h. \tag{7}$$

The claim is the converse.

Claim (D_k), for $\tanh$ and GELU blocks in the cases (n) and (+). If a $d_{\text{model}} \times d_{\text{model}}$ matrix Z and a direction η of the weights of the blocks $k+1, \dots, L$ and of D satisfy (7), then $Z \in \mathfrak{g}$, and with $Z_k = Z$ there are $Z_{k+1}, \dots, Z_L \in \mathfrak{g}$ such that

$$\eta \;=\; \Big(\big\{-Z_{\ell-1}W^\ell_{\text{read}},\;\; W^\ell_{\text{write}}Z_\ell\big\}_{\ell=k+1}^{L},\;\; -Z_L D\Big).$$

Appendix D.2 states the claim in all cases: for gated blocks η also holds the directions of the rescalings, and in the case (0) only the antisymmetric part of $Z_{\ell-1}$ acts on W^ℓ_{read}.

The claim (D_L) concerns the readout alone and is linear algebra (Lemma D.4). The claim (D_0) gives the theorem. If a direction $\eta \in \mathbb{R}^P$ has $\mathrm{D}_\theta F(\theta, x)\,\eta = 0$ at every input x, then its entry at E is EZ for a matrix Z, which satisfies (7) for $k = 0$ with the rest of η (Lemma D.2). By (D_0), η is then a direction of the changes of basis of Proposition 3.2, and the Jacobian has the rank of the bound on some finite batch (Proposition D.3). So the proof is the step from (D_{k+1}) to (D_k).

The identity for one block. Let block $k+1$ be a $\tanh$ or GELU block with q hidden units, with the read W, whose columns are $c_1, \ldots, c_q$, and the write V, in the case (n), and write $\mathcal{B} = \mathcal{B}_{>k+1}$ and $\eta = (\eta_W, \eta_V, \eta')$, where η_W and η_V are the entries of η at W and V. By the chain rule, (7) at the row n reads

$$\mathrm{D}\mathcal{B}\big(\tilde{w}V\big)\,\Big(\big(\tilde{w}' \odot n\,(ZW + \eta_W)\big)\,V + \tilde{w}\,\eta_V\Big) + \mathrm{D}_\theta\mathcal{B}\big(\tilde{w}V\big)\,\eta' \;=\; 0, \tag{8}$$

where $\tilde{w} = \sigma(nW)$ is the row of the outputs of the hidden units of the block, $\tilde{w}' = \sigma'(nW)$ the row of their derivatives, and $\odot$ the entrywise product of two rows. In the case (+) the block reads $n = N_\varepsilon(h)$, the derivative of N_ε adds one term to $n\,(ZW + \eta_W)$, and the identity holds for the rows n of the open ball $\|n\|^2 < d_{\mathrm{model}}$. Its left side is real-analytic in n, so the identity holds at every row $n \in \mathbb{R}^{1\times d_{\mathrm{model}}}$ (Lemma D.5). This is what lets us follow n to infinity.

6.2 $\tanh$ blocks on a ray

Take $n = t\,\xi$ with a row ξ, and let $t \to \infty$. With $u_j = \xi c_j$, hidden unit j outputs

$$\tanh(t\,u_j) \;=\; \pm\,\frac{1-\zeta_j(t)}{1+\zeta_j(t)}, \qquad \zeta_j(t) = e^{-2|u_j|t},$$

with the sign of u_j: it saturates at the exponential rate $2|u_j|$. When the numbers u_j are linearly independent over $\mathbb{Q}$, the hidden units can be told apart.

Lemma 6.3 (Separation on a ray)**.** *Let $u_1, \ldots, u_q$ be real numbers that are linearly independent over $\mathbb{Q}$, and let $\mathcal{T}_0, \ldots, \mathcal{T}_K$ be real-analytic maps from $\mathbb{R}^{1\times q}$ to $\mathbb{R}^{1\times d_{\mathrm{out}}}$. If*

$$\sum_{i=0}^{K} t^i\,\mathcal{T}_i\big(\tanh(tu_1), \ldots, \tanh(tu_q)\big) \;=\; 0 \qquad \textit{for all } t > 0,$$

then $\mathcal{T}_i = 0$ on $\mathbb{R}^{1\times q}$ for every i.

In the proof, every $\mathcal{T}_i$ is expanded in powers of $\zeta_1, \ldots, \zeta_q$. The monomials decay at pairwise different rates, because the u_j are linearly independent over $\mathbb{Q}$, so the slowest term that is left must vanish. This is the argument by which Sussmann (1992, Lemma 3.1) proves that $\tanh$ hidden units with different arguments are linearly independent, applied here to analytic maps of their outputs.

For $\sigma = \tanh$ we have $\sigma' = 1 - \sigma^2$, so the left side of (8) at $n = t\,\xi$ is $t\,\mathcal{T}_1(\tilde{w}) + \mathcal{T}_0(\tilde{w})$ at $\tilde{w} = \tanh(t\,\xi W)$, with

$$\mathcal{T}_1(\tilde{w}) = \mathrm{D}\mathcal{B}(\tilde{w}V)\,\Big(\big((1-\tilde{w}^2)\odot\xi\,(ZW+\eta_W)\big)\,V\Big), \qquad \mathcal{T}_0(\tilde{w}) = \mathrm{D}\mathcal{B}(\tilde{w}V)\,(\tilde{w}\,\eta_V) + \mathrm{D}_\theta\mathcal{B}(\tilde{w}V)\,\eta'.$$

By the lemma both maps vanish at every row $\tilde{w} \in \mathbb{R}^{1\times q}$: the outputs of the hidden units have become free variables. From $\mathcal{T}_1 = 0$ we get $\eta_W = -ZW$, since $\mathrm{D}\mathcal{B}$ is injective at the rows of an open dense set (Lemma D.1) and no row of V vanishes. From $\mathcal{T}_0 = 0$ we get $\eta_V = VZ'$ for a matrix Z' that satisfies (7) for $k+1$ with η' (Lemma D.2), and (D_{k+1}) gives the rest. In the case (+) there is one more term, $t^3\,\mathcal{T}_3(\tilde{w})$, and $\mathcal{T}_3 = 0$ says that Z is antisymmetric (Lemma D.6). The step needs that no row of V vanishes and that the columns of W are linearly independent over $\mathbb{Q}$, which holds for almost all weights.

6.3 GELU and gated blocks on a ray

GELU and SiLU, the gate of a SwiGLU block, do not saturate. Both satisfy $\sigma(z) - \sigma(-z) = z$, so with $\delta(y) = y - \sigma(y)$ we have, for every $y > 0$,

$$\sigma(y) = y - \delta(y), \qquad \sigma(-y) = -\delta(y),$$

and δ is small: for $y \geq 2$,

$$\tfrac{1}{6}\, e^{-y^p/p} \;\leq\; \delta(y) \;\leq\; y\, e^{-y^p/p}, \qquad p = 2 \text{ for GELU}, \quad p = 1 \text{ for SiLU}.$$

So on a ray a GELU hidden unit, and the gate of a hidden unit of a SwiGLU or GeGLU block, differ from a linear or a vanishing one by a small term, whose rate depends on the hidden unit through $|\xi c_j|$. A GELU block has the form of a GeGLU block in which the value of every hidden unit is the number 1, and we treat the three kinds of block together (Lemma D.11).

The identity for one block is expanded in the products of these small terms and of their derivatives, and the products are removed in the order of their rates. The term without a small factor is the identity for the block with its gates *frozen*: every hidden unit that is active on the ray is linear in a GELU block and quadratic in a gated block, and the others vanish. A term with one small factor belongs to one hidden unit. For GELU and GeGLU its coefficient holds the Mills ratio of the normal distribution, which agrees with no function that is meromorphic at infinity (Lemma D.8), and for SwiGLU a parity argument takes the place of this lemma. The terms with two or more small factors follow by induction. At the end the direction does not move what the gates read, and the identity holds with free variables in the place of the outputs of the hidden units, as for $\tanh$.

The later blocks. The coefficients of the expansion come from the network after the block. Under RMSNorm the later blocks read a normalized row, which stays bounded along the ray, and the coefficients are meromorphic at infinity (Lemma D.9). Without normalization the inputs of the later blocks grow along the ray, and their hidden units add small terms of their own. A SwiGLU or GeGLU block is quadratic along the ray up to its small terms, so the inputs of the later blocks grow like the square of its own input. Their small terms are then of a smaller order, and the network after the block can be replaced by a polynomial map (Lemma D.10). A GELU block is linear along the ray, the small terms of all blocks are of the same order, and the block cannot be removed on its own. This is why Theorem 6.1(ii) has no case without normalization.

The parameters. For $\tanh$ blocks the step holds at almost every θ. For GELU and gated blocks we prove the claim at one explicit choice of parameters $\theta^\dagger$, at which every hidden state stays in the positive orthant and the frozen blocks are explicit (Lemma D.12). At $\theta^\dagger$ the Jacobian then has the rank of the bound on some finite batch. The network is analytic in θ, so the Jacobian has this rank on the same batch at generic parameters.

6.4 The case $\varepsilon = 0$: a great circle

For $\varepsilon = 0$ the inputs of a block lie on the sphere $\|n\|^2 = d_{\text{model}}$, and the identity for one block holds only there. A ray leaves the sphere, so we follow a great circle and continue it to complex angles. For orthonormal rows ξ, ξ' let $\nu = \xi + \mathrm{i}\xi'$ and

$$n(t) \;=\; \frac{\sqrt{d_{\text{model}}}}{2}\left(t\,\nu + \frac{1}{t}\,\bar{\nu}\right), \qquad t \in \mathbb{C} \setminus \{0\}.$$

Then $n(t)\, n(t)^\top = d_{\text{model}}$ for all t, without complex conjugation. For $|t| = 1$ the row $n(t)$ is real and runs through the great circle of ξ and ξ', and for real $t \to \infty$ it is a complex row that grows like t. The network after the block extends to a holomorphic map on a complex cone around the real rows, so the identity continues along the path, and there the hidden units of a $\tanh$ or GELU block separate as on a ray (Appendix D.5).

7 What is open, and where the count fails

Theorem 6.1 leaves cases open, and for some kinds of block the count is false at depth. This section says which.

Numerical ranks. The rank of the Jacobian on a large batch of inputs is a lower bound on the functional dimension, and where it meets the bound of Theorem 4.3, the functional dimension is known. We computed it for 96 networks of two and three $\tanh$, GELU, SwiGLU or GeGLU blocks at random weights, without normalization and under pre-RMSNorm with $\varepsilon = 0.1$ and with $\varepsilon = 0$. In every one of them the rank is the bound (Appendix F). This includes the two cases that Theorem 6.1 leaves out for these blocks: GELU blocks without normalization, and SwiGLU and GeGLU blocks with $\varepsilon = 0$. We state the first as a conjecture.

Conjecture 1 (GELU blocks without normalization)**.** *Without normalization, networks of GELU blocks, each with at least d_{model} hidden units, satisfy the equality condition.*

Attention blocks. For attention blocks the count is proved for one block (Theorem 5.1) and is open at depth. For networks of attention blocks with one head, Henry et al. (2025) conjecture that, at any depth, the function determines the weights up to the changes of basis inside and between the blocks. In the networks of two and three attention blocks for which we computed it, the rank is the bound (Appendix F).

Where the count fails. For ReLU, ReLU^2 and ReGLU blocks, the count holds for one block and fails at depth on an open set of weights: a hidden unit of the second block can be inactive at every input, and its weights are then redundant beyond its rescaling (Proposition E.1). ReLU networks lose dimensions in other ways as well, on sets of parameters of positive measure (Grigsby et al., 2023).

8 Discussion

A change of basis at every hidden state removes d_{model}^2 parameters from a network of blocks, and $d_{\text{model}}(d_{\text{model}} - 1)/2$ under pre-RMSNorm, whatever the blocks are. For $\tanh$, SwiGLU and GeGLU blocks, and for GELU blocks under pre-RMSNorm, nothing more can be removed but the rescalings inside the blocks: the redundancy of the network is its symmetries, and the network is locally identifiable up to them. Whether these networks are identifiable, and not only locally, is open. So are the cases of Section 7, and networks with biases.

Skip connections. Most networks in use add the input of a block to its output. The reduction of Section 3 stops there: a skip connection ties the changes of basis of two consecutive hidden states, and one change of basis is left for the whole network (Appendix G). Mijoski and Karbevski (2026) show that, for most kinds of block, almost every network with skip connections computes a function that no network of such blocks without skip connections computes. We do not determine here the functional dimension of networks with skip connections.

Use of AI

The conception of the study is the author's own. The author obtained the results and proofs working with AI assistants, through continuous conversation; the assistants were used to find, verify and simplify proofs.

Code availability

The scripts that compute the numbers of Appendices F and G will be released in a public Git repository, coming soon.

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A Changes of basis and the exact reduction

This appendix proves the statements of Section 3. The tokens of a hidden state are its rows, and a normalization acts on each row, so it suffices to follow one row $h \in \mathbb{R}^{1\times d_{\text{model}}}$.

A.1 RMSNorm, and the proof of Proposition 3.2

Under normalization a read sees $N_\varepsilon(hG)$ in place of hG. The next lemma determines which changes of basis a read can undo.

Lemma A.1 (RMSNorm and changes of basis)**.** *Let $G \in \mathrm{GL}(d_{\text{model}})$. There is a matrix M with $N_\varepsilon(hG) = N_\varepsilon(h)\,M$ for all rows $h \neq 0$ if and only if $G \in \mathrm{O}(d_{\text{model}})$ when $\varepsilon > 0$, and $G \in \mathrm{CO}(d_{\text{model}})$ when $\varepsilon = 0$. Then $M = Q$, where $G = cQ$ with $c > 0$ and Q orthogonal.*

Proof. The case $\varepsilon = 0$. Here $N_\varepsilon(h) = \sqrt{d_{\text{model}}}\,h/\|h\|$. If $G = cQ$, the factor c cancels in $hG/\|hG\|$, so $N_\varepsilon(hG) = N_\varepsilon(h)\,Q$. Conversely, if $hG/\|hG\| = hM/\|h\|$ for all $h \neq 0$, taking norms gives $\|hM\| = \|h\|$, so M is orthogonal, and $hGM^\top = (\|hG\|/\|h\|)\,h$: every nonzero row is a left eigenvector of $GM^\top$, so $GM^\top = cI$ with $c = \|hG\|/\|h\| > 0$, and $G = cM$.

The case $\varepsilon > 0$. For $G = cQ$,

$$N_\varepsilon(h\,cQ) = \frac{c\,hQ}{\sqrt{c^2\|h\|^2/d_{\text{model}} + \varepsilon}}, \qquad N_\varepsilon(h)\,Q = \frac{hQ}{\sqrt{\|h\|^2/d_{\text{model}} + \varepsilon}},$$

and these agree for all h if and only if $c = 1$. If $N_\varepsilon(hG) = N_\varepsilon(h)\,M$ for all h, replace h by $t\,h$ and let $t \to \infty$: both sides tend to their values for $\varepsilon = 0$, so by the first case $G = cM$ with M orthogonal, and the computation just made gives $c = 1$. □

We recall Proposition 3.2.

Proposition 3.2 (Changes of basis preserve the function)**.** *Without normalization or under pre-RMSNorm, let $G_0, \dots, G_L \in \mathcal{G}$. The parameters*

$$\Big(EG_0,\ \ \{G_{\ell-1}^{-1} W_{\text{read}}^{\ell},\ \ W_{\text{write}}^{\ell} G_\ell\}_{\ell=1}^{L},\ \ G_L^{-1} D\Big)$$

give the same function as θ, with the hidden states $h_\ell G_\ell$: one matrix of $\mathcal{G}$ per hidden state. For $\varepsilon = 0$ and $G_\ell = c_\ell Q_\ell$, the same holds with $Q_\ell^\top$ in the place of G_ℓ^{-1}.

Proof. Let h_ℓ be the hidden states at θ, and h'_ℓ those at the new parameters. Then $h'_0 = xEG_0 = h_0 G_0$. Suppose that $h'_{\ell-1} = h_{\ell-1} G_{\ell-1}$. Without normalization, block ℓ reads

$$h'_{\ell-1}\, G_{\ell-1}^{-1} W_{\text{read}}^{\ell} \ = \ h_{\ell-1} W_{\text{read}}^{\ell},$$

as at θ, and it writes $h'_\ell = f_\ell(h_{\ell-1} W_{\text{read}}^{\ell})\, W_{\text{write}}^{\ell} G_\ell = h_\ell G_\ell$. Under pre-RMSNorm, $G_{\ell-1} = Q$ is orthogonal and the block reads $N_\varepsilon(h_{\ell-1} Q)\, Q^\top W_{\text{read}}^{\ell} = N_\varepsilon(h_{\ell-1})\, W_{\text{read}}^{\ell}$, by Lemma A.1. For $\varepsilon = 0$ and $G_{\ell-1} = cQ$, it reads $N_\varepsilon(h_{\ell-1}\, cQ)\, Q^\top W_{\text{read}}^{\ell} = N_\varepsilon(h_{\ell-1})\, W_{\text{read}}^{\ell}$, by the same lemma. So $h'_\ell = h_\ell G_\ell$ for every ℓ, and the readout computes $N(h_L G_L)\, G_L^{-1} D = N(h_L)\, D$, with $Q_L^\top$ in the place of G_L^{-1} for $\varepsilon = 0$. □

The change of basis at h_ℓ changes only the writes into h_ℓ and the reads of h_ℓ. So the changes of basis at different hidden states commute, and each weight matrix of the network is multiplied on each side by the change of basis at one hidden state only.

A.2 The exact reduction (Proposition 3.3)

We recall the proposition, and prove it together with its form for the rows of a write.

Proposition 3.3 (Exact reduction)**.** *Let d_{model} columns of the read of h_ℓ ($W_{\text{read}}^{\ell+1}$, or D if $\ell = L$) form an invertible matrix W_1.*
(i) *Without normalization, the change of basis $G = W_1$ at h_ℓ preserves the function and turns W_1 into $G^{-1} W_1 = I$.*
(ii) *Under pre-RMSNorm, let $W_1 = QR$ with Q orthogonal and R upper triangular. The change of basis Q at h_ℓ preserves the function and turns W_1 into $Q^\top W_1 = R$.*
At generic parameters every hidden state has such columns, and these changes of basis at all $L+1$ hidden states remove $(L{+}1)\dim \mathcal{G}$ parameters: $(L{+}1)d_{\text{model}}^2$ in (i), and $(L{+}1)d_{\text{model}}(d_{\text{model}}{-}1)/2$ in (ii).

Proof. By Proposition 3.2, a change of basis $G \in \mathcal{G}$ at h_ℓ preserves the function and replaces every read W of h_ℓ by $G^{-1}W$ and every write W into h_ℓ by WG. So the columns of a read that form W_1 become $G^{-1}W_1$, and the rows of a write that form a matrix W_1 become $W_1 G$. Without normalization $\mathcal{G} = \mathrm{GL}(d_{\text{model}})$, and $G = W_1$ for a read, or $G = W_1^{-1}$ for a write, turns W_1 into I. Under pre-RMSNorm, $\mathcal{G} = \mathrm{O}(d_{\text{model}})$. For a read, let $W_1 = QR$ be a QR factorization, with R upper triangular: the change of basis Q gives $Q^\top W_1 = R$. For a write, let $W_1^\top = QR$; then $W_1 = R^\top Q^\top$, and the change of basis Q gives $W_1 Q = R^\top$, which is lower triangular. The identity fixes all d_{model}^2 entries of W_1, and the triangular form fixes the $d_{\text{model}}(d_{\text{model}} - 1)/2$ entries on one side of its diagonal. The changes of basis at different hidden states act on different sides of each weight matrix, so the reductions at all $L+1$ hidden states can be made together, one submatrix per hidden state. At generic parameters every d_{model} columns of a read, and every d_{model} rows of a write, form an invertible matrix. □

The next example shows the reduction on the rows of a write.

Example A.2 (Choosing rows)**.** Let $d_{\text{model}} = 2$ and $d_{\text{hidden}} = 3$, and let W_1 consist of rows 1 and 3 of W_{write}^{ℓ}, outlined in red. The change of basis $G = W_1^{-1}$ at h_ℓ maps

$$W_{\text{write}}^{\ell} = \begin{bmatrix} \boxed{2 & 1} \\ 0 & 3 \\ \boxed{1 & 1} \end{bmatrix} \quad \longmapsto \quad W_{\text{write}}^{\ell} G = \begin{bmatrix} \boxed{1 & 0} \\ -3 & 6 \\ \boxed{0 & 1} \end{bmatrix}, \qquad G = W_1^{-1} = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix},$$

and the next read $W_{\text{read}}^{\ell+1}$ to $G^{-1}W_{\text{read}}^{\ell+1} = W_1 W_{\text{read}}^{\ell+1}$. The two outlined rows now hold $d_{\text{model}}^2 = 4$ fixed entries, and the other row has changed. Permuting the hidden units of the block moves the chosen rows next to each other. At generic parameters any d_{model} rows form an invertible matrix, so in practice one takes a $d_{\text{model}} \times d_{\text{model}}$ block of the matrix, for example its first d_{model} rows.

A.3 LayerNorm, and the count for a ViT-B

The reduction of LayerNorm to RMSNorm is standard: the centering folds into the adjacent weights (Elhage et al., 2021), and centering every write gives an RMSNorm network whose hidden states have mean zero (Ashkboos et al., 2024; Theus et al., 2025). We state it with the constants that the counts need.

Proposition A.3 (LayerNorm reduces to RMSNorm)**.** *Let* $\mathbf{1}$ *be the column of* d_{model} *ones,* $Z = I - \frac{1}{d_{\text{model}}}\mathbf{1}\mathbf{1}^\top$*, and* $\mathcal{V}$ *the space of the rows* h *with* $h\mathbf{1} = 0$*. Consider a pre-LayerNorm network, whose reads and readout see* $\mathrm{LN}(h) = \gamma \odot N_\varepsilon(hZ) + \beta$*, with rows* γ *and* β *and the entrywise product* $\odot$*.*
(i) *Replacing each read* W *by* $\mathrm{diag}(\gamma)W$ *with the added bias* βW*, and each write* W*, write bias* b *and embedding* E *by* WZ*,* bZ *and* EZ*, gives a network that computes the same function, whose hidden states lie in* $\mathcal{V}$*, and whose reads see* $N_\varepsilon(h)$*.*
(ii) *In an orthonormal basis of* $\mathcal{V}$*, this network is a pre-RMSNorm network of dimension* $d_{\text{model}} - 1$*, with every read multiplied by* $\sqrt{d_{\text{model}}/(d_{\text{model}} - 1)}$ *and* ε *replaced by* $\varepsilon d_{\text{model}}/(d_{\text{model}} - 1)$*.*

Proof. (i) A read satisfies $\mathrm{LN}(h)\,W = N_\varepsilon(hZ)\,\mathrm{diag}(\gamma)W + \beta W$. Every hidden state is the output of a write, so replacing every write, write bias and the embedding by its image under Z replaces every hidden state h by hZ. Since $Z^2 = Z$, every read then sees the same $N_\varepsilon(hZ)$ as before, and the function is unchanged; the hidden states lie in $\mathcal{V}$. (ii) Let the columns of $\bar{U} \in \mathbb{R}^{d_{\text{model}} \times (d_{\text{model}}-1)}$ be an orthonormal basis of $\mathcal{V}$, written as columns, and $h = y\bar{U}^\top$ with a row y. Then

$$\frac{\|h\|^2}{d_{\text{model}}} + \varepsilon \;=\; \frac{d_{\text{model}} - 1}{d_{\text{model}}}\Big(\frac{\|y\|^2}{d_{\text{model}} - 1} + \frac{\varepsilon\, d_{\text{model}}}{d_{\text{model}} - 1}\Big),$$

and therefore

$$N_\varepsilon(y\bar{U}^\top) \;=\; \sqrt{\frac{d_{\text{model}}}{d_{\text{model}} - 1}}\; N'(y)\,\bar{U}^\top,$$

where N' is RMSNorm of dimension $d_{\text{model}} - 1$ with the constant $\varepsilon d_{\text{model}}/(d_{\text{model}} - 1)$. So a read W of $N_\varepsilon(y\bar{U}^\top)$ is the read $\sqrt{d_{\text{model}}/(d_{\text{model}} - 1)}\,\bar{U}^\top W$ of $N'(y)$, and a write WZ is the write $W\bar{U}$ in the coordinates y. □

The ViT-B of Ji et al. The ViT-B that Ji et al. (2026) train without skip connections has $d_{\text{model}} = 768$ and 12 layers, so $L = 24$ blocks and 25 hidden states: one before each block and one before the readout. We assume standard pre-LayerNorm blocks, which reduce to RMSNorm of dimension 767 by Proposition A.3. By Proposition 3.3(ii), one triangular 767×767 submatrix of a read per hidden state removes

$$25 \cdot \frac{767 \cdot 766}{2} \;=\; 7{,}344{,}025$$

parameters, 8.5 percent of the 86.6M parameters of the model.

B Symmetries inside the blocks, and the upper bound

The symmetries inside the blocks, with their sources. Permutations of hidden units or heads and sign flips of $\tanh$ hidden units are finite and remove no parameters (Fefferman, 1994; Sussmann,

1992). The continuous symmetries inside a block are the following, where c_i is the read column of hidden unit i, or its gate column in a gated block, e_i its value column, and a_i its write row:

ReLU:	$(c_i, a_i) \mapsto (\lambda c_i,\ a_i/\lambda)$,	$\lambda > 0$;
ReLU^2:	$(c_i, a_i) \mapsto (\lambda c_i,\ a_i/\lambda^2)$,	$\lambda > 0$;
SwiGLU, GeGLU, ReGLU:	$(e_i, a_i) \mapsto (e_i/\lambda,\ \lambda a_i)$,	$\lambda \neq 0$;
ReGLU, in addition:	$(c_i, a_i) \mapsto (\lambda c_i,\ a_i/\lambda)$,	$\lambda > 0$;
attention, head i:	$(W_{Q,i}, W_{K,i}) \mapsto (W_{Q,i}K,\ W_{K,i}K^{-\top})$,	$K \in \mathrm{GL}(d_h)$;
	$(W_{V,i}, W_{O,i}) \mapsto (W_{V,i}K',\ K'^{-1}W_{O,i})$,	$K' \in \mathrm{GL}(d_h)$.

Here $W_{Q,i}, W_{K,i}, W_{V,i} \in \mathbb{R}^{d_{\text{model}} \times d_h}$ and $W_{O,i} \in \mathbb{R}^{d_h \times d_{\text{model}}}$ are the matrices of head i. The rescaling of a ReLU hidden unit is that of Rolnick and Kording (2020) and Godfrey et al. (2022), and the changes of basis inside a head are those of Elhage et al. (2021) and Theus et al. (2025). So a block has $s = d_{\text{hidden}}$ for ReLU, ReLU^2, SwiGLU and GeGLU, $s = 2d_{\text{hidden}}$ for ReGLU, $s = 2Hd_h^2$ for attention, and $s = 0$ for $\tanh$ and GELU. For a coordinate-wise activation σ, Godfrey et al. (2022, Theorem E.14) determine the matrices A with $\sigma \circ A = B \circ \sigma$ for some matrix B, and for GELU they are the permutation matrices (their Appendix E.2.1).

The changes of basis of Proposition 3.2 and the symmetries inside the blocks act *locally freely* at θ if no continuous family of them, other than the identity, fixes θ.

Lemma B.1 (Locally free action)**.** *Without normalization or under pre-RMSNorm, consider a network whose blocks are* $\tanh$*, GELU, ReLU, ReLU^2 or gated blocks with at least d_{model} hidden units, or attention blocks whose heads have width $d_h \leq d_{\text{model}}$ and total width $Hd_h \geq d_{\text{model}}$. If $d_{\text{in}} \geq d_{\text{model}}$, then at generic parameters the changes of basis and the symmetries inside the blocks act locally freely.*

Proof. Let changes of basis $G_0, \ldots, G_L \in \mathcal{G}$ and symmetries inside the blocks, all close to the identity, fix a generic θ. The embedding becomes $EG_0 = E$, and E has rank d_{model}, so $G_0 = I$. Suppose that $G_{\ell-1} = I$. In an MLP block, the read then becomes $W^\ell_{\text{read}} \operatorname{diag}(\lambda)$, where λ holds the numbers by which the symmetries inside the block multiply its columns, with entries 1 on the columns that they leave alone. No column of a generic W^ℓ_{read} vanishes, so every entry of λ is 1: for a ReGLU block this fixes both rescalings of every hidden unit, and for the other kinds the one rescaling. The write then becomes $W^\ell_{\text{write}} G_\ell = W^\ell_{\text{write}}$, and W^ℓ_{write} has rank d_{model} because the block has at least d_{model} hidden units, so $G_\ell = I$. In an attention block, the changes of basis inside the heads map $W_{Q,i} \mapsto W_{Q,i}K_i$ and $W_{V,i} \mapsto W_{V,i}K'_i$. At generic parameters these matrices have full column rank, as $d_h \leq d_{\text{model}}$, so $K_i = K'_i = I$. The write, which stacks the $W_{O,i}$, then becomes $W_O G_\ell = W_O$ and has rank d_{model}, as $Hd_h \geq d_{\text{model}}$, so $G_\ell = I$. By induction only the identity fixes θ. For $\mathcal{G} = \mathrm{O}(d_{\text{model}})$ the argument is the same. □

Remark B.2 (The width condition). The condition on the width is needed. At a hidden state that is written and read at rank $m < d_{\text{model}}$, the changes of basis that fix the parameters form a family of dimension $(d_{\text{model}} - m)^2$, so they move the parameters in only $2d_{\text{model}}m - m^2$ independent directions, as Zhao et al. (2023, Proposition C.3) compute for one hidden layer.

We recall Theorem 4.3.

Theorem 4.3 (Upper bound)**.** *Without normalization or under pre-RMSNorm, at generic parameters, the functional dimension of a network of L blocks is at most $P - (L+1)\dim \mathcal{G} - s$.*

Proof. If a continuous family of maps of the parameters preserves the function, then the function does not change to first order along any direction in which the family moves θ. So these directions lie in the kernel of the Jacobian on every batch, and the functional dimension at θ is at most P minus the dimension of their span. The changes of basis of Proposition 3.2, one copy of $\mathcal{G}$ per hidden state, and the symmetries inside the blocks commute with each other: the first multiply W^ℓ_{write} on the right and $W^{\ell+1}_{\text{read}}$ on the left, and the second multiply W^ℓ_{read} on the right and W^ℓ_{write} on the left. Together

they form a group of dimension $(L+1)\dim\mathcal{G}+s$, which preserves the function and acts locally freely at generic parameters (Lemma B.1). So at generic parameters it moves θ in $(L+1)\dim\mathcal{G}+s$ independent directions. □

We recall Proposition 4.6.

Proposition 4.6 (The equality condition and local identifiability)**.** *Let the blocks be* tanh*, GELU, SwiGLU, GeGLU or attention blocks, without normalization or under pre-RMSNorm. A network satisfies the equality condition if and only if, at generic parameters, it is locally identifiable up to its symmetries. Every symmetry that is needed is then close to the identity.*

Proof. Let $k=(L+1)\dim\mathcal{G}+s$ be the dimension of the symmetries of the network. Let r be the largest rank that the Jacobian of F_X attains on any batch X at any parameters, and fix a batch X that attains it. The outputs are analytic in θ, so the rank on X is r outside the zero set of the sum of the squares of the minors of size r, a set of measure zero. Hence r is the functional dimension at generic parameters, $r\leq P-k$ by Theorem 4.3, and the equality condition is $r=P-k$.

Take θ at which the rank on X is r and the symmetries act locally freely, as they do at generic parameters (Lemma B.1). By lower semicontinuity the rank on X is r on a neighborhood U of θ. By the constant rank theorem, after shrinking U, the set $\mathcal{Z}=\{\theta'\in U : F_X(\theta')=F_X(\theta)\}$ is a connected manifold of dimension $P-r$, whose tangent space at each point is the kernel of the Jacobian. For any further input x, the Jacobian on $X\cup\{x\}$ has rank r on U as well, so it has the same kernel, and $\theta'\mapsto F_{\theta'}(x)$ is constant on $\mathcal{Z}$. So $\mathcal{Z}=\{\theta'\in U : F_{\theta'}=F_\theta\}$. The changes of basis of Proposition 3.2, the rescalings and the changes of basis inside the heads form a Lie group of dimension k, which acts smoothly. It acts locally freely at θ, so the map that sends an element of the group to the image of θ under it is an immersion near the identity, and it maps a neighborhood of the identity onto a manifold $\mathcal{O}\subset\mathcal{Z}$ of dimension k.

If $r=P-k$, then $\mathcal{O}$ is open in $\mathcal{Z}$, since the two have the same dimension. So every θ' close to θ with $F_{\theta'}=F_\theta$ is the image of θ under a symmetry close to the identity. Conversely, let every θ' in a neighborhood of θ in $\mathcal{Z}$ be the image of θ under a symmetry. With the permutations and the sign flips, the symmetries form a Lie group of dimension k with finitely many components, and the images of θ under it form the image of a manifold of dimension k under a smooth map. By Sard's theorem this image contains no open subset of a manifold of larger dimension, so $P-r\leq k$, which with $r\leq P-k$ is the equality condition. □

C The one-block theorem

This appendix states and proves the general form of Theorem 5.1. With one block and without normalization, the network is $x\mapsto f(xEW_{\text{read}})\,W_{\text{write}}D$.

Theorem C.1 (One block)**.** *Let $d_{\text{model}}\geq 2$ and $d_{\text{in}}, d_{\text{out}}\geq d_{\text{model}}$, let inputs range over all of $\mathbb{R}^{T\times d_{\text{in}}}$, and consider networks with one block, without normalization and without biases. The block is a* tanh*, GELU, ReLU, ReLU^2, ReGLU, SwiGLU or GeGLU block with at least d_{model} hidden units, or an attention block of H heads of width $d_h\leq d_{\text{model}}$ with $Hd_h\geq d_{\text{model}}$, causal or not, on sequences of at least two tokens. At generic parameters the functional dimension is*

$$P-2d_{\text{model}}^2-s,$$

where s is the dimension of the symmetries inside the block (Appendix B).

Notation and flat directions. We write $F(\theta,x)$ for the output of the network at the parameters $\theta\in\mathbb{R}^P$ and the input x, and $\mathrm{D}_\theta F(\theta,x)$ for the Fréchet derivative of $F(\cdot,x)$ at θ, a linear map on $\mathbb{R}^P$. The derivative is written with an upright D so that it is not confused with the readout matrix D of the network. For a direction $\eta\in\mathbb{R}^P$, $\mathrm{D}_\theta F(\theta,x)\,\eta$ is the derivative of $\tau\mapsto F(\theta+\tau\eta,x)$ at $\tau=0$. We write η_E, η_D, η_{read} and η_{write} for the entries of η at the positions of E, D, W_{read} and W_{write}, and $\mathrm{D}_\theta Q\,\eta$ for the derivative along η of a matrix Q that depends on the parameters;

for instance $\mathrm{D}_\theta C\,\eta = \eta_E W_{\mathrm{read}} + E\,\eta_{\mathrm{read}}$ for $C = EW_{\mathrm{read}}$, by the product rule. Call η *flat* if $\mathrm{D}_\theta F(\theta, x)\,\eta = 0$ for every input x, and let $\mathcal{K}(\theta)$ be the space of flat directions. For ReLU, ReLU^2 and ReGLU blocks, $F(\theta, x)$ is differentiable in θ at the inputs x at which no hidden unit sits at its kink, which form an open set of full measure, and "every input" means every such input. For a batch of inputs $x_1, \ldots, x_n$, the Jacobian of $\theta \mapsto (F(\theta, x_1), \ldots, F(\theta, x_n))$ is the matrix of the linear map $\eta \mapsto (\mathrm{D}_\theta F(\theta, x_1)\,\eta, \ldots, \mathrm{D}_\theta F(\theta, x_n)\,\eta)$. It has P columns, and its kernel consists of the directions with $\mathrm{D}_\theta F(\theta, x_k)\,\eta = 0$ for the inputs of the batch. Adding inputs can only shrink this kernel, and a decreasing chain of subspaces of $\mathbb{R}^P$ stops, so some finite batch has the kernel $\mathcal{K}(\theta)$. By rank–nullity, the functional dimension at θ is therefore

$$P - \dim \mathcal{K}(\theta).$$

The theorem thus amounts to finding the flat directions.

Overview. For an MLP block, the parameters enter the function only through $C = EW_{\mathrm{read}}$ and $A = W_{\mathrm{write}}D$. Step 1 finds the directions that keep these two matrices fixed: they are the directions in which the changes of basis at h_0 and at h_1 move the parameters. Step 2 shows that a flat direction changes C and A only along the symmetries of the hidden units: each hidden unit is isolated, by its kink for ReLU, ReLU^2 and ReGLU, and for $\tanh$, GELU, SwiGLU and GeGLU by how fast it approaches its limit on rays $x = tn$ with $t \to \infty$. Step 3 counts. For an attention block the same steps run with one factorization per head, and inputs of two tokens turn Step 2 into the case of $\tanh$.

MLP blocks

Reduction. An MLP block acts on each token separately, so we take $T = 1$. The network computes $F(\theta, x) = f(xC)\,A$. For a $\tanh$, GELU, ReLU or ReLU^2 block let c_i be the columns of C, one per hidden unit, a_i the rows of A, and $z_i = xc_i$, so that $F(\theta, x) = \sum_i \sigma(z_i)\,a_i$. For a gated block, $C = [\,EW_g \mid EW_v\,]$, hidden unit i has the gate column c_i, a column of EW_g, and the value column e_i, a column of EW_v, and $F(\theta, x) = \sum_i \sigma(xc_i)\,(xe_i)\,a_i$. For generic parameters E, D, W_{read} and W_{write} have rank d_{model}, every a_i is nonzero, and the c_i are nonzero and pairwise non-collinear: they lie in the column space of E, of dimension $d_{\mathrm{model}} \geq 2$, where two generic vectors are not collinear. In a gated block, in the same way, every e_i is nonzero and is not proportional to c_i.

Step 1: changes of basis. We need the following lemma.

> **Lemma C.2** (The directions that keep C and A)**.** *Let E, D, W_{read} and W_{write} have rank d_{model}. A direction η keeps C and A fixed to first order, $\mathrm{D}_\theta C\,\eta = 0$ and $\mathrm{D}_\theta A\,\eta = 0$, if and only if*
>
> $$\eta_E = EX, \qquad \eta_{\mathrm{read}} = -XW_{\mathrm{read}}, \qquad \eta_{\mathrm{write}} = W_{\mathrm{write}}Y, \qquad \eta_D = -YD$$
>
> *for some $X, Y \in \mathbb{R}^{d_{\mathrm{model}} \times d_{\mathrm{model}}}$. The matrices X and Y are determined by the direction, so these directions form a space of dimension $2d_{\mathrm{model}}^2$.*
>
> *Proof.* Since W_{read} has rank d_{model}, it has a right inverse R, with $W_{\mathrm{read}}R = I$, and E, of rank d_{model} with d_{model} columns, has a left inverse. From $\mathrm{D}_\theta C\,\eta = \eta_E W_{\mathrm{read}} + E\eta_{\mathrm{read}} = 0$, multiplying on the right by R gives $\eta_E = EX$ with $X = -\eta_{\mathrm{read}}R$; then $E\eta_{\mathrm{read}} = -EXW_{\mathrm{read}}$, and cancelling E on the left gives $\eta_{\mathrm{read}} = -XW_{\mathrm{read}}$. In the same way, D has a right inverse and W_{write} a left inverse, and $\mathrm{D}_\theta A\,\eta = \eta_{\mathrm{write}}D + W_{\mathrm{write}}\eta_D = 0$ gives $\eta_{\mathrm{write}} = W_{\mathrm{write}}Y$ and $\eta_D = -YD$. Conversely these directions give $\mathrm{D}_\theta C\,\eta = EXW_{\mathrm{read}} - EXW_{\mathrm{read}} = 0$ and $\mathrm{D}_\theta A\,\eta = 0$. Finally $\eta_E = EX$ determines X, and $\eta_D = -YD$ determines Y. □

These are the directions in which the changes of basis $I + \tau X$ at h_0 and $I + \tau Y$ at h_1 move the parameters.

Step 2: hidden units. We need a lemma on the function

$$G(\phi, x) = xM + \gamma + \textstyle\sum_i \sigma(z_i)\,a_i, \qquad z_i = xc_i + \beta_i,$$

of the matrices $\phi = \big(M, \gamma, (c_i, \beta_i, a_i)_i\big)$ themselves. For a direction ψ of ϕ, with entries ψ_M, ψ_γ, ψ_{c_i}, ψ_{β_i} and ψ_{a_i}, the Fréchet derivative of $G(\cdot, x)$ at ϕ gives

$$\mathrm{D}_\phi G(\phi, x)\,\psi = x\psi_M + \psi_\gamma + \sum_i \big[\sigma(z_i)\,\psi_{a_i} + \sigma'(z_i)\,(x\psi_{c_i} + \psi_{\beta_i})\,a_i\big]. \tag{9}$$

A gated unit computes $\sigma(z_i)\,w_i\,a_i$ with a gate $z_i = xc_i + \beta_i$ and a value $w_i = xe_i + \beta'_i$, where $\sigma(z) = \max(z, 0)$ for ReGLU, $z/(1 + e^{-z})$ for SwiGLU and $z\Phi(z)$ for GeGLU, so that ϕ also contains the e_i and β'_i, and the terms of the unit in $\mathrm{D}_\phi G(\phi, x)\,\psi$ are $\sigma(z_i)\,w_i\,\psi_{a_i} + \sigma'(z_i)\,(x\psi_{c_i} + \psi_{\beta_i})\,w_i\,a_i + \sigma(z_i)\,(x\psi_{e_i} + \psi_{\beta'_i})\,a_i$.

Lemma C.3 (A flat direction only rescales units)**.** *Let every $a_i \neq 0$ and let the c_i be nonzero and pairwise non-collinear; for the gated blocks let also every $e_i \neq 0$, with (e_i, β'_i) not proportional to (c_i, β_i). If $\mathrm{D}_\phi G(\phi, x)\,\psi = 0$ for all x, then $\psi_M = 0$, $\psi_\gamma = 0$, and for every unit i:*
for $\tanh$ *and GELU, $\psi_{a_i} = 0$, $\psi_{c_i} = 0$ and $\psi_{\beta_i} = 0$;*
for ReLU, $(\psi_{c_i}, \psi_{\beta_i}) = \lambda_i(c_i, \beta_i)$ and $\psi_{a_i} = -\lambda_i a_i$ for some number λ_i;
for ReLU^2, $(\psi_{c_i}, \psi_{\beta_i}) = \lambda_i(c_i, \beta_i)$ and $\psi_{a_i} = -2\lambda_i a_i$;
for ReGLU, $(\psi_{c_i}, \psi_{\beta_i}) = \lambda_i(c_i, \beta_i)$, $(\psi_{e_i}, \psi_{\beta'_i}) = -\alpha_i(e_i, \beta'_i)$ and $\psi_{a_i} = (\alpha_i - \lambda_i)a_i$ for some numbers λ_i and α_i;
for SwiGLU and GeGLU, $\psi_{c_i} = 0$, $\psi_{\beta_i} = 0$, $(\psi_{e_i}, \psi_{\beta'_i}) = -\alpha_i(e_i, \beta'_i)$ and $\psi_{a_i} = \alpha_i a_i$ for some number α_i.

Proof. Let $H_i = \{x : z_i = 0\}$, a hyperplane since $c_i \neq 0$. Since the c_i are pairwise non-collinear, the H_i are pairwise non-parallel, so every H_i contains points that lie on no other H_k. We use one fact from linear algebra: an affine function that vanishes on an open piece of H_i is a multiple of z_i, since it vanishes on all of H_i and H_i has codimension one. We write $J(x) = \mathrm{D}_\phi G(\phi, x)\,\psi$, a function of x alone, which vanishes by assumption.

ReLU. Here $\sigma' = \mathbf{1}[z > 0]$ jumps at 0. Near a point x_0 of H_i that lies on no other H_k, every term of (9) is continuous except $\mathbf{1}[z_i > 0](x\psi_{c_i} + \psi_{\beta_i})\,a_i$, which jumps by $(x_0\psi_{c_i} + \psi_{\beta_i})\,a_i$ as x crosses H_i at x_0. As $J = 0$ is continuous, this jump vanishes, and since $a_i \neq 0$, the affine function $x\psi_{c_i} + \psi_{\beta_i}$ vanishes on an open piece of H_i, so $(\psi_{c_i}, \psi_{\beta_i}) = \lambda_i(c_i, \beta_i)$. The terms of unit i then add up to $\max(z_i, 0)\,(\psi_{a_i} + \lambda_i a_i)$, since $z\,\mathbf{1}[z > 0] = \max(z, 0)$. Across H_i, the gradient of J with respect to x jumps by the matrix $c_i(\psi_{a_i} + \lambda_i a_i)$, which must vanish, so $\psi_{a_i} = -\lambda_i a_i$. With every unit removed in this way, $J(x) = x\psi_M + \psi_\gamma = 0$ for all x gives $\psi_M = 0$ and $\psi_\gamma = 0$.

ReLU^2. The same argument one derivative higher: $\sigma'(z) = 2\max(z, 0)$ is continuous, and across H_i the gradient of $2\max(z_i, 0)(x\psi_{c_i} + \psi_{\beta_i})\,a_i$ jumps by $2c_i(x_0\psi_{c_i} + \psi_{\beta_i})\,a_i$, which forces $(\psi_{c_i}, \psi_{\beta_i}) = \lambda_i(c_i, \beta_i)$. The terms of unit i then add up to $\max(z_i, 0)^2(\psi_{a_i} + 2\lambda_i a_i)$, whose second derivative jumps across H_i, so $\psi_{a_i} = -2\lambda_i a_i$.

ReGLU. As (e_i, β'_i) is nonzero and not proportional to (c_i, β_i), the affine function w_i does not vanish on all of H_i, so it is nonzero on an open piece of H_i that meets no other H_k; we take x_0 there. Across H_i the terms of unit i jump by $(x_0\psi_{c_i} + \psi_{\beta_i})\,w_i(x_0)\,a_i$, so $(\psi_{c_i}, \psi_{\beta_i}) = \lambda_i(c_i, \beta_i)$ as before. The terms of unit i then add up to $\max(z_i, 0)\,V_i(x)$ with the affine function $V_i(x) = w_i(x)\,(\psi_{a_i} + \lambda_i a_i) + (x\psi_{e_i} + \psi_{\beta'_i})\,a_i$. The jump of their gradient across H_i is $c_i V_i(x_0)$, so V_i vanishes on an open piece of H_i, and $V_i(x) = z_i\,\nu_i$ for a row vector ν_i. The terms of unit i then add up to $\max(z_i, 0)^2\,\nu_i$, whose second derivative jumps across H_i, so $\nu_i = 0$ and $V_i = 0$ for all x: $w_i(x)\,(\psi_{a_i} + \lambda_i a_i) = -(x\psi_{e_i} + \psi_{\beta'_i})\,a_i$. Both sides are products of an affine function of x and a fixed row vector, and $a_i \neq 0$, so $\psi_{a_i} + \lambda_i a_i = \alpha_i a_i$ and $(\psi_{e_i}, \psi_{\beta'_i}) = -\alpha_i(e_i, \beta'_i)$ for some number α_i. Then $\psi_M = 0$ and $\psi_\gamma = 0$ as for ReLU.

$\tanh$. We restrict (9) to a ray $x = tn$, $t > 0$, with a direction n such that the numbers $u_k = nc_k$ are all nonzero and $|u_j| < |u_k|$ for one unit j and all $k \neq j$. For each j and each sign $s = \pm 1$, the directions n with $su_j > 0$ and this property form a nonempty open set: start from a point of the hyperplane $nc_j = 0$ where no other nc_k vanishes, and move a little to the side s. For

$z > 0$,

$$\tanh z = 1 - 2e^{-2z} + r(z), \quad 0 \le r(z) \le 2e^{-4z},$$
$$\tanh' z = 4e^{-2z} - q(z), \quad 0 \le q(z) \le 8e^{-4z},$$

from $\tanh z = (1 - e^{-2z})/(1 + e^{-2z})$ and $\tanh' = 1 - \tanh^2$, and $\tanh$ is odd. On the ray $z_k = u_k t + \beta_k$, so for large t the terms of unit k in (9) are

$$s_k \psi_{a_k} + e^{-2|u_k|t}\, e^{-2 s_k \beta_k} \big[-2 s_k \psi_{a_k} + 4\,(t\, n\psi_{c_k} + \psi_{\beta_k})\, a_k \big] + O\big((1+t)\, e^{-4|u_k|t}\big),$$

with s_k the sign of u_k. All terms of $J(tn)$ are therefore $t\, n\psi_M$, a constant, the bracket of unit j times $e^{-2|u_j|t}$, and terms that are $O\big((1+t)e^{-\rho t}\big)$ with $\rho = \min\big(4|u_j|, \min_{k \ne j} 2|u_k|\big) > 2|u_j|$. As $J(tn) = 0$ for all t, dividing by t and letting $t \to \infty$ gives $n\psi_M = 0$; then the constant vanishes; then multiplying by $e^{2|u_j|t}$ and letting $t \to \infty$ shows that the bracket of unit j, a polynomial of degree one in t, tends to zero, so both of its coefficients vanish: $(n\psi_{c_j})\, a_j = 0$ and $2\psi_{\beta_j} a_j = s\, \psi_{a_j}$. As $a_j \ne 0$, $n\psi_{c_j} = 0$ for all n in an open set, so $\psi_{c_j} = 0$; and the two signs $s = \pm 1$ give $\psi_{a_j} = 2\psi_{\beta_j} a_j = -2\psi_{\beta_j} a_j$, so $\psi_{\beta_j} = 0$ and $\psi_{a_j} = 0$. This holds for every unit j, and then $J(x) = x\psi_M + \psi_\gamma = 0$ gives $\psi_M = 0$ and $\psi_\gamma = 0$.

GELU. Here $\sigma(z) = z\Phi(z)$, with Φ the standard normal distribution function and $\varphi = \Phi'$ its density. The tail $\bar\Phi = 1 - \Phi$ satisfies $\varphi(z)(1/z - 1/z^3) \le \bar\Phi(z) \le \varphi(z)/z$ for $z > 0$, as one checks by differentiating, and $\Phi(-z) = 1 - \Phi(z)$ carries these bounds to $z < 0$. From these, as $|z| \to \infty$ on either side,

$$\sigma(z) = \max(z, 0) - \varphi(z) + O\big(\varphi(z)/z^2\big),$$
$$\sigma'(z) = \mathbf{1}[z > 0] + \varphi(z)\,(z - 1/z) + O\big(\varphi(z)/|z|^3\big).$$

Take a ray $x = tn$ as for $\tanh$. The terms of unit k in (9) split into a polynomial of degree at most one in t, from $\max(z, 0)$ and $\mathbf{1}[z > 0]$, and the Gaussian part

$$\varphi(z_k) \Big[u_k\,(n\psi_{c_k})\, t^2 + \big(u_k \psi_{\beta_k} + \beta_k\, n\psi_{c_k}\big)\, t + \beta_k \psi_{\beta_k} - (n\psi_{c_k})/u_k \Big] a_k \\ - \ \varphi(z_k)\, \psi_{a_k} \ + \ O\big(\varphi(z_k)/t\big).$$

Gaussian parts tend to zero faster than any power of t, so as $J(tn) = 0$ for all t, the polynomial part vanishes, and so does the sum of the Gaussian parts. For $k \ne j$, $\varphi(z_k)/\varphi(z_j) = \exp\big(-\frac{1}{2}(u_k^2 - u_j^2)\, t^2 + O(t)\big)$ tends to zero faster than any power of t, since $u_k^2 > u_j^2$. Dividing the sum of the Gaussian parts by $\varphi(z_j)$ and letting $t \to \infty$ therefore makes the coefficients of t^2, t and 1 in the bracket of unit j vanish, in this order: $u_j(n\psi_{c_j})\, a_j = 0$, so $n\psi_{c_j} = 0$; then $u_j \psi_{\beta_j}\, a_j = 0$, so $\psi_{\beta_j} = 0$; then $\psi_{a_j} = 0$. As for $\tanh$, $n\psi_{c_j} = 0$ on an open set gives $\psi_{c_j} = 0$, and finally $\psi_M = 0$ and $\psi_\gamma = 0$.

SwiGLU and GeGLU. Take a ray $x = tn$ as for $\tanh$, and write $v_k = ne_k$, so that $w_k = v_k t + \beta'_k$; as $e_j \ne 0$, also $v_j \ne 0$ on an open set of such n. For SwiGLU, $\sigma(z) = z - z/(1 + e^z)$ gives, as $|z| \to \infty$ with s the sign of z,

$$\sigma(z) = \max(z, 0) - |z| e^{-|z|} + O\big(|z| e^{-2|z|}\big),$$
$$\sigma'(z) = \mathbf{1}[z > 0] + s\,(|z| - 1)\, e^{-|z|} + O\big(|z| e^{-2|z|}\big).$$

The terms of unit k in (9) then split into a polynomial in t, from $\max(z, 0)$ and $\mathbf{1}[z > 0]$, and a rest that is $e^{-|z_k|}$ times a polynomial in t, up to $O(t^3 e^{-2|z_k|})$; for GeGLU, with the expansions of GELU, the rest is $\varphi(z_k)$ times a polynomial in t and $1/z_k$, up to $O(\varphi(z_k)/t)$. As for $\tanh$ and GELU, the polynomial parts vanish, and dividing the sum of the rests by $e^{-|z_j|}$, respectively $\varphi(z_j)$, and letting $t \to \infty$ shows that the rest of the slowest unit j vanishes. For SwiGLU this rest is $e^{-|z_j|}$ times

$$s\,(|z_j| - 1)\,(t\, n\psi_{c_j} + \psi_{\beta_j})\, w_j\, a_j - |z_j| \big((t\, n\psi_{e_j} + \psi_{\beta'_j})\, a_j + w_j\, \psi_{a_j} \big),$$

with s the sign of u_j and $|z_j| = |u_j|t + s\beta_j$, a polynomial of degree three in t, whose coefficients therefore vanish. The coefficient of t^3 is $s|u_j|\,(n\psi_{c_j})\,v_j\,a_j$, so $\psi_{c_j} = 0$ as for $\tanh$. The coefficient of t^2 is then $-|u_j|\big((n\psi_{e_j})\,a_j + v_j(\psi_{a_j} - s\psi_{\beta_j}a_j)\big)$. As it vanishes for all n in an open set, $\psi_{e_j} = -\alpha e_j$ and $\psi_{a_j} - s\psi_{\beta_j}a_j = \alpha a_j$ for a number α, the same for both signs since ψ_{e_j} is; the two signs then give $\psi_{\beta_j} = 0$. The coefficient of t is then $-|u_j|(\psi_{\beta'_j} + \alpha\beta'_j)\,a_j$, so $\psi_{\beta'_j} = -\alpha\beta'_j$. For GeGLU the rest of unit j is $\varphi(z_j)$ times the same bracket with $z_j - 1/z_j$ in place of $s(|z_j| - 1)$ and 1 in place of $|z_j|$, and its coefficients of t^3, t^2, t and 1 give, in this order, $\psi_{c_j} = 0$, $\psi_{\beta_j} = 0$, $(\psi_{e_j}, \psi_{a_j}) = (-\alpha e_j, \alpha a_j)$ and $\psi_{\beta'_j} = -\alpha\beta'_j$. In both cases the terms of unit j then vanish; this holds for every unit, and $\psi_M = 0$ and $\psi_\gamma = 0$ as for $\tanh$. □

The lemma is stated with a linear term xM, a constant γ and biases β_i and β'_i, which cost nothing in its proof. For the MLP blocks of this paper they all vanish, and the attention blocks below use them. For $\tanh$, the proof on rays follows that of Lemma 3.1 of Sussmann (1992). The directions left by the lemma are those of the rescalings of the hidden units (Appendix B).

Step 3: counting. Let η be flat. The network depends on θ only through the columns and rows c_i, e_i and a_i of C and A, and $F(\theta, x)$ is the function $G(\phi, x)$ of Lemma C.3 with $M = 0$, $\gamma = 0$ and all biases zero. By the chain rule, the direction ψ that holds the derivatives of the c_i, e_i and a_i along η, and zeros at M, γ and the biases, satisfies $\mathrm{D}_\phi G(\phi, x)\,\psi = \mathrm{D}_\theta F(\theta, x)\,\eta = 0$ for all x. By Lemma C.3 there are numbers λ_i and α_i such that

$$\mathrm{D}_\theta C\,\eta \;=\; \big[\, EW_g\,\mathrm{diag}(\lambda) \;\big|\; -EW_v\,\mathrm{diag}(\alpha)\,\big], \qquad \mathrm{D}_\theta A\,\eta \;=\; \mathrm{diag}(\mu)\,A$$

for a gated block, with $\mu_i = \alpha_i - \lambda_i$, and with $\lambda = 0$ for SwiGLU and GeGLU; and $\mathrm{D}_\theta C\,\eta = C\,\mathrm{diag}(\lambda)$, $\mathrm{D}_\theta A\,\eta = \mathrm{diag}(\mu)\,A$ for the other blocks, with $\mu_i = -\lambda_i$ for ReLU, $\mu_i = -2\lambda_i$ for ReLU^2, and $\lambda = 0$, $\mu = 0$ for $\tanh$ and GELU. The direction η^{sc} with the entries

$$\begin{gathered}\eta^{\mathrm{sc}}_{\mathrm{read}} = \big[\, W_g\,\mathrm{diag}(\lambda) \;\big|\; -W_v\,\mathrm{diag}(\alpha)\,\big] \;\text{ for a gated block,}\\ \eta^{\mathrm{sc}}_{\mathrm{read}} = W_{\mathrm{read}}\,\mathrm{diag}(\lambda) \;\text{ for the other blocks,}\\ \eta^{\mathrm{sc}}_{\mathrm{write}} = \mathrm{diag}(\mu)\,W_{\mathrm{write}}, \qquad \eta^{\mathrm{sc}}_E = 0, \qquad \eta^{\mathrm{sc}}_D = 0\end{gathered}$$

is the direction of a rescaling of the hidden units, and along it C and A have the same derivatives as along η. So $\eta - \eta^{\mathrm{sc}}$ keeps C and A fixed to first order, and Lemma C.2 describes it. Hence every flat direction is the sum of a direction of the changes of basis of Lemma C.2 and of a rescaling of the hidden units, and conversely all of these are flat, since they come from symmetries. The two kinds are independent: a rescaling that is also a direction of Lemma C.2 has $EX = 0$, so $X = 0$, and then its entry at the read vanishes, so every λ_i and α_i is zero, because no column of W_{read} vanishes. So

$$\dim \mathcal{K}(\theta) \;=\; 2d^2_{\mathrm{model}} + s,$$

where s is the number of the rescalings: d_{hidden} for ReLU, ReLU^2, SwiGLU and GeGLU, $2d_{\mathrm{hidden}}$ for ReGLU, and none for $\tanh$ and GELU. The functional dimension $P - \dim \mathcal{K}(\theta)$ is the value of the theorem.

Attention blocks

Reduction. Let $W_{Q,i}, W_{K,i}, W_{V,i} \in \mathbb{R}^{d_{\mathrm{model}} \times d_h}$ and $W_{O,i} \in \mathbb{R}^{d_h \times d_{\mathrm{model}}}$ be the matrices of head i, and let

$$B_i = EW_{Q,i}W_{K,i}^\top E^\top/\sqrt{d_h}, \qquad C_i = EW_{V,i}W_{O,i}D, \qquad S = \textstyle\sum_i C_i.$$

Let x_t be row t of the input x. Head i gives token t the scores $x_t B_i x_s^\top$ and the weights $a^{(i)}_{ts}$, their softmax over the tokens s that token t sees, and the output at token t is $\sum_i \sum_s a^{(i)}_{ts}\, x_s C_i$. So the parameters enter only through these matrices. We use inputs of two kinds: with a causal mask, any sequence, whose second token sees only the first two; without a mask, sequences whose tokens other than the second all equal the first. Let k be the number of tokens other than the second that the second

token sees, $k = 1$ with a mask and $k = T - 1$ without. With $u = x_2$ and $v = x_2 - x_1$, head i puts the weight $1/(1 + k\, e^{-uB_i v^\top})$ on the second token and the rest on the copies of the first, and $1/(1 + e^{-z}) = \frac{1}{2} + \frac{1}{2}\tanh(z/2)$ gives the output at the second token

$$y_2 = uS - \tfrac{1}{2}\, vS + \tfrac{1}{2} \sum_i \tanh\left(\tfrac{1}{2}(uB_i v^\top - \log k)\right) vC_i. \tag{10}$$

Step 1: the directions that keep B_i and C_i. For generic parameters E and D have rank d_{model}; every $W_{Q,i}$, $W_{K,i}$, $W_{V,i}$ and $W_{O,i}^\top$ has rank d_h, since $d_h \le d_{\text{model}}$; and the queries $[W_{Q,1}, \dots, W_{Q,H}] \in \mathbb{R}^{d_{\text{model}} \times H d_h}$, the keys $[W_{K,1}, \dots, W_{K,H}]$ and the output projection $W_O \in \mathbb{R}^{H d_h \times d_{\text{model}}}$ of all heads have rank d_{model}, since $H d_h \ge d_{\text{model}}$. Write $\eta_{Q,i}$, $\eta_{K,i}$, $\eta_{V,i}$ and $\eta_{O,i}$ for the entries of a direction η at the positions of $W_{Q,i}$, $W_{K,i}$, $W_{V,i}$ and $W_{O,i}$, and consider a direction with $\mathrm{D}_\theta B_i\, \eta = 0$ and $\mathrm{D}_\theta C_i\, \eta = 0$ for every head. Multiplying $\mathrm{D}_\theta B_i\, \eta = 0$ on the left by the projection $I - EE^+$ onto the orthogonal complement of the column space of E, where E^+ is the pseudo-inverse, leaves $(I - EE^+)\, \eta_E\, W_{Q,i} W_{K,i}^\top E^\top = 0$. As $W_{K,i}^\top E^\top$ has full row rank, $(I - EE^+)\, \eta_E\, W_{Q,i} = 0$ for every head, and as the queries of all heads have rank d_{model}, $\eta_E = EX$ with $X = E^+ \eta_E$. In the same way, multiplying $\mathrm{D}_\theta C_i\, \eta = 0$ on the right by the projection $I - D^+ D$ gives $\eta_D = -YD$ with $Y = -\eta_D\, D^+$, because $EW_{V,i}$ has full column rank and the output projection has rank d_{model}. Substituting, and cancelling E on the left and $E^\top$ or D on the right,

$$(\eta_{Q,i} + XW_{Q,i}) W_{K,i}^\top + W_{Q,i} (\eta_{K,i} + XW_{K,i})^\top = 0,$$
$$(\eta_{V,i} + XW_{V,i}) W_{O,i} + W_{V,i} (\eta_{O,i} - W_{O,i} Y) = 0.$$

If P_1 and P_2 have full column rank and $Z_1 P_2^\top + P_1 Z_2^\top = 0$, then multiplying on the right by a right inverse of $P_2^\top$ gives $Z_1 = P_1 \Gamma$ for a square matrix Γ, and then $Z_2 = -P_2 \Gamma^\top$. Hence

$$\eta_{Q,i} = -XW_{Q,i} + W_{Q,i}\Gamma_i, \qquad \eta_{K,i} = -XW_{K,i} - W_{K,i}\Gamma_i^\top,$$
$$\eta_{V,i} = -XW_{V,i} + W_{V,i}\Gamma_i', \qquad \eta_{O,i} = W_{O,i} Y - \Gamma_i' W_{O,i}.$$

These are the directions of the changes of basis $I + \tau X$ at h_0 and $I + \tau Y$ at h_1, and of $I + \tau \Gamma_i$ between the queries and keys and $I + \tau \Gamma_i'$ between the values and outputs of head i. They are independent: $EX = 0$ forces $X = 0$, $YD = 0$ forces $Y = 0$, and then $W_{Q,i}\Gamma_i = 0$ and $W_{V,i}\Gamma_i' = 0$ force $\Gamma_i = \Gamma_i' = 0$.

Step 2: two tokens. For generic parameters and generic v the columns $B_i v^\top$ are nonzero and pairwise non-collinear, since $d_{\text{model}} \ge 2$, and every row vC_i is nonzero. For such a v, (10) is, as a function of u, the function G of Lemma C.3 with one $\tanh$ hidden unit per head: unit i has the read column $\frac{1}{2} B_i v^\top$, the bias $-\frac{1}{2}\log k$ and the write row $\frac{1}{2} vC_i$, the linear term is uS, and the constant is $-\frac{1}{2} vS$. Along a flat direction η these have the derivatives $\frac{1}{2}(\mathrm{D}_\theta B_i\, \eta) v^\top$, 0, $\frac{1}{2} v(\mathrm{D}_\theta C_i\, \eta)$, $\mathrm{D}_\theta S\, \eta$ and $-\frac{1}{2} v(\mathrm{D}_\theta S\, \eta)$, and Lemma C.3 for $\tanh$ shows that all of these vanish. This holds for all v in an open set, so $\mathrm{D}_\theta B_i\, \eta = 0$ and $\mathrm{D}_\theta C_i\, \eta = 0$ for every head.

Step 3: counting. Every direction of Step 1 is flat, since it comes from a symmetry, and by Step 2 every flat direction, already on the inputs of (10), is one of them. So $\dim \mathcal{K}(\theta) = 2d_{\text{model}}^2 + 2Hd_h^2$, and $s = 2Hd_h^2$ for an attention block.

Proof of Theorem 5.1. Within the scope of Section 2, it is Theorem C.1: the functional dimension is the bound of Theorem 4.3 for $L = 1$ without normalization. □

Remark C.4 (One $\tanh$ block, from the theorem of Sussmann). For a $\tanh$ block, Theorem 5.1 also follows from Theorem 2.1 of Sussmann (1992). That theorem concerns networks with one hidden layer of $\tanh$ hidden units and one output,

$$x \;\mapsto\; \tanh\Big(\gamma + \sum_j \alpha_j\, \tanh(x c_j + \beta_j)\Big),$$

with numbers α_j, β_j and γ. It calls such a network irreducible if no α_j and no c_j vanishes and $(c_i, \beta_i) \neq \pm(c_j, \beta_j)$ for $i \neq j$. It says that two irreducible networks with the same function have the same number of hidden units, and differ by a permutation of the hidden units and by sign flips $(c_j, \beta_j, \alpha_j) \mapsto (-c_j, -\beta_j, -\alpha_j)$.

The outer $\tanh$ is injective. So each of the d_{out} outputs of the network $x \mapsto \sum_i \tanh(xc_i)\, a_i$ of this appendix is the function of such a network, with $\beta_i = 0$, $\gamma = 0$ and with α_i an entry of a_i. At generic parameters these networks are irreducible, and so are those of all nearby parameters. So nearby parameters with the same function have the same columns c_i and rows a_i, up to a permutation and sign flips of the hidden units, which are the identity for parameters that are close enough. They then have the same $C = EW_{\text{read}}$ and $A = W_{\text{write}}D$. A matrix of rank d_{model} has only one factorization into two factors of rank d_{model}, up to an invertible matrix between them, so the parameters with the same C and A are

$$EG_0, \qquad G_0^{-1}W_{\text{read}}, \qquad W_{\text{write}}G_1, \qquad G_1^{-1}D,$$

with invertible $d_{\text{model}} \times d_{\text{model}}$ matrices G_0 and G_1. So the network is locally identifiable up to its symmetries at generic parameters, which is the equality condition (Proposition 4.6).

The proof of that theorem restricts the hidden units to a ray $t \mapsto t\,x$ and separates them by the exponential rates at which they approach their limits. The proof of Lemma C.3 for $\tanh$ and Lemma 6.3 use this argument. Lemma C.3 is the statement for the derivative of the network, with a linear term, which the attention blocks need, and for the other activations.

Remark C.5 (One attention block, from earlier work). For an attention block, Theorem 5.1 also follows from earlier work: from Theorem 3.9 of Henry et al. (2025) for one head, and from Theorem 2.1 of Henry (2026) for several heads and sequences of at least three tokens. With B_i and C_i as above, the network gives token t the output $y_t = \sum_{i=1}^{H} \sum_{s=1}^{T} a_{ts}^{(i)}\, x_s C_i$, where $(a_{t1}^{(i)}, \dots, a_{tT}^{(i)})$ is the softmax of the scores $(x_t B_i x_1^\top, \dots, x_t B_i x_T^\top)$. The two theorems say that the map $x \mapsto y$ determines the pairs (B_i, C_i), up to the order of the heads: the first for one head and all matrices B_1 and $C_1 \neq 0$ of these sizes, the second at generic parameters. The second is stated for $d_{\text{in}} = d_{\text{out}}$, and padding B_i and C_i with zeros reduces the other cases to it.

At generic parameters, B_i and C_i are products of two factors of rank d_h each: $\sqrt{d_h}\, B_i = (EW_{Q,i})(EW_{K,i})^\top$ and $C_i = (EW_{V,i})(W_{O,i}D)$. A factorization of a matrix of rank r into two factors of rank r is unique up to an invertible matrix between the factors. The heads have total width d_{model} (Section 2), so the factors $EW_{Q,i}$ of all heads, side by side, form a matrix of rank d_{model} with the factors E and W_Q, with the same E for the keys and the values. The same holds for the factors W_O and D. So the weights with the same pairs (B_i, C_i) are exactly the following, with G_0 and G_1 any invertible $d_{\text{model}} \times d_{\text{model}}$ matrices and K_i and K_i' any invertible $d_h \times d_h$ matrices:

$$\begin{aligned} &EG_0, \qquad G_0^{-1}W_{Q,i}K_i, \qquad G_0^{-1}W_{K,i}K_i^{-\top}, \\ &G_0^{-1}W_{V,i}K_i', \qquad K_i'^{-1}W_{O,i}G_1, \qquad G_1^{-1}D. \end{aligned}$$

Here G_0 and G_1 change the basis of h_0 and of h_1, and K_i and K_i' act inside head i. These are symmetries of the network, and so is a change in the order of the heads. So the network is identifiable up to its symmetries at generic parameters, which implies the equality condition (Proposition 4.6). The proof of Theorem C.1 does not use these theorems, and it also covers sequences of two tokens and causal attention.

D Proof of Theorem 6.1

We recall the theorem.

Theorem 6.1 (The bound is attained at any depth)**.** *Let $d_{\text{in}}, d_{\text{out}} \geq d_{\text{model}} \geq 2$, and consider networks without biases whose blocks are all of one kind,* $\tanh$*, GELU, SwiGLU or GeGLU, each with at least d_{model} hidden units. In the following cases the functional dimension at generic parameters is*

$$P - (L+1)\dim \mathcal{G} - s,$$

so the network satisfies the equality condition.
(i) $\tanh$ *blocks: without normalization, and under pre-RMSNorm with* $\varepsilon > 0$ *and with* $\varepsilon = 0$.
(ii) *GELU blocks: under pre-RMSNorm with* $\varepsilon > 0$ *and with* $\varepsilon = 0$.
(iii) *SwiGLU blocks, and GeGLU blocks: without normalization, and under pre-RMSNorm with* $\varepsilon > 0$.

Appendix D.1 fixes the notation of the proof. Appendix D.2 states Claim (D_k) and reduces the theorem to the step from (D_{k+1}) to (D_k). Appendices D.3 and D.4 make this step on rays, in the cases (n) and (+), and Appendix D.5 on a great circle, in the case (0). Appendix D.6 puts the parts together. Gated blocks are treated in the cases (n) and (+) only.

D.1 Notation: blocks, the three cases, symmetries

Blocks. A hidden state is a row $h \in \mathbb{R}^{1\times d_{\text{model}}}$: an MLP block acts on each token of a sequence separately, so we take inputs with one token. As in Section 2, block ℓ has the read W^ℓ_{read} and the write W^ℓ_{write}. For two rows of the same length, $\odot$ is their entrywise product. Let block ℓ have q_ℓ hidden units, and let $a_1, \dots, a_{q_\ell}$ be the rows of its write $W^\ell_{\text{write}} \in \mathbb{R}^{q_\ell \times d_{\text{model}}}$. A $\tanh$ or GELU block has the activation σ, and its read $W^\ell_{\text{read}} \in \mathbb{R}^{d_{\text{model}}\times q_\ell}$ has the columns $c_1, \dots, c_{q_\ell}$. A SwiGLU or GeGLU block, a *gated* block, has the read $W^\ell_{\text{read}} = [\, W^\ell_g \mid W^\ell_v \,]$, where $W^\ell_g \in \mathbb{R}^{d_{\text{model}}\times q_\ell}$ holds the gate columns $c_1, \dots, c_{q_\ell}$ and $W^\ell_v \in \mathbb{R}^{d_{\text{model}}\times q_\ell}$ the value columns $e_1, \dots, e_{q_\ell}$. Its gate σ is GELU in a GeGLU block and SiLU, $\sigma(z) = z/(1+e^{-z})$, in a SwiGLU block. Block ℓ maps its input n to

$$g_\ell(n) \;=\; \sigma(n\, W^\ell_{\text{read}})\, W^\ell_{\text{write}} \;=\; \sum_{j=1}^{q_\ell} \sigma(nc_j)\, a_j \qquad \text{for a } \tanh \text{ or GELU block,}$$

$$g_\ell(n) \;=\; \big(\sigma(n\, W^\ell_g) \odot n\, W^\ell_v\big)\, W^\ell_{\text{write}} \;=\; \sum_{j=1}^{q_\ell} \sigma(nc_j)\,(ne_j)\, a_j \qquad \text{for a gated block.}$$

The three cases. The three cases (n), (+) and (0) are those of Section 2. For a row h, the table gives N, the set of its values, which are the inputs of the blocks, the group $\mathcal{G}$ of the changes of basis, and its Lie algebra $\mathfrak{g}$: the set of the matrices Z with $\exp(\tau Z) \in \mathcal{G}$ for all numbers τ.

Case	$N(h)$	Values of N	$\mathcal{G}$	$\mathfrak{g}$
(n)	h	all rows $n \in \mathbb{R}^{1\times d_{\text{model}}}$	$\mathrm{GL}(d_{\text{model}})$	all matrices Z
(+)	$h\left(\lVert h\rVert^2/d_{\text{model}} + \varepsilon\right)^{-1/2}$	the ball $\lVert n\rVert^2 < d_{\text{model}}$	$\mathrm{O}(d_{\text{model}})$	antisymmetric Z
(0)	$\sqrt{d_{\text{model}}}\, h/\lVert h\rVert$, for $h \neq 0$	the sphere $\mathbb{S}$: $\lVert n\rVert^2 = d_{\text{model}}$	$\mathrm{CO}(d_{\text{model}})$	$Z = Y + \beta I$

In the last row Y is antisymmetric and β is a number, and $\mathbb{S}$ is the set of the rows $n \in \mathbb{R}^{1\times d_{\text{model}}}$ with $\lVert n\rVert^2 = d_{\text{model}}$. In the case (0) a network is defined at the inputs x for which no hidden state vanishes.

Symmetries. We write $F_\theta(x)$ for the output of the network at the input $x \in \mathbb{R}^{1\times d_{\text{in}}}$, and $\mathrm{D}_\theta F(\theta, x)\,\eta$ for its derivative along a direction $\eta \in \mathbb{R}^P$ of the parameters. The changes of basis of Proposition 3.2 take the following form. For $G \in \mathcal{G}$ let $\widehat{G} = G$ in the cases (n) and (+), and $\widehat{G} = Q$ for $G = \lambda Q$ in the case (0), with a number $\lambda > 0$ and an orthogonal Q. Then $N(hG) = N(h)\,\widehat{G}$ in all three cases. So for $G_0, \dots, G_L \in \mathcal{G}$ the parameters

$$\Big(EG_0,\;\; \big\{\widehat{G}^{-1}_{\ell-1} W^\ell_{\text{read}},\;\; W^\ell_{\text{write}} G_\ell\big\}_{\ell=1}^{L},\;\; \widehat{G}^{-1}_L D\Big)$$

give the same function, with the hidden states $h_\ell G_\ell$: one matrix of $\mathcal{G}$ per hidden state. For $Z \in \mathfrak{g}$ let $\widehat{Z} = Z$ in the cases (n) and (+), and $\widehat{Z} = Y$ for $Z = Y + \beta I$ in the case (0). For $Z_0, \dots, Z_L \in \mathfrak{g}$ let

$$\eta^Z \;=\; \Big(EZ_0,\;\; \big\{-\widehat{Z}_{\ell-1} W^\ell_{\text{read}},\;\; W^\ell_{\text{write}} Z_\ell\big\}_{\ell=1}^{L},\;\; -\widehat{Z}_L D\Big) \in \mathbb{R}^P, \tag{11}$$

the derivative of these parameters at $\tau = 0$ for $G_\ell = \exp(\tau Z_\ell)$. So $\mathrm{D}_\theta F(\theta, x)\, \eta^Z = 0$ at every input at which F_θ is defined. When E and every W^ℓ_{write} have rank d_{model}, the map $(Z_0, \dots, Z_L) \mapsto \eta^Z$ is injective.

A gated block has, in addition, one rescaling per hidden unit. For diagonal matrices $\Delta_1, \dots, \Delta_L$, with Δ_ℓ of size q_ℓ, the parameters

$$\Big(EG_0,\ \big\{G_{\ell-1}^{-1} W_g^\ell,\ G_{\ell-1}^{-1} W_v^\ell\, e^{-\Delta_\ell},\ e^{\Delta_\ell}\, W^\ell_{\mathrm{write}} G_\ell\big\}_{\ell=1}^{L},\ G_L^{-1} D\Big)$$

give the same function as well: the diagonal matrix e^{Δ_ℓ} multiplies the write row of every hidden unit of block ℓ by a positive number and divides its value column by the same number. For $Z_0, \dots, Z_L \in \mathfrak{g}$ let

$$\eta^{Z,\Delta} \;=\; \Big(EZ_0,\ \big\{-Z_{\ell-1}W_g^\ell,\ -Z_{\ell-1}W_v^\ell - W_v^\ell \Delta_\ell,\ \Delta_\ell W^\ell_{\mathrm{write}} + W^\ell_{\mathrm{write}} Z_\ell\big\}_{\ell=1}^{L},\ -Z_L D\Big) \in \mathbb{R}^P, \tag{12}$$

the derivative of these parameters at $\tau = 0$ for $G_\ell = \exp(\tau Z_\ell)$ and with $\tau\Delta_\ell$ in the place of Δ_ℓ. So $\mathrm{D}_\theta F(\theta, x)\, \eta^{Z,\Delta} = 0$ at every input x. When E and every W^ℓ_{write} have rank d_{model} and no value column vanishes, $\eta^{Z,\Delta} = 0$ only if every Z_ℓ and every Δ_ℓ vanishes: $EZ_0 = 0$ gives $Z_0 = 0$, then $W_v^1 \Delta_1 = 0$ gives $\Delta_1 = 0$, then $W^1_{\mathrm{write}} Z_1 = 0$ gives $Z_1 = 0$, and so on.

Let $s = 0$ for $\tanh$ and GELU blocks and $s = q_1 + \cdots + q_L$ for gated blocks. Under these conditions the directions of the symmetries form a space of dimension $(L+1)\dim \mathcal{G} + s$, so on every batch the Jacobian has rank at most

$$P - (L+1)\dim \mathcal{G} - s. \tag{13}$$

In the cases (n) and (+) this is the bound of Theorem 4.3, and in the case (0) it is the bound with $\mathcal{G} = \mathrm{CO}(d_{\mathrm{model}})$ that Section 4 names.

D.2 The claim, and the reduction to one step

Notation. For rows u, v, real or complex, $\langle u, v\rangle = uv^\top$, without complex conjugation. For $k = 0, \dots, L$ and a row h, with $h \neq 0$ in the case (0), let $\mathcal{B}_{>k}(h)$ be the output of the network when its hidden state h_k is set to h:

$$\mathcal{B}_{>L}(h) = N(h)\, D, \qquad \mathcal{B}_{>k-1}(h) = \mathcal{B}_{>k}\big(g_k(N(h))\big), \qquad \text{so that} \quad F_\theta(x) = \mathcal{B}_{>0}(xE).$$

$\mathcal{B}_{>k}$ depends on the weights of the blocks $k+1, \dots, L$ and on D. We write $\mathrm{D}\mathcal{B}_{>k}(h)\, v$ for its derivative at h along a row v, and $\mathrm{D}_\theta \mathcal{B}_{>k}(h)\, \eta$ for its derivative at fixed h along a direction η of these weights. In the cases (+) and (0), for a $d_{\mathrm{model}} \times d_{\mathrm{model}}$ matrix Z and $n = N(h)$,

$$\mathrm{D}N(h)\,(hZ) \;=\; nZ - \frac{\langle nZ, n\rangle}{d_{\mathrm{model}}}\, n. \tag{14}$$

In the case (0), $\mathcal{B}_{>k}(\lambda h) = \mathcal{B}_{>k}(h)$ for $\lambda > 0$, so

$$\mathrm{D}\mathcal{B}_{>k}(h)\, h = 0. \tag{15}$$

We call θ *regular* if

- (R1) E, D and every W^ℓ_{write} have rank d_{model};
- (R2) in the case (0), for every ℓ and every $n \in \mathbb{S}$: $g_\ell(n) \neq 0$;
- (R3) for every ℓ: in the cases (n) and (+), $\mathrm{D}g_\ell(n)$ is invertible at some row n; in the case (0), there is an $n \in \mathbb{S}$ such that no row $v \neq 0$ with $\langle v, n\rangle = 0$ has $\mathrm{D}g_\ell(n)\, v \in \mathbb{R}\, g_\ell(n)$.

In the cases (n) and (+), every $\mathcal{B}_{>k}$ is real-analytic on all rows h. In the case (0), by (R2), it is defined and real-analytic on all rows $h \neq 0$.

Lemma D.1 (The derivative of $\mathcal{B}_{>k}$)**.** *Let θ be regular. For every k there is an open dense set $\mathcal{U}_k$ of rows such that, for $h \in \mathcal{U}_k$, $\mathrm{D}\mathcal{B}_{>k}(h)\, v = 0$ only if $v = 0$, in the cases (n) and (+), and only if $v \in \mathbb{R}\, h$, in the case (0). In the case (0), $\mathcal{U}_k$ consists of rows $h \neq 0$ and is closed under multiplication by positive numbers.*

Proof. The cases (n) and (+). N is the identity, or a diffeomorphism of $\mathbb{R}^{1\times d_{\text{model}}}$ onto the ball of the rows n with $\|n\|^2 < d_{\text{model}}$, with the inverse $n \mapsto \sqrt{\varepsilon}\, n \left(1 - \|n\|^2/d_{\text{model}}\right)^{-1/2}$. So $\mathrm{D}N(h)$ is invertible at every h. As D is injective on rows, $\mathrm{D}\mathcal{B}_{>L}(h)\, v = \big(\mathrm{D}N(h)\, v\big)D$ vanishes only for $v = 0$: $\mathcal{U}_L$ is the set of all rows. Let $k < L$. The determinant of $\mathrm{D}g_{k+1}(n)$ is real-analytic in n and does not vanish identically, by (R3). So g_{k+1} is a local diffeomorphism on an open dense set of rows n, and $g_{k+1} \circ N$ on an open dense set of rows h. Let $\mathcal{U}_k$ be the set of the h of this set with $g_{k+1}(N(h)) \in \mathcal{U}_{k+1}$. It is open and dense, and for $h \in \mathcal{U}_k$,

$$\mathrm{D}\mathcal{B}_{>k}(h)\, v \;=\; \mathrm{D}\mathcal{B}_{>k+1}\big(g_{k+1}(N(h))\big)\, \Big(\mathrm{D}\big(g_{k+1}\circ N\big)(h)\, v\Big)$$

vanishes only for $v = 0$.

The case (0). For a row $h \neq 0$, $\mathrm{D}N_\varepsilon(h)\, v = \frac{\sqrt{d_{\text{model}}}}{\|h\|}\big(v - \frac{\langle v,h\rangle}{\|h\|^2}\, h\big)$, which vanishes exactly for $v \in \mathbb{R}\, h$. For $k = L$, $\mathrm{D}\mathcal{B}_{>L}(h)\, v = \big(\mathrm{D}N_\varepsilon(h)\, v\big)D$ and D is injective on rows, so $\mathcal{U}_L$ is the set of all rows $h \neq 0$. Let $k < L$ and $g = g_{k+1}$. For $n \in \mathbb{S}$ and a row v,

$$\mathrm{D}\mathcal{B}_{>k}(n)\, v \;=\; \mathrm{D}\mathcal{B}_{>k+1}\big(g(n)\big)\, \big(\mathrm{D}g(n)\, v^\perp\big), \qquad v^\perp = v - \frac{\langle v, n\rangle}{d_{\text{model}}}\, n,$$

where $v^\perp$ is orthogonal to n and vanishes exactly for $v \in \mathbb{R}\, n$. For $n \in \mathbb{S}$ consider the linear map $v \mapsto \mathrm{D}g(n)\, v^\perp + \frac{\langle v,n\rangle}{d_{\text{model}}}\, g(n)$ of $\mathbb{R}^{1\times d_{\text{model}}}$. By (R2) it is injective exactly when n is as in (R3), and its determinant is real-analytic in n. By (R3) the determinant does not vanish identically on the connected set $\mathbb{S}$, so the n as in (R3) form an open dense subset of $\mathbb{S}$. On this subset the map $n \mapsto N_\varepsilon(g(n))$ from $\mathbb{S}$ to $\mathbb{S}$ has an injective derivative, so it is a local diffeomorphism, and the n of the subset with $g(n) \in \mathcal{U}_{k+1}$ form an open dense subset of $\mathbb{S}$ again. For such n, $\mathrm{D}\mathcal{B}_{>k}(n)\, v = 0$ gives $\mathrm{D}g(n)\, v^\perp \in \mathbb{R}\, g(n)$, so $v^\perp = 0$ and $v \in \mathbb{R}\, n$. Let $\mathcal{U}_k$ be the set of the positive multiples of these n. □

Lemma D.2 (A direction at a write)**.** *Let θ be regular, $0 \le k \le L$, and let A be a matrix with d_{model} columns and rank d_{model}. Let η_A be a matrix of the size of A and η a direction of the weights of the blocks $k+1, \dots, L$ and of D. If*

$$\mathrm{D}\mathcal{B}_{>k}(wA)\, (w\, \eta_A) + \mathrm{D}_\theta\mathcal{B}_{>k}(wA)\, \eta \;=\; 0 \qquad \textit{for all rows } w, \textit{ with } wA \neq 0 \textit{ in the case (0)},$$

then $\eta_A = AZ$ for a $d_{\text{model}} \times d_{\text{model}}$ matrix Z, and

$$\mathrm{D}\mathcal{B}_{>k}(h)\, (hZ) + \mathrm{D}_\theta\mathcal{B}_{>k}(h)\, \eta \;=\; 0 \qquad \textit{for all rows } h, \textit{ with } h \neq 0 \textit{ in the case (0)}. \tag{16}$$

Proof. Let $w_1 A = 0$. Subtracting the hypothesis at w and at $w + w_1$ gives $\mathrm{D}\mathcal{B}_{>k}(wA)\, (w_1\eta_A) = 0$. For $wA \in \mathcal{U}_k$, Lemma D.1 gives $w_1\eta_A = 0$ in the cases (n) and (+), and $w_1\eta_A \in \mathbb{R}\, wA$ in the case (0). In the case (0), as $\mathcal{U}_k$ is open and $d_{\text{model}} \ge 2$, wA takes two linearly independent values in $\mathcal{U}_k$, so $w_1\eta_A = 0$ as well. Let A^+ be a matrix with $A^+A = I$. For every row w, $(w - wAA^+)\, A = 0$, so $w\, \eta_A = wAA^+\eta_A$: $\eta_A = AZ$ with $Z = A^+\eta_A$. The hypothesis is now (16) at $h = wA$, which takes every value. □

Claim (D_k), for $k = 0, \dots, L$. Let Z be a $d_{\text{model}} \times d_{\text{model}}$ matrix and η a direction of the weights of the blocks $k+1, \dots, L$ and of D. If (16) holds, then $Z \in \mathfrak{g}$, and with $Z_k = Z$ there are $Z_{k+1}, \dots, Z_L \in \mathfrak{g}$ such that, for $\tanh$ and GELU blocks,

$$\eta \;=\; \Big(\big\{-\widehat{Z}_{\ell-1} W^\ell_{\text{read}},\;\; W^\ell_{\text{write}} Z_\ell\big\}_{\ell=k+1}^{L},\;\; -\widehat{Z}_L D\Big). \tag{17}$$

In the case (0), with $Z_\ell = Y_\ell + \beta_\ell I$ for antisymmetric matrices $Y_k, \dots, Y_L$ and numbers $\beta_k, \dots, \beta_L$, this reads

$$Z = Y_k + \beta_k I, \qquad \eta = \Big(\big\{-Y_{\ell-1} W^\ell_{\text{read}},\;\; W^\ell_{\text{write}}(Y_\ell + \beta_\ell I)\big\}_{\ell=k+1}^{L},\;\; -Y_L D\Big). \tag{18}$$

For gated blocks there are, in addition, diagonal matrices $\Delta_{k+1}, \dots, \Delta_L$, with Δ_ℓ of size q_ℓ, such that

$$\eta \;=\; \Big(\big\{ -Z_{\ell-1} W_g^\ell,\;\; -Z_{\ell-1} W_v^\ell - W_v^\ell \Delta_\ell,\;\; \Delta_\ell W_{\text{write}}^\ell + W_{\text{write}}^\ell Z_\ell \big\}_{\ell=k+1}^{L},\;\; -Z_L D \Big). \tag{19}$$

Conversely, every $Z = Z_k \in \mathfrak{g}$ and η of the form (17), or (19) for gated blocks, satisfy (16): it is the derivative at $\tau = 0$ of the identity

$$\mathcal{B}_{>k}\big(h \exp(\tau Z_k)\big) \;=\; \mathcal{B}_{>k}(h),$$

in which the left side is taken at the weights moved by the symmetries with $G_\ell = \exp(\tau Z_\ell)$ for $\ell \geq k$, and for gated blocks with $\tau \Delta_\ell$ in the place of Δ_ℓ.

Proposition D.3 (The claim gives the rank)**.** *Let θ be regular, with no vanishing value column if the blocks are gated, and let (D_0) hold at θ. Then the directions $\eta \in \mathbb{R}^P$ with $\mathrm{D}_\theta F(\theta, x)\, \eta = 0$ for all x, with $xE \neq 0$ in the case (0), are the directions η^Z of (11), or $\eta^{Z,\Delta}$ of (12) for gated blocks, and the Jacobian of the network has the rank (13) on some finite batch of such inputs.*

Proof. Let $\eta = (\eta_E, \eta')$ be such a direction, with η_E its entry at E. As $F_\theta(x) = \mathcal{B}_{>0}(xE)$,

$$\mathrm{D}_\theta F(\theta, x)\, \eta \;=\; \mathrm{D}\mathcal{B}_{>0}(xE)\, (x\, \eta_E) + \mathrm{D}_\theta \mathcal{B}_{>0}(xE)\, \eta'.$$

By Lemma D.2 with $A = E$, $\eta_E = EZ$ and (16) holds for $k = 0$. By (D_0), $\eta = \eta^Z$ for some $Z_0, \dots, Z_L \in \mathfrak{g}$ with $Z_0 = Z$, or $\eta = \eta^{Z,\Delta}$ for gated blocks. The kernels of the Jacobian on finite batches are subspaces of $\mathbb{R}^P$ that can only shrink when the batch grows, so on some finite batch the kernel is the space of the directions of the symmetries, which has dimension $(L+1) \dim \mathcal{G} + s$ by the hypotheses on θ. □

Lemma D.4 (The readout)**.** *If D has rank d_{model}, then (D_L) holds.*

Proof. The case (n). Here (16) is $hZD + h\, \eta_D = 0$ for all rows h, so $\eta_D = -ZD$, and $\mathfrak{g}$ contains every Z.

The case (+). By (14), (16) is $\big(nZ - \frac{\langle nZ, n\rangle}{d_{\text{model}}}\, n\big) D + n\, \eta_D = 0$ for all n in the ball, hence as a polynomial in n. Its part of degree 1 gives $\eta_D = -ZD$. Its part of degree 3 gives $\langle nZ, n\rangle\, nD = 0$, and $nD \neq 0$ for $n \neq 0$, so $\langle nZ, n\rangle = 0$ for all n: Z is antisymmetric.

The case (0). Here (16) is $\big(\mathrm{D}N_\varepsilon(h)\, (hZ)\big) D + N_\varepsilon(h)\, \eta_D = 0$. Multiplying by $\|h\|^3 / \sqrt{d_{\text{model}}}$ gives

$$\|h\|^2\, h\, (ZD + \eta_D) \;=\; \langle hZ, h\rangle\, hD \qquad \text{for all rows } h.$$

So $h\,(ZD + \eta_D) = \mu(h)\, hD$ with a number $\mu(h)$, for every $h \neq 0$. As $h \mapsto hD$ is injective and $d_{\text{model}} \geq 2$, comparing h, h' and $h + h'$ for linearly independent h, h' shows that μ is a constant β, so $ZD + \eta_D = \beta D$. Then $\langle hZ, h\rangle = \beta \|h\|^2$ for all h, so the symmetric part of Z is βI: $Z = Y_L + \beta I$ with Y_L antisymmetric, and $\eta_D = \beta D - ZD = -Y_L D$. □

What remains is the step from (D_{k+1}) to (D_k). For $\tanh$ blocks in the cases (n) and (+) we prove it at almost all parameters (Lemma D.6). For GELU blocks in the case (+) and for gated blocks we prove it at explicit parameters (Lemmas D.11 and D.12), and in the case (0) at suitable parameters (Lemmas D.17 and D.18).

The identity for one block. Fix $k < L$, let block $k+1$ be a $\tanh$ or GELU block, and write $W = W_{\text{read}}^{k+1}$, with the columns $c_1, \dots, c_q$, $V = W_{\text{write}}^{k+1}$, with the rows $a_1, \dots, a_q$, and $\mathcal{B} = \mathcal{B}_{>k+1}$. Let Z and $\eta = (\eta_W, \eta_V, \eta')$ be as in (D_k), where η_W and η_V are the entries of η at W and V and η' is the rest. For rows n, $\tilde{w}$ and $\tilde{w}'$ let

$$\dot{z}(n) = \begin{cases} nZ\, W + n\, \eta_W & \text{in the case (n),} \\ \Big(nZ - \dfrac{\langle nZ, n\rangle}{d_{\text{model}}}\, n\Big)\, W + n\, \eta_W & \text{in the cases (+) and (0),} \end{cases} \tag{20}$$

$$\mathcal{L}(\tilde{w}, \tilde{w}'; n) = \mathrm{D}\mathcal{B}(\tilde{w}V)\, \Big(\big(\tilde{w}' \odot \dot{z}(n)\big)\, V + \tilde{w}\, \eta_V \Big) + \mathrm{D}_\theta \mathcal{B}(\tilde{w}V)\, \eta'. \tag{21}$$

By the chain rule and (14), the left side of (16) at h is $\mathcal{L}\big(\sigma(nW), \sigma'(nW); n\big)$ at $n = N(h)$: $\tilde{w} = \sigma(nW)$ is the row of the outputs of the hidden units of block $k+1$, $\tilde{w}' = \sigma'(nW)$ the row of their derivatives, and $\dot{z}(n)$ the derivative of the row nW that they read. So (16) for k is equivalent to

$$\mathcal{L}\big(\sigma(nW),\ \sigma'(nW);\ n\big)\ =\ 0 \qquad \text{for all values } n \text{ of } N, \tag{22}$$

that is, for all rows n in the case (n), for the rows n with $\|n\|^2 < d_{\text{model}}$ in the case (+), and for $n \in \mathbb{S}$ in the case (0).

Lemma D.5 (The identity at every real row)**.** *In the cases (n) and (+),* (16) *for k holds if and only if*

$$\mathcal{L}\big(\sigma(nW),\ \sigma'(nW);\ n\big)\ =\ 0 \qquad \textit{for every row } n \in \mathbb{R}^{1\times d_{\text{model}}}. \tag{23}$$

Proof. In the case (n) this is (22). In the case (+), the left side of (23) is real-analytic in n on all of $\mathbb{R}^{1\times d_{\text{model}}}$, because $\mathcal{B}$ is real-analytic on all rows. So it vanishes on the ball if and only if it vanishes everywhere. □

In the case (0) the identity holds only on the sphere $\mathbb{S}$, which contains no ray. This is where the cases part: we follow rays in the cases (n) and (+), and a great circle at complex angles in the case (0).

D.3 Rays: $\tanh$ blocks

Rays: no normalization, and $\varepsilon > 0$. In the cases (n) and (+) we take $n = t\,\xi$ in (23), with a row ξ, and let $t \to \infty$. The next lemma separates the hidden units of a $\tanh$ block.

Lemma 6.3 (Separation on a ray)**.** *Let $u_1, \dots, u_q$ be real numbers that are linearly independent over $\mathbb{Q}$, and let $\mathcal{T}_0, \dots, \mathcal{T}_K$ be real-analytic maps from $\mathbb{R}^{1\times q}$ to $\mathbb{R}^{1\times d_{\text{out}}}$. If*

$$\sum_{i=0}^{K} t^i\ \mathcal{T}_i\big(\tanh(tu_1), \dots, \tanh(tu_q)\big)\ =\ 0 \qquad \textit{for all } t > 0,$$

then $\mathcal{T}_i = 0$ on $\mathbb{R}^{1\times q}$ for every i.

Proof. No u_j is zero. Let s_j be the sign of u_j and $\zeta_j(t) = e^{-2|u_j|t} \in (0,1)$, so that $\tanh(tu_j) = s_j\,(1-\zeta_j(t))/(1+\zeta_j(t))$. The maps $\zeta \mapsto \mathcal{T}_i\big(s_1\frac{1-\zeta_1}{1+\zeta_1}, \dots, s_q\frac{1-\zeta_q}{1+\zeta_q}\big)$ are real-analytic near $\zeta = 0$, so for some $\rho \in (0,1)$ they are sums of power series $\sum_\alpha b_{i,\alpha}\,\zeta^\alpha$, over the multi-indices $\alpha = (\alpha_1, \dots, \alpha_q)$ of nonnegative integers, that converge absolutely for $|\zeta_j| \le \rho$. Choose $t_0 \ge 1$ with $\zeta_j(t_0) \le \rho$ for all j. For $t \ge t_0$ the hypothesis reads

$$\sum_\alpha p_\alpha(t)\, e^{-\mu_\alpha t} = 0, \qquad p_\alpha(t) = \sum_{i=0}^{K} t^i\, b_{i,\alpha}, \qquad \mu_\alpha = 2\sum_{j=1}^{q} \alpha_j\,|u_j|.$$

The numbers μ_α are distinct for distinct α, because the u_j are linearly independent over $\mathbb{Q}$, and only finitely many of them lie below any bound, because every $|u_j| > 0$. Suppose that some p_α is not zero, and among these let α^* have the smallest μ_α, say μ^*. Let $\gamma > 0$ be the distance from μ^* to the next larger μ_α. For $t \ge t_0$,

$$|p_{\alpha^*}(t)|\ \le \sum_{\mu_\alpha > \mu^*} |p_\alpha(t)|\, e^{-(\mu_\alpha - \mu^*)t}\ \le\ t^K\, e^{-\gamma(t-t_0)}\, e^{\mu^* t_0} \sum_{i=0}^{K}\sum_\alpha |b_{i,\alpha}|\,\zeta(t_0)^\alpha,$$

which tends to 0 as $t \to \infty$. A polynomial that tends to 0 is zero, a contradiction. So every $b_{i,\alpha} = 0$, and every $\mathcal{T}_i$ vanishes on the image of a neighborhood of $\zeta = 0$, which is a neighborhood of the row $(s_1, \dots, s_q)$. A real-analytic map on $\mathbb{R}^{1\times q}$ that vanishes on an open set vanishes everywhere. □

Consider, for a block ℓ with $W = W^\ell_{\text{read}}$ and $V = W^\ell_{\text{write}}$:

(T1) no row of V vanishes;

(T2) the columns of W are linearly independent over $\mathbb{Q}$.

Lemma D.6 (The step for $\tanh$ on rays)**.** *Let $\sigma = \tanh$, in the case (n) or (+), let θ be regular, and let block $k+1$ satisfy (T1) and (T2). Then (D_{k+1}) implies (D_k).*

Proof. Let Z and η satisfy (16) for k, so that (23) holds, with the notation above. Take $n = t\,\xi$ with a row ξ and $t > 0$. Then $\tilde{w} = \tanh(t\,\xi W)$ and $\tilde{w}' = 1 - \tilde{w}^2$, the row with the entries $1 - \tilde{w}_j^2$. By (20),

$$\dot{z}(t\,\xi) \;=\; t\,\xi\,(ZW + \eta_W) \;-\; t^3\,\frac{\langle \xi Z, \xi\rangle}{d_{\text{model}}}\,\xi W$$

in the case (+), and without the term in t^3 in the case (n). So (23) at $n = t\,\xi$ reads

$$t^3\,\mathcal{T}_3(\tilde{w}) + t\,\mathcal{T}_1(\tilde{w}) + \mathcal{T}_0(\tilde{w}) = 0 \qquad \text{at } \tilde{w} = \tanh(t\,\xi W),$$

with the maps

$$\begin{aligned}
\mathcal{T}_3(\tilde{w}) &= -\frac{\langle \xi Z, \xi\rangle}{d_{\text{model}}}\,\mathrm{D}\mathcal{B}(\tilde{w}V)\,\Big(\big((1-\tilde{w}^2)\odot \xi W\big)\,V\Big),\\
\mathcal{T}_1(\tilde{w}) &= \mathrm{D}\mathcal{B}(\tilde{w}V)\,\Big(\big((1-\tilde{w}^2)\odot \xi\,(ZW+\eta_W)\big)\,V\Big),\\
\mathcal{T}_0(\tilde{w}) &= \mathrm{D}\mathcal{B}(\tilde{w}V)\,(\tilde{w}\,\eta_V) + \mathrm{D}_\theta\mathcal{B}(\tilde{w}V)\,\eta',
\end{aligned}$$

where $\mathcal{T}_3 = 0$ in the case (n). They are defined and real-analytic for every row $\tilde{w} \in \mathbb{R}^{1\times q}$. By (T2), for integers $\kappa_1, \dots, \kappa_q$ that are not all zero, $\sum_j \kappa_j\,\xi c_j = \xi \sum_j \kappa_j c_j$ vanishes only for ξ in a hyperplane. So for ξ outside countably many hyperplanes the numbers $\xi c_1, \dots, \xi c_q$ are linearly independent over $\mathbb{Q}$, and Lemma 6.3 gives $\mathcal{T}_3 = \mathcal{T}_1 = \mathcal{T}_0 = 0$ on $\mathbb{R}^{1\times q}$.

We use the following fact: if a row $b \in \mathbb{R}^{1\times q}$ has $\mathrm{D}\mathcal{B}(\tilde{w}V)\big(((1-\tilde{w}^2)\odot b)\,V\big) = 0$ for all rows $\tilde{w}$, then $b = 0$. Indeed, by Lemma D.1 the row $\sum_j b_j\,(1-\tilde{w}_j^2)\,a_j$ vanishes for all $\tilde{w}$ with $\tilde{w}V \in \mathcal{U}_{k+1}$. These $\tilde{w}$ form an open dense set, because V has rank d_{model}, so the row vanishes as a polynomial in $\tilde{w}$. Its coefficient of $\tilde{w}_j^2$ is $-b_j a_j$, and $a_j \neq 0$ by (T1).

(a) $\mathcal{T}_3 = 0$ gives Z. In the case (n) every matrix Z lies in $\mathfrak{g}$. In the case (+), $\xi W \neq 0$, so the fact gives $\langle \xi Z, \xi\rangle = 0$ for a dense set of rows ξ, hence for all: Z is antisymmetric. In both cases $Z \in \mathfrak{g}$ and $\widehat{Z} = Z$.

(b) $\mathcal{T}_1 = 0$ gives η_W. The fact gives $\xi\,(ZW+\eta_W) = 0$ for a dense set of rows ξ, so $\eta_W = -ZW$.

(c) $\mathcal{T}_0 = 0$ gives the rest. $\mathrm{D}\mathcal{B}(\tilde{w}V)\,(\tilde{w}\,\eta_V) + \mathrm{D}_\theta\mathcal{B}(\tilde{w}V)\,\eta' = 0$ for all rows $\tilde{w}$. By Lemma D.2 with $A = V$, $\eta_V = VZ'$ and (16) holds for $k+1$ with Z' and η'. By (D_{k+1}), $Z' \in \mathfrak{g}$ and η' is of the form (17) with $Z_{k+1} = Z'$. With (a) and (b) this is (17) for k. □

D.4 Rays: GELU and gated blocks

GELU and the Mills ratio. GELU is $\sigma(z) = \frac{z}{2}\big(1 + \operatorname{erf}(z/\sqrt{2})\big)$, which is entire. For $z \in \mathbb{C}$ let

$$\varphi(z) = \frac{e^{-z^2/2}}{\sqrt{2\pi}}, \qquad \operatorname{mr}(z) \;=\; \sqrt{\frac{\pi}{2}}\,e^{z^2/2}\,\Big(1 - \operatorname{erf}\frac{z}{\sqrt{2}}\Big),$$

the normal density and the Mills ratio.

Lemma D.7 (The Mills ratio)**. (i)** *For all $z \in \mathbb{C}$: $\sigma(z) = z - \varphi(z)\,z\,\operatorname{mr}(z)$ and $\sigma'(z) = 1 - \varphi(z)\,\big(\operatorname{mr}(z) - z\big)$.*
(ii) *For $\operatorname{Re} z > 0$ and every integer $m \geq 0$, with $(-1)!! = 1$,*

$$\Big|\,\operatorname{mr}(z) - \sum_{l=0}^{m-1} \frac{(-1)^l\,(2l-1)!!}{z^{2l+1}}\,\Big| \;\leq\; \frac{(2m-1)!!}{(\operatorname{Re} z)^{2m+1}}.$$

Proof. (i) follows from $\varphi(z)\,\mathrm{mr}(z) = \frac{1}{2}\big(1-\mathrm{erf}(z/\sqrt{2})\big)$ and $\sigma'(z) = \frac{1}{2}\big(1+\mathrm{erf}(z/\sqrt{2})\big) + z\,\varphi(z)$. For (ii): for real z, $\mathrm{mr}(z) = e^{z^2/2}\int_z^\infty e^{-y^2/2}\,dy$, and the substitution $y = z+u$ gives $\mathrm{mr}(z) = \int_0^\infty e^{-zu-u^2/2}\,du$. Both sides are entire, so this holds for all z. For a number $y \geq 0$, $\big|e^{-y} - \sum_{l<m}\frac{(-y)^l}{l!}\big| \leq \frac{y^m}{m!}$. With $y = u^2/2$, and with $\int_0^\infty e^{-zu}u^{2l}\,du = \frac{(2l)!}{z^{2l+1}}$ and $\frac{(2l)!}{2^l\,l!} = (2l-1)!!$, this gives (ii). □

We say that a map Θ on an interval (t_0,∞) is *meromorphic at infinity* if $\Theta(t) = t^l\,\Theta_0(1/t)$ for an integer l and a map Θ_0 that is holomorphic near 0. Such a map vanishes identically if it tends to 0 faster than every power of $1/t$.

Lemma D.8 (The Mills ratio at infinity)**.** *Let $z(t) = \omega\,t + \omega'/t$ with $\omega,\omega' \in \mathbb{C}$ and $\mathrm{Re}\,\omega > 0$, and let Θ_1 and Θ_2 be meromorphic at infinity, with values in $\mathbb{C}^{1\times d_{\mathrm{out}}}$. If $\Theta_1(t)\;\mathrm{mr}(z(t)) + \Theta_2(t)$ tends to 0 faster than every power of $1/t$ as $t\to+\infty$, then $\Theta_1 = 0$ and $\Theta_2 = 0$.*

Proof. Suppose that a component of Θ_1 is not zero, and let Θ_1,Θ_2 stand for this component of both. Then $\mathrm{mr}(z(t)) = -\Theta_2(t)/\Theta_1(t) + O(t^{-m})$ for every m. For large t, $t = \frac{z}{2\omega}\big(1+\sqrt{1-4\omega\omega'/z^2}\big)$ with $z = z(t)$, so $1/t$ is a convergent power series in $1/z$ without constant term. Hence $-\Theta_2/\Theta_1$ is, for large t, a convergent series in powers of $1/z$ with finitely many positive powers of z, and this series is an asymptotic expansion of $\mathrm{mr}(z)$ along the curve $z = z(t)$. On this curve $\mathrm{Re}\,z$ is at least a positive constant times $|z|$ for large t, so by Lemma D.7(ii) the series $\sum_l (-1)^l(2l-1)!!\,z^{-2l-1}$ is one as well. Asymptotic expansions are unique, so the two series have the same coefficients. But the numbers $(2l-1)!!$ grow faster than every geometric sequence, so the second series converges for no z, a contradiction. Hence $\Theta_1 = 0$, then Θ_2 tends to 0 faster than every power, and $\Theta_2 = 0$. □

A gate on a ray. GELU and SiLU both satisfy $\sigma(z) - \sigma(-z) = z$, and so $\sigma'(z) + \sigma'(-z) = 1$. Let $\delta(y) = y - \sigma(y)$, with the derivative $\delta'(y) = 1 - \sigma'(y)$. For every real number $y > 0$,

$$\sigma(y) = y - \delta(y), \qquad \sigma(-y) = -\delta(y), \qquad \sigma'(y) = 1 - \delta'(y), \qquad \sigma'(-y) = \delta'(y). \tag{24}$$

By Lemma D.7(i), for GELU $\delta(y) = \varphi(y)\,y\;\mathrm{mr}(y)$ and $\delta'(y) = \varphi(y)\,\big(\mathrm{mr}(y) - y\big)$, and for SiLU

$$\delta(y) = \frac{y}{1+e^y}, \qquad \delta'(y) = \frac{1+e^y - y\,e^y}{(1+e^y)^2}.$$

Let $p = 2$ for GELU and $p = 1$ for SiLU. For every real number $y \geq 2$,

$$\tfrac{1}{6}\,e^{-y^p/p} \;\leq\; \delta(y) \;\leq\; y\,e^{-y^p/p}, \qquad |\delta'(y)| \;\leq\; 4\,y\,\delta(y). \tag{25}$$

For GELU, Lemma D.7(ii) gives $\mathrm{mr}(y) \leq 1/y$ and $|\mathrm{mr}(y) - 1/y| \leq 1/y^3$, so $\frac{3}{4} \leq y\,\mathrm{mr}(y) \leq 1$ and $|\mathrm{mr}(y) - y| \leq 2y$. For SiLU, $\frac{1}{2}\,y\,e^{-y} \leq \delta(y) \leq y\,e^{-y}$ and $|\delta'(y)| \leq (2+y)\,e^{-y}$. So on a ray a GELU hidden unit, and the gate of a hidden unit of a gated block, differ from a linear or a vanishing one by the small term δ.

Lemma D.9 (The rest of the network on a ray)**.** *In the case (+), let $k \leq L$, let $h^\circ \neq 0$ be a row, let $v_1,\dots,v_i$ be rows, and let η be a direction of the weights of the blocks $k+1,\dots,L$ and of D. With D^i the derivative of order i in h, the maps*

$$t \;\mapsto\; t^i\;\mathrm{D}^i\mathcal{B}_{>k}(t\,h^\circ)\,(v_1,\dots,v_i) \qquad \textit{and} \qquad t \;\mapsto\; t^i\;\mathrm{D}^i\big(\mathrm{D}_\theta\mathcal{B}_{>k}(\cdot)\,\eta\big)(t\,h^\circ)\,(v_1,\dots,v_i)$$

are real-analytic functions of $1/t$ near $1/t = 0$, so they are meromorphic at infinity. Moreover, there are numbers $\varrho, t_0 > 0$ such that the maps $u \mapsto \mathcal{B}_{>k}\big(t\,(h^\circ+u)\big)$ and $u \mapsto \mathrm{D}_\theta\mathcal{B}_{>k}\big(t\,(h^\circ+u)\big)\,\eta$ extend to holomorphic maps on the rows $u \in \mathbb{C}^{1\times d_{\mathrm{model}}}$ with $|u_j| \leq \varrho$ for all j, which are bounded uniformly in $t \geq t_0$.

Proof. $\mathcal{B}_{>k} = \mathcal{B}^\sharp \circ N_\varepsilon$, with $\mathcal{B}^\sharp(n) = nD$ if $k = L$ and $\mathcal{B}^\sharp = \mathcal{B}_{>k+1}\circ g_{k+1}$ otherwise, which is real-analytic on all rows. For $t > 0$ and a row u,

$$N_\varepsilon\big(t\,(h^\circ+u)\big) \;=\; \frac{h^\circ+u}{\sqrt{\|h^\circ+u\|^2/d_{\mathrm{model}} + \varepsilon\,t^{-2}}},$$

and the right side is a real-analytic function of $(1/t, u)$ near $(0,0)$, also at $1/t = 0$, because $h^\circ \neq 0$. So $\mathcal{B}_{>k}\big(t\,(h^\circ + u)\big)$ is a real-analytic function of $(1/t, u)$ near $(0,0)$. It extends to a holomorphic and bounded function on a complex neighborhood of $(0,0)$, which gives ϱ and t_0. A derivative of order i in $v = t\,u$ at $v = 0$ is t^{-i} times the derivative of order i in u at $u = 0$. The weights enter analytically as well, which gives the same for $\mathrm{D}_\theta \mathcal{B}_{>k}(\cdot)\,\eta$. □

GELU and gated blocks in one form. The rest of the treatment of rays is for GELU, SwiGLU and GeGLU blocks. A GELU block has the form of a GeGLU block in which the value ne_j of every hidden unit is replaced by the number 1. So in a GELU block let $W_g^\ell = W_{\mathrm{read}}^\ell$, and call $c_1, \dots, c_{q_\ell}$ its gate columns. For a row n let $\iota^\ell(n) \in \mathbb{R}^{1\times q_\ell}$ be the row of the values of its hidden units:

$$\iota^\ell(n) = n\,W_v^\ell \text{ in a gated block,} \qquad \iota^\ell(n) = (1,\dots,1) \text{ in a GELU block,}$$
$$g_\ell(n) \;=\; \big(\sigma(n\,W_g^\ell) \odot \iota^\ell(n)\big)\,W_{\mathrm{write}}^\ell \text{ in both.}$$

Let $m = 2$ for gated blocks and $m = 1$ for GELU blocks, so that $\iota^\ell(\lambda n) = \lambda^{m-1}\,\iota^\ell(n)$ for every number $\lambda > 0$. The exponent p of (25) is 2 for GELU and GeGLU blocks and 1 for SwiGLU blocks.

Cones and good parameters. A *cone* is a set of rows that contains λh for each of its rows h and every number $\lambda > 0$. Let $\mathcal{C}_0, \dots, \mathcal{C}_L$ be open convex cones in $\mathbb{R}^{1\times d_{\mathrm{model}}}$ that do not contain 0, one for each hidden state. We call θ *good* for these cones if (R1) holds and every block ℓ, with its gate columns c_j, its write rows a_j and the entries $\iota_j(n)$ of $\iota^\ell(n)$, satisfies:

(K1) $nc_j \neq 0$ for every $n \in \mathcal{C}_{\ell-1}$ and every j. So every gate has one sign on $\mathcal{C}_{\ell-1}$. Let $\mathcal{A}_\ell$ be the set of the hidden units j with $nc_j > 0$ on $\mathcal{C}_{\ell-1}$, the *active* ones, and let

$$g_\ell^\circ(n) \;=\; \sum_{j\in\mathcal{A}_\ell} (nc_j)\,\iota_j(n)\,a_j$$

be the block with its gates *frozen*. It is a linear map for a GELU block and a quadratic map for a gated block: $g_\ell^\circ(\lambda n) = \lambda^m\,g_\ell^\circ(n)$.

(K2) $g_\ell^\circ(n) \in \mathcal{C}_\ell$ for every $n \in \mathcal{C}_{\ell-1}$, and $\mathrm{D}g_\ell^\circ(n)$ is invertible at some row n.

(K3) No a_j vanishes, and in a gated block no value column e_j vanishes. For GELU and GeGLU blocks the matrices $c_1c_1^\top, \dots, c_{q_\ell}c_{q_\ell}^\top$ are linearly independent over $\mathbb{Q}$, and for SwiGLU blocks the columns $c_1, \dots, c_{q_\ell}$ are.

The determinant of $\mathrm{D}g_\ell^\circ(n)$ is a polynomial in n, so by (K2) g_ℓ° is a local diffeomorphism on an open dense set of rows. A good θ is regular. Indeed, take $\xi \in \mathcal{C}_{\ell-1}$ with $\mathrm{D}g_\ell^\circ(\xi)$ invertible. For a row v, by (24) and (25), in a gated block

$$\frac{1}{t}\,\mathrm{D}g_\ell(t\,\xi)\,v \;=\; \sum_j \Big[\sigma'(t\,\xi c_j)\,(vc_j)\,(\xi e_j) + \frac{\sigma(t\,\xi c_j)}{t}\,(ve_j)\Big]\,a_j$$
$$\longrightarrow \sum_{j\in\mathcal{A}_\ell} \Big[(vc_j)\,(\xi e_j) + (\xi c_j)\,(ve_j)\Big]\,a_j \;=\; \mathrm{D}g_\ell^\circ(\xi)\,v \qquad \text{as } t\to\infty,$$

and in a GELU block $\mathrm{D}g_\ell(t\,\xi)\,v = \sum_j \sigma'(t\,\xi c_j)\,(vc_j)\,a_j$ tends to $\sum_{j\in\mathcal{A}_\ell}(vc_j)\,a_j = \mathrm{D}g_\ell^\circ(\xi)\,v$. So $\mathrm{D}g_\ell(t\,\xi)$ is invertible for large t, which is (R3).

The frozen network. In the case (n), for gated blocks and a good θ, let $\mathcal{B}_{>k}^\circ$ be the network after h_k with the gates of all its blocks frozen:

$$\mathcal{B}_{>L}^\circ(h) = hD, \qquad \mathcal{B}_{>k-1}^\circ(h) = \mathcal{B}_{>k}^\circ\big(g_k^\circ(h)\big).$$

It is a polynomial map of h and of the weights. Lemmas D.1 and D.2 hold for $\mathcal{B}_{>k}^\circ$ in the place of $\mathcal{B}_{>k}$, with an open dense set $\mathcal{U}_k^\circ$ of rows and with the same proofs, in which g_{k+1}° takes the place of g_{k+1}.

The identity for one GELU or gated block. Fix $k < L$ and write $W_g = W_g^{k+1}$, with the columns $c_1, \dots, c_q$, $V = W_{\mathrm{write}}^{k+1}$, with the rows $a_1, \dots, a_q$, $\iota = \iota^{k+1}$ and $\mathcal{B} = \mathcal{B}_{>k+1}$, and in a gated block $W_v = W_v^{k+1}$, with the columns $e_1, \dots, e_q$. Let Z and $\eta = (\eta_g, \eta_v, \eta_V, \eta')$ be as in (D_k), where

η_g, η_v and η_V are the entries of η at W_g, W_v and V and η' is the rest; in a GELU block there is no η_v. For a row n let $\tilde{n} = nZ$ in the case (n) and $\tilde{n} = nZ - \frac{\langle nZ, n\rangle}{d_{\text{model}}}\, n$ in the case (+). For rows n and $\tilde{w}, \tilde{w}' \in \mathbb{R}^{1\times q}$ let

$$\dot{z}^g(n) = \tilde{n}\, W_g + n\, \eta_g, \qquad \dot{z}^v(n) = \begin{cases} \tilde{n}\, W_v + n\, \eta_v & \text{in a gated block,} \\ 0 & \text{in a GELU block,} \end{cases} \tag{26}$$

$$\mathcal{L}(\tilde{w}, \tilde{w}'; n) = \mathrm{D}\mathcal{B}\big(\hat{w}V\big)\, \Big(\big(\tilde{w}' \odot \dot{z}^g(n) \odot \iota(n) + \tilde{w} \odot \dot{z}^v(n)\big)\, V + \hat{w}\, \eta_V \Big) + \mathrm{D}_\theta \mathcal{B}\big(\hat{w}V\big)\, \eta', \tag{27}$$

where $\hat{w} = \tilde{w} \odot \iota(n)$, and $\dot{z}^g_j(n)$ and $\dot{z}^v_j(n)$ are the entries of the rows (26). Here $\tilde{w}$ stands for the row $\sigma(nW_g)$ of the gates, $\tilde{w}'$ for the row $\sigma'(nW_g)$ of their derivatives, and $\hat{w}$ for the row of the outputs of the hidden units, and $\dot{z}^g(n)$ and $\dot{z}^v(n)$ are the derivatives of the rows nW_g and $\iota(n)$. In a GELU block $\hat{w} = \tilde{w}$, and these are the maps (20) and (21). By the chain rule and (14), the left side of (16) at h is $\mathcal{L}\big(\sigma(nW_g), \sigma'(nW_g); n\big)$ at $n = N(h)$. As in Lemma D.5, (16) for k therefore holds if and only if

$$\mathcal{L}\big(\sigma(nW_g),\ \sigma'(nW_g);\ n\big) \ = \ 0 \qquad \text{for every row } n \in \mathbb{R}^{1\times d_{\text{model}}}. \tag{28}$$

We use two properties of $\mathcal{L}$. The row $\tilde{w}'$ enters only through the terms $\tilde{w}'_j\, \dot{z}^g_j(n)\, \iota_j(n)\, a_j$, so

$$\frac{\partial \mathcal{L}}{\partial \tilde{w}'_j}(\tilde{w}, \tilde{w}'; n) \ = \ \dot{z}^g_j(n)\ \iota_j(n)\ \mathrm{D}\mathcal{B}\big(\hat{w}V\big)\, a_j. \tag{29}$$

And in a gated block $\dot{z}^g$ and $\dot{z}^v$ are odd in n, while $\hat{w}$ does not change when $\tilde{w}$ and n both change their sign, so

$$\mathcal{L}(-\tilde{w}, \tilde{w}'; -n) \ = \ \mathcal{L}(\tilde{w}, \tilde{w}'; n) \qquad \text{in a gated block.} \tag{30}$$

Lemma D.10 (The network after h_k far out in its cone)**.** *In the case (n), for gated blocks, let θ be good, $k \le L$, $h^\circ \in \mathcal{C}_k$, and let η be a direction of the weights of the blocks $k+1, \dots, L$ and of D. There are numbers $\varrho, \kappa, t_0 > 0$ such that, for $t \ge t_0$, the maps $u \mapsto \mathcal{B}_{>k}\big(t\,(h^\circ + u)\big)$ and $u \mapsto \mathrm{D}_\theta \mathcal{B}_{>k}\big(t\,(h^\circ + u)\big)\,\eta$ extend to holomorphic maps on the rows $u \in \mathbb{C}^{1\times d_{\text{model}}}$ with $|u_j| \le \varrho$ for all j, and for these u, at $h = t\,(h^\circ + u)$,*

$$\|\mathcal{B}_{>k}(h) - \mathcal{B}^\circ_{>k}(h)\| \ \le \ e^{-\kappa t^p}, \qquad \|\mathrm{D}_\theta \mathcal{B}_{>k}(h)\,\eta - \mathrm{D}_\theta \mathcal{B}^\circ_{>k}(h)\,\eta\| \ \le \ e^{-\kappa t^p}.$$

Proof. One gate. Let $z \in \mathbb{C}$ with $\operatorname{Re} z \ge 2$ and $|\operatorname{Im} z| \le \frac{1}{2} \operatorname{Re} z$, so that $\operatorname{Re}(z^2) \ge \frac{3}{4}(\operatorname{Re} z)^2$ and $|z| \le 2 \operatorname{Re} z$. For GELU, by Lemma D.7, $\sigma(z) - z = -\varphi(z)\, z\, \mathrm{mr}(z)$ and $|z\, \mathrm{mr}(z)| \le |z| / \operatorname{Re} z \le 2$, so $|\sigma(z) - z| \ \le \ e^{-\frac{3}{8}(\operatorname{Re} z)^2}$. For SiLU, $\sigma(z) - z = -z/(1+e^z)$ and $|1 + e^z| \ \ge \ \frac{1}{2}\, e^{\operatorname{Re} z}$, so $|\sigma(z) - z| \le 4 \operatorname{Re} z\, e^{-\operatorname{Re} z}$. Both gates are holomorphic at z and at $-z$, and $\sigma(-z) = \sigma(z) - z$. So there is a number $\kappa_1 > 0$ such that, for large $\operatorname{Re} z$,

$$|\sigma(z) - z| \ \le \ e^{-\kappa_1 (\operatorname{Re} z)^p} \qquad \text{and} \qquad |\sigma(-z)| \ \le \ e^{-\kappa_1 (\operatorname{Re} z)^p}. \tag{31}$$

The first bound, by induction on $L - k$. We prove it at the weights $\theta + \tau\eta$ as well, for all complex numbers τ with $|\tau| \le \tau_0$ and a small $\tau_0 > 0$, where the hidden units whose gates are frozen are those of θ. For $k = L$, $\mathcal{B}_{>L}(h) = hD = \mathcal{B}^\circ_{>L}(h)$. Let the bound hold for some $k \ge 1$, and let $h^\circ \in \mathcal{C}_{k-1}$. By (K1), $h^\circ c_j \ne 0$ for the gate columns c_j of block k at θ. Let ϱ and τ_0 be small, and let $h = t\,(h^\circ + u)$. For every j, with c_j taken at the weights $\theta + \tau\eta$, the number $z = hc_j$ or $z = -hc_j$ with the sign of $h^\circ c_j$ satisfies $\operatorname{Re} z \ge \frac{t}{2}\, |h^\circ c_j|$ and $|\operatorname{Im} z| \le \frac{1}{2} \operatorname{Re} z$. By (31), g_k is holomorphic at these h, and

$$\|g_k(h) - g^\circ_k(h)\| \ \le \ e^{-\kappa_2 t^p}, \qquad g^\circ_k(h) \ = \ t^2\, \big(g^\circ_k(h^\circ) + u'\big),$$

with a number $\kappa_2 > 0$, where g_k and g°_k are taken at the weights $\theta + \tau\eta$ and $g^\circ_k(h^\circ)$ at θ, and where the entries of the row u' are as small as we like when ϱ and τ_0 are small. So $g_k(h)$ is $t^2\, \big(g^\circ_k(h^\circ) + u''\big)$ with a row u'' of the same kind. By (K2), $g^\circ_k(h^\circ) \in \mathcal{C}_k$. So the bound for k, at the row $g^\circ_k(h^\circ)$ and with t^2 in the place of t, gives $\|\mathcal{B}_{>k}(g_k(h)) - \mathcal{B}^\circ_{>k}(g_k(h))\| \le e^{-\kappa t^{2p}}$. The derivative of the polynomial map $\mathcal{B}^\circ_{>k}$ is at most a fixed power of t at the rows $t^2\, (g^\circ_k(h^\circ) + u')$, so $\|\mathcal{B}^\circ_{>k}(g_k(h)) - \mathcal{B}^\circ_{>k}(g^\circ_k(h))\|$

is at most a fixed power of t times $e^{-\kappa_2 t^p}$. As $\mathcal{B}_{>k-1} = \mathcal{B}_{>k} \circ g_k$ and $\mathcal{B}^\circ_{>k-1} = \mathcal{B}^\circ_{>k} \circ g^\circ_k$, this gives the bound for $k-1$, with a smaller κ.

The second bound. At $h = t\,(h^\circ + u)$ the difference of $\mathcal{B}_{>k}(h)$ and $\mathcal{B}^\circ_{>k}(h)$ at the weights $\theta + \tau\eta$ is holomorphic in τ for $|\tau| \le \tau_0$ and at most $e^{-\kappa t^p}$. By Cauchy's estimate its derivative at $\tau = 0$, which is $\mathrm{D}_\theta \mathcal{B}_{>k}(h)\,\eta - \mathrm{D}_\theta \mathcal{B}^\circ_{>k}(h)\,\eta$, is at most $e^{-\kappa t^p}/\tau_0$. □

Lemma D.11 (The step for GELU and gated blocks on rays)**.** *Let the blocks be GELU blocks in the case (+), or gated blocks in the case (n) or (+). Let θ be good for the cones $\mathcal{C}_0, \dots, \mathcal{C}_L$, and let $k < L$. Then (D_{k+1}) implies (D_k).*

Proof. Let Z and η satisfy (16) for k, so that (28) holds, with the notation above. Let $\mathcal{A} = \mathcal{A}_{k+1}$ and $g^\circ = g^\circ_{k+1}$. Let s_j be the sign of nc_j on $\mathcal{C}_k$ and $\bar{s} = (s_1, \dots, s_q)$, let $I_\mathcal{A}$ be the diagonal matrix with the entries 1 at the hidden units of $\mathcal{A}$ and 0 elsewhere, and $1_\mathcal{A}$ the row with these entries. Let

$$\mathcal{B}^\infty = \mathcal{B}_{>k+1} \text{ in the case (+)}, \qquad \mathcal{B}^\infty = \mathcal{B}^\circ_{>k+1} \text{ in the case (n)},$$

and let $\mathcal{L}^\infty$ be (27) with $\mathcal{B}^\infty$ in the place of $\mathcal{B}$. It is real-analytic in $(\tilde{w}, \tilde{w}', n)$ on all rows, a polynomial in the case (n), and it satisfies (29), and (30) in a gated block, with $\mathcal{B}^\infty$. Lemmas D.1 and D.2 hold for $\mathcal{B}^\infty$, and we write $\mathcal{U}^\infty$ for its open dense set $\mathcal{U}_{k+1}$ or $\mathcal{U}^\circ_{k+1}$.

The ray. Let $\xi \in \mathcal{C}_k$ and $n = t\,\xi$ with $t > 0$. Let

$$r_j = |\xi c_j|^p, \qquad \delta_j(t) = \delta\big(t\,|\xi c_j|\big), \qquad \delta'_j(t) = \delta'\big(t\,|\xi c_j|\big),$$

and let $\delta(t)$ and $\delta'(t)$ be the rows with these entries. By (25), with a constant κ, for large t

$$\tfrac{1}{6}\, e^{-r_j t^p/p} \;\le\; \delta_j(t) \;\le\; \kappa\, t\, e^{-r_j t^p/p}, \qquad |\delta'_j(t)| \;\le\; \kappa\, t\, \delta_j(t). \tag{32}$$

By (24), $\sigma(nW_g) = n\,W_g I_\mathcal{A} - \delta(t)$ and $\sigma'(nW_g) = 1_\mathcal{A} - \bar{s} \odot \delta'(t)$. So (28) at $n = t\,\xi$ is

$$\mathcal{L}\big(t\,\xi W_g I_\mathcal{A} - \delta(t),\;\; 1_\mathcal{A} - \bar{s} \odot \delta'(t);\;\; t\,\xi\big) \;=\; 0 \qquad \text{for all } t > 0. \tag{33}$$

Without the small terms the hidden units of the block give the row $t\,\xi W_g I_\mathcal{A} \odot \iota(t\,\xi)$, and the block gives $g^\circ(t\,\xi) = t^m\, h^\circ$ with $h^\circ = g^\circ(\xi)$, which lies in $\mathcal{C}_{k+1}$ by (K2), so $h^\circ \ne 0$. With a row δ in the place of $\delta(t)$ the block gives $t^m\,(h^\circ + u)$ with $u = -\frac{1}{t}\big(\delta \odot \iota(\xi)\big)\,V$.

The expansion. For a multi-index $\alpha = (\alpha_1, \dots, \alpha_q)$ of nonnegative integers let $\delta^\alpha = \prod_j \delta_j^{\alpha_j}$, $|\alpha| = \sum_j \alpha_j$, $\alpha! = \prod_j \alpha_j!$ and $r_\alpha = \sum_j \alpha_j r_j$, and let 1_j be the multi-index with the entry 1 at j and 0 elsewhere. Let

$$\mathcal{L}_\alpha(t) = \frac{1}{\alpha!}\,\frac{\partial^{|\alpha|}\mathcal{L}^\infty}{\partial \tilde{w}^\alpha}, \qquad \mathcal{L}'_{\alpha,i}(t) = \frac{1}{\alpha!}\,\frac{\partial^{|\alpha|}}{\partial \tilde{w}^\alpha}\,\frac{\partial \mathcal{L}^\infty}{\partial \tilde{w}'_i}, \qquad \text{both at } \big(t\,\xi W_g I_\mathcal{A},\; 1_\mathcal{A};\; t\,\xi\big).$$

We write $\mathcal{L}_j = \mathcal{L}_{1_j}$ and $\mathcal{L}'_j = \mathcal{L}'_{0,j}$. As $\mathcal{L}^\infty$ is real-analytic in $\tilde{w}$ and affine in $\tilde{w}'$,

$$\mathcal{L}^\infty\big(t\,\xi W_g I_\mathcal{A} - \delta,\;\; 1_\mathcal{A} - \bar{s} \odot \delta';\;\; t\,\xi\big) \;=\; \sum_\alpha (-1)^{|\alpha|}\,\delta^\alpha \Big[\mathcal{L}_\alpha(t) - \sum_{i=1}^q s_i\,\mathcal{L}'_{\alpha,i}(t)\,\delta'_i\Big] \tag{34}$$

for small rows δ and all rows δ'. By (32), at $\delta = \delta(t)$ and $\delta' = \delta'(t)$ the term of δ^α is of the order of $e^{-r_\alpha t^p/p}$, up to powers of t, and the term of $\delta^\alpha \delta'_i$ of the order of $e^{-(r_\alpha + r_i)t^p/p}$. We call r_α and $r_\alpha + r_i$ the *exponents* of these terms. We use three facts.

(a) Every $\mathcal{L}_\alpha$ and $\mathcal{L}'_{\alpha,i}$ is a real-analytic function of $t \in \mathbb{R}$ that is meromorphic at infinity. In the case (n) it is a polynomial in t. In the case (+) it is built from derivatives of $\mathcal{B}$ and of $\mathrm{D}_\theta \mathcal{B}(\cdot)\,\eta'$ at $t^m\,h^\circ$ and from polynomials in t, so this is Lemma D.9 with t^m in the place of t.

(b) At $\delta = \delta(t)$ and $\delta' = \delta'(t)$, the left side of (34) vanishes in the case (+), by (33). In the case (n), where the blocks are gated and $m = 2$, it is at most $e^{-\kappa' t^{2p}}$ for large t, with a number $\kappa' > 0$. Indeed, by Lemma D.10 for $k+1$, at the row h° and with t^2 in the place of t, the maps $\mathcal{B} - \mathcal{B}^\infty$ and $\mathrm{D}_\theta \mathcal{B}(\cdot)\,\eta' - \mathrm{D}_\theta \mathcal{B}^\infty(\cdot)\,\eta'$ are holomorphic and at most $e^{-2\kappa' t^{2p}}$ at the rows $t^2\,(h^\circ + u)$ with $|u_j| \le \varrho$, and by Cauchy's estimate so is the derivative of the first, for large t, where $|u_j| \le \varrho/2$. At $\delta = \delta(t)$ the block gives such a row, with $u \to 0$, and the rows to which these derivatives are applied in (27) are at most a fixed power of t. So $\mathcal{L}$ and $\mathcal{L}^\infty$ differ at the point of (33) by at most a fixed power of t times $e^{-2\kappa' t^{2p}}$, and $\mathcal{L}$ vanishes there.

(c) For large t the series (34) converges absolutely at $\delta(t)$, $\delta'(t)$, and the sum of any set of its terms is at most a fixed power of t times $e^{-rt^p/p}$, where r is the smallest exponent of a term of the set. In the case (n) the series is a finite sum with coefficients that are polynomials in t, and this follows from (32). In the case (+), by Lemma D.9 with t^m in the place of t, the maps $u \mapsto \mathcal{B}\big(t^m(h^\circ + u)\big)$ and $u \mapsto \mathrm{D}_\theta \mathcal{B}\big(t^m(h^\circ + u)\big)\,\eta'$ are holomorphic and bounded on the rows u with $|u_j| \le \varrho$, uniformly in $t \ge t_0$, and by Cauchy's estimate so is the derivative of the first where $|u_j| \le \varrho/2$. With a row δ in the place of $\delta(t)$ the block gives $t^m\,(h^\circ + u)$, and there is a number $\varrho' > 0$ such that $|u_j| \le \varrho/2$ for the rows $\delta \in \mathbb{C}^{1\times q}$ with $|\delta_j| \le \varrho'\,t$ for all j. So the left side of (34), which is affine in δ', is holomorphic in these rows δ, and there its value at $\delta' = 0$ and its coefficients of the entries of δ' are at most a fixed power of t. By Cauchy's estimates, $\mathcal{L}_\alpha(t)$ and $\mathcal{L}'_{\alpha,i}(t)$ are at most a fixed power of t times $(\varrho'\,t)^{-|\alpha|}$. With (32), a term with the exponent r' is then at most a fixed power of t times $(\kappa/\varrho')^{|\alpha|}\,e^{-r't^p/p}$. As $r' \ge |\alpha|\,\min_j r_j$, the terms with large $|\alpha|$ are bounded by a geometric series.

A generic ray. We now take $\xi \in \mathcal{C}_k$ such that

(i) the numbers $r_1, \dots, r_q$ are linearly independent over $\mathbb{Q}$;
(ii) $\iota_j(\xi) \ne 0$ for every j;
(iii) $g^\circ(\xi) \in \mathcal{U}^\infty$.

These ξ are dense in $\mathcal{C}_k$. For (i): if integers $\kappa_1, \dots, \kappa_q$ are not all zero, $\sum_j \kappa_j r_j$ is $\xi\big(\sum_j \kappa_j c_j c_j^\top\big)\xi^\top$ for GELU and GeGLU blocks and $\xi \sum_j \kappa_j s_j c_j$ for SwiGLU blocks, and by (K3) it vanishes only for ξ on a quadric, or in a hyperplane. For (ii): in a gated block $\iota_j(\xi) = \xi e_j$, and $e_j \ne 0$ by (K3). For (iii): g° is a local diffeomorphism on an open dense set of rows, and $\mathcal{U}^\infty$ is open and dense. By (i), different multi-indices have different exponents r_α, and only finitely many exponents lie below any bound.

The induction. We show that, for every multi-index α,

$$\mathcal{L}_\alpha(t) = 0 \quad \text{and} \quad \mathcal{L}'_{\alpha - 1_i,\,i}(t) = 0 \ \text{ for every } i \text{ with } \alpha_i \ge 1, \qquad \text{for all } t \in \mathbb{R}, \tag{35}$$

by induction on the exponent r_α. These are the coefficients of the terms of (34) with the exponent r_α. Let (35) hold for all multi-indices with a smaller exponent, and let $r^+ > r_\alpha$ be the next exponent. In (34) at $\delta(t)$, $\delta'(t)$, the terms with a smaller exponent vanish, and by (c) the terms with a larger exponent add up to at most a fixed power of t times $e^{-r^+t^p/p}$. By (b), so do the terms with the exponent r_α, which are

$$(-1)^{|\alpha|}\,\delta(t)^\alpha \Big[\mathcal{L}_\alpha(t) + \sum_{i:\,\alpha_i \ge 1} s_i\,\mathcal{L}'_{\alpha - 1_i,\,i}(t)\,\frac{\delta'_i(t)}{\delta_i(t)}\Big].$$

By (32), $\delta(t)^\alpha \ge 6^{-|\alpha|}\,e^{-r_\alpha t^p/p}$. So

$$\mathcal{L}_\alpha(t) \;+\; \sum_{i:\,\alpha_i \ge 1} s_i\,\mathcal{L}'_{\alpha - 1_i,\,i}(t)\,\frac{\delta'_i(t)}{\delta_i(t)} \;\longrightarrow\; 0 \qquad \text{faster than every power of } 1/t. \tag{36}$$

No small factor, $\alpha = 0$. (36) is $\mathcal{L}_0(t) \to 0$, and by (a) $\mathcal{L}_0 = 0$.

One small factor, $\alpha = 1_j$. With $y = t\,|\xi c_j|$, (36) is $\mathcal{L}_j + s_j\,\mathcal{L}'_j\,\delta'(y)/\delta(y) \to 0$.

For GELU and GeGLU blocks, $\delta'(y)/\delta(y) = \big(\mathrm{mr}(y) - y\big)/\big(y\,\mathrm{mr}(y)\big)$ and $y\,\mathrm{mr}(y) \le 1$, so

$$\big(y\,\mathcal{L}_j + s_j\,\mathcal{L}'_j\big)\,\mathrm{mr}(y) \;-\; s_j\,y\,\mathcal{L}'_j \;\longrightarrow\; 0 \qquad \text{faster than every power of } 1/t.$$

By (a), the maps $y\,\mathcal{L}_j + s_j\mathcal{L}'_j$ and $y\,\mathcal{L}'_j$ are meromorphic at infinity. By Lemma D.8 with $z(t) = |\xi c_j|\,t$ both vanish, so $\mathcal{L}'_j = 0$ and $\mathcal{L}_j = 0$.

For SwiGLU blocks, $y\,\delta'(y)/\delta(y) = 1 - y + \delta(y)$, and $\delta(y)$ tends to 0 faster than every power of $1/t$, so

$$f(t) \;=\; |\xi c_j|\,t\,\mathcal{L}_j(t) + s_j\,\big(1 - |\xi c_j|\,t\big)\,\mathcal{L}'_j(t)$$

tends to 0 faster than every power of $1/t$ as $t \to +\infty$. By (a), f is meromorphic at infinity, so $f(t) = 0$ for large t, and then for all $t \in \mathbb{R}$, because f is real-analytic on $\mathbb{R}$. By (30), $\mathcal{L}_j$ is odd in t and $\mathcal{L}'_j$ is even. So $0 = f(t) - f(-t) = -2\,s_j\,|\xi c_j|\,t\,\mathcal{L}'_j(t)$: $\mathcal{L}'_j = 0$, and then $\mathcal{L}_j = 0$.

For all three kinds of block, by (29), $\mathcal{L}'_j(t) = \dot{z}^g_j(t\,\xi)\;\iota_j(t\,\xi)\;\mathrm{D}\mathcal{B}^\infty(t^m\,h^\circ)\,a_j$. By (iii), (K3) and Lemma D.1, $\mathrm{D}\mathcal{B}^\infty(h^\circ)\,a_j \neq 0$, so the last factor, which is real-analytic in t, vanishes only at isolated t. By (ii), $\iota_j(t\,\xi) = t^{m-1}\,\iota_j(\xi) \neq 0$ for $t > 0$. So

$$\dot{z}^g_j(t\,\xi) = 0 \qquad \text{for all } t > 0. \tag{37}$$

Two or more small factors, $|\alpha| \geq 2$. Let $\alpha_i \geq 1$. Then $r_i < r_\alpha$, so (37) holds for the hidden unit i, and by (29) the derivative of $\mathcal{L}^\infty(\tilde{w}, \tilde{w}'; t\,\xi)$ in $\tilde{w}'_i$ vanishes for all rows $\tilde{w}$. So $\mathcal{L}'_{\alpha-1_i,\,i} = 0$. Then (36) is $\mathcal{L}_\alpha(t) \to 0$, and by (a) $\mathcal{L}_\alpha = 0$. This ends the induction.

What the induction gives. For this ξ and every $t > 0$, all coefficients of the expansion of $(\tilde{w}, \tilde{w}') \mapsto \mathcal{L}^\infty(\tilde{w}, \tilde{w}'; t\,\xi)$ at $(t\,\xi W_g I_{\mathcal{A}}, 1_{\mathcal{A}})$ vanish, so this real-analytic map vanishes. The rows $t\,\xi$ of this kind are dense in $\mathcal{C}_k$. So, with (37), for all $n \in \mathcal{C}_k$ and all rows $\tilde{w} \in \mathbb{R}^{1\times q}$, with $\hat{w} = \tilde{w} \odot \iota(n)$,

$$\dot{z}^g(n) = 0, \qquad \mathrm{D}\mathcal{B}^\infty(\hat{w}V)\left(\big(\tilde{w} \odot \dot{z}^v(n)\big)\,V + \hat{w}\,\eta_V\right) + \mathrm{D}_\theta\mathcal{B}^\infty(\hat{w}V)\,\eta' = 0. \tag{38}$$

Step 1: the gates. By (26), $\dot{z}^g(n)$ is a polynomial in n that vanishes on the open set $\mathcal{C}_k$, so it vanishes for all n. In the case (n) this is $n\,(ZW_g + \eta_g) = 0$, so $\eta_g = -ZW_g$, and $\mathfrak{g}$ contains every Z. In the case (+) its part of degree 3 is $\langle nZ, n\rangle\,nW_g = 0$, so Z is antisymmetric, as $W_g \neq 0$, and its part of degree 1 gives $\eta_g = -ZW_g$. In both cases $Z \in \mathfrak{g}$, and in a gated block $\dot{z}^v(n) = n\,(ZW_v + \eta_v)$.

Step 2: the values and the write. In a gated block, fix $n \in \mathcal{C}_k$ with $ne_j \neq 0$ for every j, and let $\Delta(n)$ be the diagonal matrix with the entries $-\dot{z}^v_j(n)/(ne_j)$, so that $\tilde{w} \odot \dot{z}^v(n) = -\hat{w}\,\Delta(n)$. In a GELU block, fix $n \in \mathcal{C}_k$ and let $\Delta(n) = 0$. The row $\hat{w}$ takes every value in $\mathbb{R}^{1\times q}$, and (38) reads

$$\mathrm{D}\mathcal{B}^\infty(\hat{w}V)\left(\hat{w}\left(\eta_V - \Delta(n)\,V\right)\right) + \mathrm{D}_\theta\mathcal{B}^\infty(\hat{w}V)\,\eta' = 0 \qquad \text{for all rows } \hat{w}.$$

By Lemma D.2 for $\mathcal{B}^\infty$, with $A = V$, there is a matrix $Z'(n)$ with

$$\eta_V - \Delta(n)\,V = V\,Z'(n), \qquad \mathrm{D}\mathcal{B}^\infty(h)\left(hZ'(n)\right) + \mathrm{D}_\theta\mathcal{B}^\infty(h)\,\eta' = 0 \quad \text{for all rows } h. \tag{39}$$

For two such rows n and n_1, subtracting gives $\mathrm{D}\mathcal{B}^\infty(h)\left(h\,(Z'(n) - Z'(n_1))\right) = 0$ for all h, so $Z'(n) = Z'(n_1)$ by Lemma D.1. Let Z' be this matrix. Then $\Delta(n)\,V = \eta_V - VZ'$ does not depend on n, and no row of V vanishes, so $\Delta(n)$ is one diagonal matrix Δ_{k+1} for all these n, with $\Delta_{k+1} = 0$ in a GELU block. So $\eta_V = \Delta_{k+1}V + VZ'$. In a gated block, $\dot{z}^v(n) = -n\,W_v\Delta_{k+1}$ on an open set of rows n, that is, $n\,(ZW_v + \eta_v + W_v\Delta_{k+1}) = 0$: $\eta_v = -ZW_v - W_v\Delta_{k+1}$.

Step 3: the rest. In the case (+), $\mathcal{B}^\infty = \mathcal{B}_{>k+1}$, and the second part of (39) is (16) for $k+1$, with Z' and η'. By (D_{k+1}), $Z' \in \mathfrak{g}$ and η' is of the form (19), or (17) for GELU blocks, with $Z_{k+1} = Z'$. With Steps 1 and 2 this is (19), or (17), for k.

In the case (n), where the blocks are gated, (39) is an identity for the frozen network. The frozen network does not change along some directions along which the network does, for example along every direction of the weights of a hidden unit that is not active, so (39) does not determine η'. Let $\eta^\flat$ be the direction (19) for k with $Z_k = Z$, with Δ_{k+1} as in Step 2, with $Z_{k+1} = Z'$, and with $Z_\ell = 0$ and $\Delta_\ell = 0$ for all later ℓ; here $\mathfrak{g}$ contains every matrix. By the converse of the claim, Z and $\eta^\flat$ satisfy (16) for k, and so do Z and η. So the matrix 0 and the direction $\eta - \eta^\flat$ satisfy it. By Steps 1 and 2, $\eta - \eta^\flat$ vanishes at the weights of block $k+1$. Let η'' be its entries at the later weights. Then (16) for k reads

$$\mathrm{D}_\theta\mathcal{B}_{>k+1}\big(g_{k+1}(h)\big)\,\eta'' = 0 \qquad \text{for all rows } h.$$

By (R3) the values of g_{k+1} contain an open set, and the left side is real-analytic, so $\mathrm{D}_\theta\mathcal{B}_{>k+1}(h)\,\eta'' = 0$ for all rows h. This is (16) for $k+1$ with the matrix 0. By (D_{k+1}), η'' is of the form (19) for $k+1$ with $Z_{k+1} = 0$. So $\eta = \eta^\flat + \eta''$ is of the form (19) for k. □

The parameters $\theta^\dagger$. Let $\mathbf{1}$ be the column of d_{model} ones. Choose numbers $\mu_1, \mu_2, \cdots > 0$ such that $1, \mu_1, \mu_2, \ldots$ are linearly independent over $\mathbb{Q}$, for SwiGLU blocks, or such that $\mu_1^2, \mu_2^2, \ldots$ are, for GELU and GeGLU blocks. Almost every choice of finitely many numbers is of this kind. Let $\theta^\dagger$ be the parameters with

$$E = \begin{bmatrix} I \\ 0 \end{bmatrix} \in \mathbb{R}^{d_{\mathrm{in}}\times d_{\mathrm{model}}}, \qquad D = \begin{bmatrix} I & 0 \end{bmatrix} \in \mathbb{R}^{d_{\mathrm{model}}\times d_{\mathrm{out}}},$$

and, in every block ℓ, with $q = q_\ell$ hidden units,

$$W_g^\ell = \big[\, I \,\big|\, -\mu_1 \mathbf{1} \,\big|\, \cdots \,\big|\, -\mu_{q-d_{\text{model}}} \mathbf{1} \,\big], \qquad W_{\text{write}}^\ell = \begin{bmatrix} I \\ \mathbf{1}^\top \\ \vdots \\ \mathbf{1}^\top \end{bmatrix}, \qquad W_v^\ell = \big[\, \mathbf{1} \,\big|\, \cdots \,\big|\, \mathbf{1} \,\big],$$

where only a gated block has W_v^ℓ. So the gate column c_j is column j of I for $j \le d_{\text{model}}$ and $c_j = -\mu_{j-d_{\text{model}}}\, \mathbf{1}$ for $j > d_{\text{model}}$, the write row a_j is row j of I for $j \le d_{\text{model}}$ and $a_j = \mathbf{1}^\top$ for $j > d_{\text{model}}$, and in a gated block every value column is $e_j = \mathbf{1}$. Let $\mathbb{R}^{1\times d_{\text{model}}}_{>0}$ be the positive orthant, the set of the rows n with $n_i > 0$ for every i. It is an open convex cone that does not contain 0.

Lemma D.12 ($\theta^\dagger$ is good)**.** *$\theta^\dagger$ is good for the cones $\mathcal{C}_0 = \cdots = \mathcal{C}_L = \mathbb{R}^{1\times d_{\text{model}}}_{>0}$. For every block, $\mathcal{A}_\ell = \{1, \dots, d_{\text{model}}\}$, and $g_\ell^\circ(n) = n$ for a GELU block and $g_\ell^\circ(n) = (n\mathbf{1})\, n$ for a gated block, where $n\mathbf{1} = n_1 + \cdots + n_{d_{\text{model}}}$.*

Proof. (R1): E, D and every W_{write}^ℓ contain the identity matrix. (K1): for $n \in \mathbb{R}^{1\times d_{\text{model}}}_{>0}$, $nc_j = n_j > 0$ for $j \le d_{\text{model}}$ and $nc_j = -\mu_{j-d_{\text{model}}}\, n\mathbf{1} < 0$ for $j > d_{\text{model}}$. So $\mathcal{A}_\ell = \{1, \dots, d_{\text{model}}\}$, and $g_\ell^\circ(n) = \sum_{j\le d_{\text{model}}} n_j\, \iota_j(n)\, a_j$ is n for a GELU block and $(n\mathbf{1})\, n$ for a gated block, where $\iota_j(n) = n\mathbf{1}$. (K2): $g_\ell^\circ(n)$ is a positive multiple of n. Its derivative is the identity for a GELU block, and for a gated block the map $v \mapsto (n\mathbf{1})\, v + (v\mathbf{1})\, n$, with the determinant $2\,(n\mathbf{1})^{d_{\text{model}}} \neq 0$. (K3): no a_j and no e_j vanishes. For SwiGLU blocks, $\sum_j \kappa_j c_j = 0$ with integers κ_j says that $\kappa_j = \sum_{i>d_{\text{model}}} \kappa_i\, \mu_{i-d_{\text{model}}}$ for every $j \le d_{\text{model}}$, so every κ_j vanishes, by the choice of the μ_i. For GELU and GeGLU blocks, the entries of $\sum_j \kappa_j c_j c_j^\top = 0$ off the diagonal, which exist because $d_{\text{model}} \ge 2$, give $\sum_{i>d_{\text{model}}} \kappa_i\, \mu_{i-d_{\text{model}}}^2 = 0$, so $\kappa_i = 0$ for $i > d_{\text{model}}$, and then the diagonal gives $\kappa_j = 0$ for $j \le d_{\text{model}}$. □

D.5 The great circle: $\varepsilon = 0$

The great circle: $\varepsilon = 0$. We now treat the case (0), for $\tanh$ and GELU blocks. The identity (22) then holds only on the sphere $\mathbb{S}$. We continue it along a great circle of $\mathbb{S}$ to complex angles, on which the inputs of the hidden units go to infinity.

RMSNorm at complex rows. For $h \in \mathbb{C}^{1\times d_{\text{model}}}$ with $\operatorname{Re}\langle h, h\rangle > 0$ let

$$N_\varepsilon(h) \;=\; \frac{\sqrt{d_{\text{model}}}\, h}{\sqrt{\langle h, h\rangle}},$$

with the principal square root. It is holomorphic, $N_\varepsilon(\lambda h) = N_\varepsilon(h)$ for $\lambda > 0$, and

$$\langle n, n\rangle = d_{\text{model}} \quad\Longrightarrow\quad N_\varepsilon(n) = n, \qquad \mathrm{D}N_\varepsilon(n)\, v = v - \frac{\langle v, n\rangle}{d_{\text{model}}}\, n. \tag{40}$$

For $\varkappa \in (0, 1]$ let

$$\mathbb{C}_\varkappa \;=\; \big\{ u + \mathrm{i}u' \;:\; u, u' \in \mathbb{R}^{1\times d_{\text{model}}},\ \|u'\| < \varkappa \|u\| \big\},$$

an open cone in $\mathbb{C}^{1\times d_{\text{model}}}$ that contains every real row $h \neq 0$. On $\mathbb{C}_1$, $\operatorname{Re}\langle h, h\rangle = \|u\|^2 - \|u'\|^2 > 0$.

Lemma D.13 ($\mathcal{B}_{>k}$ on a complex cone)**.** *Let θ satisfy (R1) and (R2). For every k there is a $\varkappa_k \in (0, 1]$ such that $\mathcal{B}_{>k}$, with its derivatives $\mathrm{D}\mathcal{B}_{>k}$ and $\mathrm{D}_\theta\mathcal{B}_{>k}$, extends to a holomorphic map on $\mathbb{C}_{\varkappa_k}$ with $\mathcal{B}_{>k}(\lambda h) = \mathcal{B}_{>k}(h)$ for $\lambda > 0$. The number $\varkappa_k$ depends only on the weights of the blocks $k{+}1, \dots, L$ and on D.*

Proof. $\mathcal{B}_{>L}(h) = N_\varepsilon(h)\, D$ is holomorphic on $\mathbb{C}_1$: $\varkappa_L = 1$. Let $k < L$. The map g_{k+1} is holomorphic near $\mathbb{S}$ and maps the compact set $\mathbb{S}$ into the real rows $h \neq 0$, by (R2), hence into the open set $\mathbb{C}_{\varkappa_{k+1}}$. So there is a $\rho > 0$ such that $n \mapsto \mathcal{B}_{>k+1}(g_{k+1}(n))$ is holomorphic on the set of the

$n \in \mathbb{C}^{1\times d_{\text{model}}}$ at distance less than ρ from $\mathbb{S}$. On the compact set of the $u + \mathrm{i}u'$ with $\|u\| = 1$ and $\|u'\| \le \frac{1}{2}$, N_ε is uniformly continuous, and $N_\varepsilon(u) \in \mathbb{S}$. So there is a $\varkappa_k$ such that N_ε maps $\mathbb{C}_{\varkappa_k}$ into the points at distance less than ρ from $\mathbb{S}$, and $\mathcal{B}_{>k} = \big(\mathcal{B}_{>k+1} \circ g_{k+1}\big) \circ N_\varepsilon$ is holomorphic on $\mathbb{C}_{\varkappa_k}$. The weights enter holomorphically as well. □

The path. For orthonormal rows $\xi, \xi' \in \mathbb{R}^{1\times d_{\text{model}}}$ let $\nu = \xi + \mathrm{i}\xi'$, $\bar\nu = \xi - \mathrm{i}\xi'$ and

$$n(t) \;=\; \frac{\sqrt{d_{\text{model}}}}{2}\Big(t\,\nu + \frac{1}{t}\,\bar\nu\Big), \qquad t \in \mathbb{C}\setminus\{0\}.$$

As $\langle\nu,\nu\rangle = \langle\bar\nu,\bar\nu\rangle = 0$ and $\langle\nu,\bar\nu\rangle = 2$, we have $\langle n(t), n(t)\rangle = d_{\text{model}}$ for all t. For $|t| = 1$ the row $n(t)$ is real and runs through the great circle of ξ and ξ' in $\mathbb{S}$, with $n(1) = \sqrt{d_{\text{model}}}\,\xi$. For real $t \ge 1$,

$$n(t) \;=\; \sqrt{d_{\text{model}}}\Big(\frac{t+t^{-1}}{2}\,\xi + \mathrm{i}\,\frac{t-t^{-1}}{2}\,\xi'\Big). \tag{41}$$

The identity on the path. With the notation of (20) and (21), let $\varkappa = \varkappa_{k+1}$. The same formulas define $\dot z(n) \in \mathbb{C}^{1\times q}$ for $n \in \mathbb{C}^{1\times d_{\text{model}}}$, and $\mathcal{L}(\tilde w, \tilde w'; n)$ for rows $\tilde w, \tilde w' \in \mathbb{C}^{1\times q}$ with $\tilde w V \in \mathbb{C}_\varkappa$. $\mathcal{L}$ is holomorphic in $\tilde w$, affine in $\tilde w'$ and polynomial in n.

Lemma D.14 (Continuation)**.** *In the case (0), let (22) hold, and let ξ, ξ' be orthonormal. Suppose that for every real $t \ge 1$ the function σ is holomorphic at every $n(t)\,c_j$ and $\sigma(n(t)W)\,V \in \mathbb{C}_\varkappa$. Then*

$$\mathcal{L}\big(\sigma(n(t)W),\ \sigma'(n(t)W);\ n(t)\big) \;=\; 0 \qquad \textit{for all real } t \ge 1.$$

Proof. The left side is a holomorphic function of t on an open set that contains $[1,\infty)$, hence on a connected open neighborhood of $[1,\infty)$. This neighborhood contains an arc of the unit circle around $t = 1$, on which $n(t) \in \mathbb{S}$ and the function vanishes by (22). So it vanishes on the whole neighborhood. □

The step for $\tanh$ **on the path.** Let $\sigma = \tanh$. Consider, for a block ℓ with $W = W^\ell_{\text{read}}$, its columns c_j, $V = W^\ell_{\text{write}}$ and $\varkappa_\ell$ of Lemma D.13:

(S1) any two rows of V are linearly independent;
(S2) the columns of W are linearly independent over $\mathbb{Q}$;
(S3) there are a unit row ξ_0 and a number $\rho \in (0,1]$ such that every $\xi_0 c_j \ne 0$ and, with s_j the sign of $\xi_0 c_j$ and $\bar s = (s_1,\dots,s_q)$: the rows $\tilde w V$ with $|\tilde w_j - s_j| \le \rho$ for all j lie in $\mathbb{C}_{\varkappa_\ell}$, and $e^{-2\sqrt{d_{\text{model}}}\,|\xi_0 c_j|} < \rho/3$ for every j.

(S3) asks for large reads: on the row $\sqrt{d_{\text{model}}}\,\xi_0$ every hidden unit of the block is close to saturation. It then stays so on the whole path (41), and the outputs of the block stay in a small set, on which the rest of the network is holomorphic.

Lemma D.15 (Separation on the path)**.** *Let $u_1,\dots,u_q > 0$ be linearly independent over $\mathbb{Q}$, let $u'_1,\dots,u'_q$ be real, and for $t \ge 1$ let*

$$\zeta_j(t) \;=\; \exp\Big(-\big(t+t^{-1}\big)\,u_j - \mathrm{i}\,\big(t-t^{-1}\big)\,u'_j\Big).$$

Let $\rho_0 > \max_j e^{-2u_j}$, and let $\mathcal{T}_{-K},\dots,\mathcal{T}_K$ be holomorphic maps from the polydisc $\{\zeta \in \mathbb{C}^q : |\zeta_j| < \rho_0\}$ to $\mathbb{C}^{1\times d_{\text{out}}}$. If

$$\sum_{i=-K}^{K} t^i\,\mathcal{T}_i\big(\zeta_1(t),\dots,\zeta_q(t)\big) \;=\; 0 \qquad \textit{for all } t \ge 1,$$

then $\mathcal{T}_i = 0$ for every i.

Proof. This is the proof of Lemma 6.3, with a phase and with negative powers. On the polydisc $\mathcal{T}_i(\zeta) = \sum_\alpha b_{i,\alpha}\,\zeta^\alpha$, over the multi-indices α of nonnegative integers, and the series converges absolutely at $|\zeta_j| = e^{-2u_j}$. Let $\mu_\alpha = \sum_j \alpha_j u_j$ and $p_\alpha(t) = \sum_i t^i\, b_{i,\alpha}$. Then $|\zeta(t)^\alpha| = e^{-(t+t^{-1})\mu_\alpha}$, and the hypothesis reads $\sum_\alpha p_\alpha(t)\,\zeta(t)^\alpha = 0$. The numbers μ_α are pairwise different, and finitely many lie below any bound. Suppose that some p_α is not zero, let α^* be the one among these with the smallest μ_α, say μ^*, and let $\gamma > 0$ be the distance from μ^* to the next larger μ_α. For $t \geq 1$, as $t + t^{-1} \geq 2$,

$$|p_{\alpha^*}(t)| \;\leq\; \sum_{\mu_\alpha > \mu^*} |p_\alpha(t)|\, e^{-(t+t^{-1})(\mu_\alpha - \mu^*)} \;\leq\; t^K\, e^{-(t+t^{-1}-2)\gamma}\, e^{2\mu^*} \sum_i \sum_\alpha |b_{i,\alpha}|\, e^{-2\mu_\alpha},$$

which tends to 0 faster than every power of $1/t$. But $p_{\alpha^*}(t) = \sum_{i=-K}^{K} t^i\, b_{i,\alpha^*}$ is not zero, so $|p_{\alpha^*}(t)|$ is at least a positive constant times t^{-K} for large t, a contradiction. □

Lemma D.16 (A change of the reads moves the output)**.** *In the case (0), let θ be regular with (S1) for block $k+1$, let $V = W_{\text{write}}^{k+1}$ have the rows $a_1, \ldots, a_q$, and let $b \in \mathbb{C}^{1\times q}$. If*

$$\mathrm{D}\mathcal{B}_{>k+1}(\tilde{w}V)\,\Big(\big((1-\tilde{w}^2)\odot b\big)\,V\Big) \;=\; 0$$

for all $\tilde{w}$ in a nonempty open subset of $\mathbb{R}^{1\times q}$, where $1-\tilde{w}^2$ is the row with the entries $1-\tilde{w}_j^2$, then $b = 0$.

Proof. At real $\tilde{w}$ the derivative is a real linear map, so the real and the imaginary part of b satisfy the hypothesis: let b be real. By Lemma D.1, for $\tilde{w}$ in an open dense subset of the open set, the row $\sum_j b_j(1-\tilde{w}_j^2)\,a_j$ is a multiple of the row $\sum_l \tilde{w}_l\,a_l$. So all 2×2 minors of these two rows vanish on an open set, hence as polynomials in $\tilde{w}$. Their part of degree 1 in $\tilde{w}$ shows that bV is a multiple of every a_l, so $bV = 0$ by (S1). The coefficient of $\tilde{w}_j^2\,\tilde{w}_l$ with $l \neq j$ shows that $b_j a_j$ is a multiple of a_l, so $b_j = 0$ by (S1). □

Lemma D.17 (The step for $\tanh$ on the path)**.** *Let $\sigma = \tanh$, in the case (0), let θ be regular, and let block $k+1$ satisfy (S1) to (S3). Then (D_{k+1}) implies (D_k).*

Proof. Let Z and η satisfy (16) for k, so that (22) holds, with the notation above. By (S3) there is an open set $\mathcal{P}$ of unit rows around ξ_0 such that ξc_j has the sign s_j and $e^{-2\sqrt{d_{\text{model}}}\,|\xi c_j|} < \rho/3$ for every $\xi \in \mathcal{P}$ and every j. By (S2), for integers κ_j that are not all zero, $\sum_j \kappa_j\,\xi c_j = \xi \sum_j \kappa_j c_j$ vanishes only for ξ in a hyperplane. Let $\mathcal{P}' \subset \mathcal{P}$ be the dense set of the ξ for which $\xi c_1, \ldots, \xi c_q$ are linearly independent over $\mathbb{Q}$.

The path. Fix $\xi \in \mathcal{P}'$ and a unit row ξ' orthogonal to ξ, and let $n(t)$ be the path (41). Let

$$\zeta_j(t) \;=\; e^{-2s_j\,n(t)\,c_j} \;=\; \exp\Big(-\big(t+t^{-1}\big)\sqrt{d_{\text{model}}}\,|\xi c_j| - \mathrm{i}\,\big(t-t^{-1}\big)\sqrt{d_{\text{model}}}\,s_j\,\xi' c_j\Big),$$

so that $|\zeta_j(t)| < \rho/3$ for $t \geq 1$, and

$$\tanh\big(n(t)\,c_j\big) = s_j\,\frac{1-\zeta_j(t)}{1+\zeta_j(t)}, \qquad \tanh'\big(n(t)\,c_j\big) = 1 - \tanh^2\big(n(t)\,c_j\big).$$

For ζ in the polydisc $\mathbb{D} = \{|\zeta_j| < \rho/3\}$ let $\tilde{w}(\zeta)$ be the row with the entries $s_j\frac{1-\zeta_j}{1+\zeta_j}$. As $|\tilde{w}_j(\zeta) - s_j| = \frac{2|\zeta_j|}{|1+\zeta_j|} \leq 3|\zeta_j| < \rho$, (S3) gives $\tilde{w}(\zeta)\,V \in \mathbb{C}_{\mathscr{H}}$. The function $\tanh$ is holomorphic at $n(t)\,c_j$, as $\zeta_j(t) \neq -1$. So Lemma D.14 applies:

$$\mathcal{L}\big(\tilde{w}(\zeta(t)),\; 1-\tilde{w}(\zeta(t))^2;\; n(t)\big) \;=\; 0 \qquad \text{for all } t \geq 1. \tag{42}$$

The powers of t. By (20), $\dot{z}(n(t)) = t^3\dot{z}^{(3)} + t\,\dot{z}^{(1)} + t^{-1}\dot{z}^{(-1)} + t^{-3}\dot{z}^{(-3)}$ with rows $\dot{z}^{(i)} \in \mathbb{C}^{1\times q}$ that do not depend on t, and

$$\begin{aligned}
\dot{z}^{(3)} &= -\frac{\sqrt{d_{\text{model}}}}{8}\,\langle \nu Z, \nu\rangle\,\nu W,\\
\dot{z}^{(1)} &= \frac{\sqrt{d_{\text{model}}}}{2}\,\Big(\nu\,(ZW + \eta_W) - \frac{1}{4}\Big(\big(\langle \nu Z, \bar{\nu}\rangle + \langle \bar{\nu} Z, \nu\rangle\big)\,\nu + \langle \nu Z, \nu\rangle\,\bar{\nu}\Big) W\Big).
\end{aligned} \tag{43}$$

$\mathcal{L}$ is affine in $\dot z$, so (42) reads $\sum_i t^i\,\mathcal{T}_i(\zeta(t)) = 0$ with the maps

$$\mathcal{T}_i(\zeta) = \mathrm{D}\mathcal{B}\big(\tilde w(\zeta)V\big)\,\Big(\big((1-\tilde w(\zeta)^2)\odot \dot z^{(i)}\big)\,V\Big) \qquad (i=\pm1,\pm3),$$
$$\mathcal{T}_0(\zeta) = \mathrm{D}\mathcal{B}\big(\tilde w(\zeta)V\big)\,\big(\tilde w(\zeta)\,\eta_V\big) + \mathrm{D}_\theta\mathcal{B}\big(\tilde w(\zeta)V\big)\,\eta',$$

which are holomorphic on $\mathbb{D}$. By Lemma D.15, with $u_j = \sqrt{d_{\mathrm{model}}}\,|\xi c_j|$ and $\rho_0 = \rho/3$, every $\mathcal{T}_i$ vanishes on $\mathbb{D}$. For real $\zeta \in \mathbb{D}$ the rows $\tilde w(\zeta)$ fill a neighborhood of $\bar s$ in $\mathbb{R}^{1\times q}$.

(a) $\mathcal{T}_3 = 0$ gives Z. Here $\dot z^{(3)}$ is the number $-\frac{\sqrt{d_{\mathrm{model}}}}{8}\langle \nu Z, \nu\rangle$ times the row νW, which is not zero. By Lemma D.16, $\langle \nu Z, \nu\rangle = 0$. Its imaginary part is $\xi\,(Z+Z^\top)\,\xi'^\top = 0$. This holds for every unit row ξ' orthogonal to ξ, so $\xi\,(Z+Z^\top)$ is a multiple of ξ, for every $\xi \in \mathcal{P}'$ and then, by continuity, for every ξ in the open set $\mathcal{P}$. A matrix for which every row of an open set is an eigenvector is a multiple of the identity. So $Z + Z^\top = 2\beta_k I$, and $Z = Y_k + \beta_k I$ with Y_k antisymmetric.

(b) $\mathcal{T}_1 = 0$ gives η_W. Now $\langle \nu Z, \nu\rangle = 0$ and $\langle \nu Z, \bar\nu\rangle + \langle \bar\nu Z, \nu\rangle = 2\beta_k\langle \nu, \bar\nu\rangle = 4\beta_k$, so by (43) $\dot z^{(1)} = \frac{\sqrt{d_{\mathrm{model}}}}{2}\,\nu\,(Y_k W + \eta_W)$. By Lemma D.16, $\dot z^{(1)} = 0$, so $\xi\,(Y_k W + \eta_W) = 0$ for every $\xi \in \mathcal{P}'$, and $\eta_W = -Y_k W$.

(c) $\mathcal{T}_0 = 0$ gives the rest. $\mathrm{D}\mathcal{B}(\tilde w V)\,(\tilde w\,\eta_V) + \mathrm{D}_\theta\mathcal{B}(\tilde w V)\,\eta' = 0$ for real $\tilde w$ near $\bar s$. The left side is real-analytic on the rows $\tilde w \in \mathbb{R}^{1\times q}$ with $\tilde w V \neq 0$. These form a connected set, the complement of a subspace of codimension $d_{\mathrm{model}} \geq 2$, so the identity holds on all of it. By Lemma D.2 with $A = V$, $\eta_V = VZ'$ and (16) holds for $k+1$ with Z' and η'. By (D_{k+1}), $Z' = Y_{k+1} + \beta_{k+1} I$ and η' is as in (18). With (a) and (b) this is (18) for k. □

The step for GELU on the path. Consider, for a block ℓ with $W = W^\ell_{\mathrm{read}}$, its columns c_j, $V = W^\ell_{\mathrm{write}}$, its rows a_j, and $\varkappa_\ell$ of Lemma D.13:

(P1) the columns of W are pairwise different, and the matrix WV is invertible;

(P2) there is a unit row ξ_0 such that, for every unit row ξ' orthogonal to ξ_0,

$$|\xi' c_j| < \xi_0 c_j \ \text{ for every } j, \qquad \|\xi' WV\| < \frac{\varkappa_\ell}{2}\,\|\xi_0 WV\|,$$
$$\sum_j e^{-d_{\mathrm{model}}(\xi_0 c_j)^2/2}\,\|a_j\| \ < \ \frac{\varkappa_\ell\sqrt{d_{\mathrm{model}}}}{8}\,\|\xi_0 WV\|.$$

(P2) asks for reads that are large and close to a matrix of rank one: every column of W is close to a positive multiple of $\xi_0^\top$. Then every hidden unit is close to linear on the whole path, the output of the block is close to $n(t)\,WV$, and its imaginary part stays small against its real part, so that the rest of the network is holomorphic along the path (Lemma D.13).

Lemma D.18 (The step for GELU on the path)**.** *Let σ be GELU, in the case (0), let θ be regular, and let block $k+1$ satisfy (P1) and (P2). Then (D_{k+1}) implies (D_k).*

Proof. Let Z and η satisfy (16) for k, so that (22) holds, with the notation above. Both sides of every inequality of (P2), the left sides as suprema over ξ', are continuous in ξ_0. So there is an open set $\mathcal{P}$ of unit rows around ξ_0 such that (P2) holds for every $\xi \in \mathcal{P}$ in the place of ξ_0.

The path. Fix $\xi \in \mathcal{P}$ and a unit row ξ' orthogonal to ξ, let $n(t)$ be the path (41), and

$$z_j(t) = n(t)\,c_j, \qquad r_j = (\xi c_j)^2 - (\xi' c_j)^2 > 0, \qquad \delta_j(t) = z_j - \sigma(z_j), \qquad \delta_j'(t) = 1 - \sigma'(z_j).$$

Let $z(t), \delta(t), \delta'(t)$ be the rows with these entries. For $t \geq 1$, by (41) and (P2),

$$\begin{aligned} &\mathrm{Re}\,z_j \geq \sqrt{d_{\mathrm{model}}}\,\xi c_j > 0, \qquad |\mathrm{Im}\,z_j| \leq \mathrm{Re}\,z_j, \\ &\mathrm{Re}\,\big(z_j^2\big) = d_{\mathrm{model}}\,(\xi c_j)^2 + \frac{d_{\mathrm{model}}}{4}\,\big(t - t^{-1}\big)^2\,r_j. \end{aligned} \tag{44}$$

By Lemma D.7, $\delta_j = \varphi(z_j)\,z_j\ \mathrm{mr}(z_j)$ and $\delta_j' = \varphi(z_j)\,\big(\mathrm{mr}(z_j) - z_j\big)$, with $|\mathrm{mr}(z_j)| \leq 1/\,\mathrm{Re}\,z_j$ and $|z_j| \leq \sqrt{2}\ \mathrm{Re}\,z_j$. So, with a constant κ,

$$|\delta_j(t)| \ \leq \ e^{-d_{\mathrm{model}}(\xi c_j)^2/2}\,e^{-\frac{d_{\mathrm{model}}}{8}(t-t^{-1})^2 r_j}, \qquad |\delta_j'(t)| \ \leq \ \kappa\,t\,e^{-\frac{d_{\mathrm{model}}}{8}(t-t^{-1})^2 r_j}. \tag{45}$$

The real and the imaginary part of $n(t)\,WV$ are $\sqrt{d_{\text{model}}}\,\frac{t+t^{-1}}{2}\,\xi WV$ and $\sqrt{d_{\text{model}}}\,\frac{t-t^{-1}}{2}\,\xi' WV$. By (P2), the second is shorter than $\frac{\varkappa}{2}$ times the first, and by (45) $\|\delta(t)\,V\| \le \sum_j |\delta_j|\,\|a_j\|$ is at most $\frac{\varkappa}{8}$ times the norm of the first. Let $\varrho = \varkappa\sqrt{d_{\text{model}}}\,\|\xi WV\| \big/ \big(16\sum_j \|a_j\|\big)$. For a row $\delta \in \mathbb{C}^{1\times q}$,

$$|\delta_j| \le \varrho\, t \ \text{ for all } j, \ \text{ or } \delta = \delta(t) \qquad \Longrightarrow \qquad \big(z(t)-\delta\big)\,V = n(t)\,WV - \delta V \ \in \mathbb{C}_{\varkappa}, \tag{46}$$

because then $\|\delta V\|$ is at most $\frac{\varkappa}{8}$ times the norm of the real part of $n(t)\,WV$, and $\frac{\varkappa/2+\varkappa/8}{1-\varkappa/8} < \varkappa$. As $\sigma(z) = z - \delta(t)$ and $\sigma'(z) = 1 - \delta'(t)$ on the path, Lemma D.14 applies:

$$\mathcal{L}\big(z(t)-\delta(t),\ 1-\delta'(t);\ n(t)\big) \ = \ 0 \qquad \text{for all } t \ge 1. \tag{47}$$

The expansion. For fixed t, by (46) the map $(\delta, \delta') \mapsto \mathcal{L}\big(z(t)-\delta, 1-\delta'; n(t)\big)$ is holomorphic in the rows δ with $|\delta_j| \le \varrho\, t$ and affine in the row δ'. We expand it in powers of δ and δ'. Its value at 0 and its first derivatives are

$$\mathcal{L}_0(t) = \mathcal{L}\big(z(t), 1; n(t)\big), \qquad \mathcal{L}_j(t) = \frac{\partial \mathcal{L}}{\partial \tilde{w}_j}\big(z(t), 1; n(t)\big), \qquad \mathcal{L}'_j(t) = \frac{\partial \mathcal{L}}{\partial \tilde{w}'_j}\big(z(t), 1; n(t)\big),$$

so that (47) is

$$\mathcal{L}_0(t) \ - \ \sum_j \Big[\mathcal{L}_j(t)\,\delta_j(t) + \mathcal{L}'_j(t)\,\delta'_j(t)\Big] \ + \ \big(\text{terms with two or more factors}\big) \ = \ 0, \tag{48}$$

where the factors are entries of $\delta(t)$ and $\delta'(t)$. We use two facts about the coefficients. First, every coefficient is meromorphic at infinity. Indeed, it is built from derivatives of $\mathcal{B}$ at $z(t)\,V = n(t)\,WV$ and from polynomials in $n(t)$, and

$$n(t)\,WV \ = \ \sqrt{d_{\text{model}}}\ \frac{t+t^{-1}}{2}\ \Big(\xi\,WV + \mathrm{i}\,\frac{1-t^{-2}}{1+t^{-2}}\ \xi' WV\Big),$$

where the row in the bracket is holomorphic in $1/t$ near 0, with values in $\mathbb{C}_{\varkappa/2}$, and a derivative of order i of $\mathcal{B}$ is homogeneous of degree $-i$. Second, on the rows $u + \mathrm{i}u' \in \mathbb{C}_{3\varkappa/4}$ with $\|u\| = 1$, $\mathrm{D}\mathcal{B}$ and $\mathrm{D}_\theta\mathcal{B}$ are bounded. So by Cauchy's estimates on the polydisc of (46), the coefficient of a product of i factors δ_j and of at most one factor δ'_j is at most a fixed power of t times $(\varrho\, t)^{-i}$. With (45), the series (48) therefore converges absolutely for large t, and the sum of any set of its terms is at most a power of t times $e^{-\frac{d_{\text{model}}}{8}(t-t^{-1})^2 r}$, where r is the smallest sum of the numbers r_j of the factors of a term of the set.

Step 1: the term without δ gives η'. By (48), $\mathcal{L}_0(t)$ tends to 0 faster than every power of $1/t$. It is meromorphic at infinity, so $\mathcal{L}_0(t) = 0$ for large t, and then for all $t \ge 1$, because it is holomorphic near $[1, \infty)$. At $t = 1$, where $n(t)$ is the row $n = \sqrt{d_{\text{model}}}\,\xi$, this reads, by (21), (20) and (15),

$$0 \ = \ \mathcal{L}\big(nW, 1; n\big) \ = \ \mathrm{D}\mathcal{B}(n\,WV)\,\Big(n\,\big(Z\,WV + \eta_W V + W\eta_V\big)\Big) + \mathrm{D}_\theta\mathcal{B}(n\,WV)\,\eta'.$$

The right side is real-analytic and homogeneous of degree 0 in the row $n \ne 0$, and it vanishes for ξ in the open set $\mathcal{P}$, hence for all $n \ne 0$. With $h = n\,WV$, which takes every value by (P1), this is (16) for $k+1$ with

$$Z' = (WV)^{-1}\,\big(Z\,WV + \eta_W V + W\eta_V\big) \tag{49}$$

and η'. By (D_{k+1}), $Z' = Y_{k+1} + \beta' I$ and η' is as in (18), for antisymmetric $Y_{k+1}, \dots, Y_L$ and numbers $\beta', \beta_{k+2}, \dots, \beta_L$.

Step 2: the slowest hidden unit gives Z. On an open dense set of pairs (ξ, ξ') the numbers $r_1, \dots, r_q$ are pairwise different: $r_i = r_j$ on an open set of pairs would make $c_i c_i^\top - c_j c_j^\top$ a multiple of the identity, which is impossible for $c_i \ne \pm c_j$, and $c_i \ne c_j$ by (P1), $c_i \ne -c_j$ by (P2). Take a hidden unit j and a nonempty open set of pairs, with $\xi \in \mathcal{P}$, on which $r_j < r_i$ for all $i \ne j$, and let $\gamma = \min\big(r_j, \min_{i\ne j}(r_i - r_j)\big)$. By Step 1, all terms of (48) other than the two of unit j are, together,

at most a power of t times $e^{-\frac{d_{\text{model}}}{8}(t-t^{-1})^2(r_j+\gamma)}$. Dividing by $\varphi(z_j)$, whose absolute value is a constant times $e^{-\frac{d_{\text{model}}}{8}(t-t^{-1})^2 r_j}$ by (44),

$$\big(z_j\,\mathcal{L}_j + \mathcal{L}_j'\big)\ \mathrm{mr}(z_j)\ -\ z_j\,\mathcal{L}_j'\ \longrightarrow\ 0 \qquad \text{faster than every power of } 1/t.$$

Here $z_j(t) = \frac{\sqrt{d_{\text{model}}}}{2}\big(t\,\nu c_j + \frac{1}{t}\,\bar{\nu} c_j\big)$ with $\mathrm{Re}(\nu c_j) = \xi c_j > 0$, and $z_j\mathcal{L}_j + \mathcal{L}_j'$ and $z_j\mathcal{L}_j'$ are meromorphic at infinity. By Lemma D.8 both vanish, so

$$\mathcal{L}_j'(t) = 0 \quad \text{and} \quad \mathcal{L}_j(t) = 0 \qquad \text{for large } t, \text{ hence for all } t \geq 1. \tag{50}$$

By (21), with $\dot{z}_j$ the entry j of $\dot{z}$,

$$\mathcal{L}_j'(t)\ =\ \dot{z}_j\big(n(t)\big)\ \mathrm{D}\mathcal{B}\big(n(t)\,WV\big)\,a_j.$$

At $t = 1$ this is $\dot{z}_j(n)\,\mathrm{D}\mathcal{B}(n\,WV)\,a_j = 0$ at $n = \sqrt{d_{\text{model}}}\,\xi$, for ξ in an open set. By Lemma D.1, $\mathrm{D}\mathcal{B}(n\,WV)\,a_j \neq 0$ on an open dense set of n. So $\dot{z}_j(n) = 0$ on an open subset of $\mathbb{S}$, that is, by (20),

$$\|n\|^2\ n\,\big(Zc_j + \eta_{c_j}\big)\ =\ \langle nZ, n\rangle\ nc_j$$

on an open cone of rows n, hence for all n, where η_{c_j} is column j of η_W. For n orthogonal to c_j this gives $n\,(Zc_j + \eta_{c_j}) = 0$, so $Zc_j + \eta_{c_j} = \beta_k c_j$ for a number β_k, and then $\langle nZ, n\rangle = \beta_k\|n\|^2$ for all n. So $Z = Y_k + \beta_k I$ with Y_k antisymmetric.

Step 3: subtracting a symmetry. Let $\beta_{k+1} = \beta' - \beta_k$. The pair $Y_k + \beta_k I$ and $\big(-Y_k W,\ V(Y_{k+1} + \beta_{k+1}I),\ \eta'\big)$ is of the form (18), so it satisfies (16) for k. Subtracting it from Z and η leaves the pair 0 and $(\eta_W^\circ, \eta_V^\circ, 0)$ with

$$\eta_W^\circ = \eta_W + Y_k W, \qquad \eta_V^\circ = \eta_V - V\,(Y_{k+1} + \beta_{k+1}I),$$

which satisfies (16) for k as well, and by (49) and Step 1

$$\eta_W^\circ V + W\eta_V^\circ\ =\ WV\,(Y_{k+1} + \beta' I) - \beta_k\,WV - WV\,(Y_{k+1} + \beta_{k+1}I)\ =\ 0. \tag{51}$$

For this pair, (21) is $\mathcal{L}^\circ(\tilde{w}, \tilde{w}'; n) = \mathrm{D}\mathcal{B}(\tilde{w}V)\,\big((\tilde{w}' \odot n\eta_W^\circ)\,V + \tilde{w}\,\eta_V^\circ\big)$. It remains to show $\eta_W^\circ = 0$ and $\eta_V^\circ = 0$.

Step 4: the hidden units in the order of their rates. Take a nonempty open set of pairs (ξ, ξ'), with $\xi \in \mathcal{P}$, on which $r_{j_1} < r_{j_2} < \cdots < r_{j_q}$ for one order $j_1, \ldots, j_q$ of the hidden units. We show by induction on i that column j_i of η_W° and row j_i of η_V° vanish. Let this hold for the units $j_1, \ldots, j_{i-1}$, and let $j = j_i$. If the rows δ and δ' vanish outside the entries $j_1, \ldots, j_{i-1}$, then $(1 - \delta') \odot n\eta_W^\circ = n\eta_W^\circ$ and $(z - \delta)\,\eta_V^\circ = nW\eta_V^\circ$, so by (51)

$$\mathcal{L}^\circ\big(z - \delta,\ 1 - \delta';\ n\big)\ =\ \mathrm{D}\mathcal{B}\big((z-\delta)V\big)\,\Big(n\,\big(\eta_W^\circ V + W\eta_V^\circ\big)\Big)\ =\ 0.$$

So in the expansion (48) of $\mathcal{L}^\circ$ every term has a factor $\delta_l(t)$ or $\delta_l'(t)$ of a unit l other than $j_1, \ldots, j_{i-1}$. Other than the two terms of unit j, such a term has a factor of a unit $j_{i'}$ with $i' > i$, or a factor of unit j and a second factor. With $\gamma = \min(r_{j_1}, r_{j_{i+1}} - r_j)$, or $\gamma = r_{j_1}$ if $i = q$, we get (50) as in Step 2: $\mathcal{L}_j^{\circ\prime}(t) = 0$ and $\mathcal{L}_j^\circ(t) = 0$ for $t \geq 1$. Here, by (51),

$$\mathcal{L}_j^{\circ\prime}(t) = \big(n\,\eta_{c_j}^\circ\big)\ \mathrm{D}\mathcal{B}(n\,WV)\,a_j, \qquad \mathcal{L}_j^\circ(t) = \mathrm{D}\mathcal{B}(n\,WV)\,\eta_{a_j}^\circ \qquad \text{at } n = n(t),$$

with $\eta_{c_j}^\circ$ column j of η_W° and $\eta_{a_j}^\circ$ row j of η_V°. At $t = 1$, for ξ in an open set and $n = \sqrt{d_{\text{model}}}\,\xi$: by Lemma D.1, $n\,\eta_{c_j}^\circ = 0$, so $\eta_{c_j}^\circ = 0$, and $\eta_{a_j}^\circ$ is a multiple of $n\,WV$ for an open set of rows n, so $\eta_{a_j}^\circ = 0$.

After $i = q$: $\eta_W = -Y_k W$ and $\eta_V = V\,(Y_{k+1} + \beta_{k+1}I)$. With Step 1 for η' and Step 2 for Z this is (18) for k. □

Parameters that satisfy all conditions, in the case (0). We choose the blocks from the last to the first. Let D and the blocks $\ell+1,\dots,L$ be chosen, regular and with (S1) to (S3) for $\tanh$, or (P1) and (P2) for GELU. This fixes $\varkappa_\ell$ (Lemma D.13). We choose $(W,V)=(W^\ell_{\text{read}},W^\ell_{\text{write}})$.

Conditions that fail on a set of measure zero of pairs (W,V). (R1), (S1), (S2) and (P1) are of this kind; (S2) fails on countably many proper subspaces. (R2): the triples (n,W,V) with $n\in\mathbb{S}$, $\sigma(nW)\neq 0$ and $\sigma(nW)\,V=0$ form a manifold of dimension one less than that of the pairs (W,V), so their projection to the pairs has measure zero; and $\sigma(nW)=0$ only if $nW=0$, which requires W of rank less than d_{model}. (R3): the determinant in the proof of Lemma D.1 is real-analytic in (n,W,V), so (R3) fails on a set of measure zero once it holds for one pair. It holds for $W=[\,I\mid 0\,]$ and V with the rows of I and zero rows, at any $n\in\mathbb{S}$ with positive entries n_i: there $g_\ell(n)=\sigma(n)$, and $\mathrm{D}g_\ell(n)\,v=\mu\,g_\ell(n)$ gives $v_i=\mu\,\sigma(n_i)/\sigma'(n_i)$, so $0=\langle v,n\rangle=\mu\sum_i n_i\,\sigma(n_i)/\sigma'(n_i)$ with a positive sum, hence $\mu=0$ and $v=0$.

Conditions that hold on a nonempty open set of pairs. (S3): take any V and W°, with the columns c°_j, and a unit row ξ_0 with $\xi_0 c^\circ_j\neq 0$ and $\bar{s}V\neq 0$ for the row $\bar{s}$ of the signs. As $\bar{s}V$ is a real nonzero row, there is a ρ as in (S3). Then $W=\lambda W^\circ$ satisfies (S3) for large $\lambda>0$, with strict inequalities and a compact set of rows $\tilde{w}V$ inside the open cone, so (S3) holds near (W,V) as well. (P2): take a row α with positive entries, any V with $\alpha V\neq 0$, and $W^\circ=\xi_0^\top\alpha$. For W° the left sides of the first two inequalities of (P2) vanish and $\xi_0 c^\circ_j=\alpha_j>0$, so these two inequalities hold for all $W=\lambda\,(W^\circ+W')$ with $\lambda>0$ and W' small. For fixed small W', the third holds for large λ: its left side decreases like $e^{-\kappa\lambda^2}$ with a constant $\kappa>0$, and its right side grows like λ. All three are strict, so (P2) holds on a nonempty open set of pairs.

A nonempty open set minus a set of measure zero is not empty. So there are parameters θ^* that are regular and whose blocks all satisfy (S1) to (S3) for $\tanh$, or (P1) and (P2) for GELU.

Lemma D.19 (The domain)**.** *In the case (0), let X be a finite batch of nonzero inputs, and let $\mathcal{O}_X\subset\mathbb{R}^P$ be the set of parameters at which the hidden states $h_0,\dots,h_L$ of the network are nonzero for every $x\in X$. Then $\mathcal{O}_X$ is open and connected, its complement has measure zero, and the outputs F_X on the batch are real-analytic on $\mathcal{O}_X$.*

Proof. Let $\mathcal{O}^0$ be the set of parameters with $xE\neq 0$ for all $x\in X$, and for $\ell\geq 1$ let $\mathcal{O}^\ell$ be the set of parameters in $\mathcal{O}^{\ell-1}$ with $h_\ell\neq 0$ for all $x\in X$. The complement of $\mathcal{O}^0$ is a finite union of subspaces of codimension $d_{\text{model}}\geq 2$. Let $\ell\geq 1$ and $x\in X$. On $\mathcal{O}^{\ell-1}$, $h_\ell=\tilde{w}_\ell\,W^\ell_{\text{write}}$ with $\tilde{w}_\ell=\sigma\big(N_\varepsilon(h_{\ell-1})\,W^\ell_{\text{read}}\big)$, and $h_{\ell-1}$ does not depend on the weights of block ℓ. For $\tanh$ and GELU, $\sigma(z)=0$ only at $z=0$. So $\tilde{w}_\ell=0$ exactly where $N_\varepsilon(h_{\ell-1})\,W^\ell_{\text{read}}=0$, a closed submanifold of $\mathcal{O}^{\ell-1}$ of codimension at least $d_{\text{model}}\geq 2$, because the derivative with respect to W^ℓ_{read} is onto. Where $\tilde{w}_\ell\neq 0$, the derivative of h_ℓ with respect to W^ℓ_{write} is onto $\mathbb{R}^{1\times d_{\text{model}}}$, so there the parameters with $h_\ell=0$ form a closed submanifold of codimension $d_{\text{model}}\geq 2$. Removing finitely many closed submanifolds of codimension at least two, one after the other, from a connected open set leaves a connected open set and removes a set of measure zero. □

D.6 The parts together

Proof of Theorem 6.1. $\tanh$ *blocks in the cases (n) and (+).* Let θ satisfy (R1), let every block satisfy (T1) and (T2), and let every $W^\ell_{\text{read}}W^\ell_{\text{write}}$ be invertible. These conditions fail only on a set of measure zero; (T2) fails on countably many proper linear subspaces. Such θ are regular: $\mathrm{D}g_\ell(0)$ is the map $v\mapsto v\,W^\ell_{\text{read}}W^\ell_{\text{write}}$. By Lemma D.4 and Lemma D.6, Claim (D_k) holds at θ for $k=L$, then for $k=L-1$, and so on down to $k=0$. By Proposition D.3, the Jacobian at θ has the rank (13) on some finite batch, and by (R1) it has at most this rank on every batch. So the functional dimension is (13) at almost every θ.

GELU blocks in the case (+), and gated blocks in the cases (n) and (+). By Lemma D.12, $\theta^\dagger$ is good, so it is regular, and no value column vanishes. (D_L) holds at $\theta^\dagger$ (Lemma D.4), and by Lemma D.11, applied for $k=L-1$, then for $k=L-2$, and so on down to $k=0$, so does (D_0). By Proposition D.3, the Jacobian at $\theta^\dagger$ has the rank (13) on some finite batch X. In the cases (n) and (+) the outputs on X are real-analytic in $\theta\in\mathbb{R}^P$, and so are the minors of the Jacobian. A minor of the size (13) does not vanish at $\theta^\dagger$, so it vanishes only on a set of measure zero. At generic parameters

the Jacobian on X therefore has at least the rank (13), and it has at most this rank on every batch, because E and every W^{ℓ}_{write} have rank d_{model} and, in gated blocks, no value column vanishes. So the functional dimension is (13).

The case (0). By Lemma D.4 and Lemma D.17 or Lemma D.18, Claim (D_k) holds at θ^* for $k = L$, then for $k = L-1$, and so on down to $k = 0$. By Proposition D.3 there is a finite batch X^*, with $xE \neq 0$ for all its inputs, on which the Jacobian at θ^* has the rank (13). By (R2), $\theta^* \in \mathcal{O}_{X^*}$. The sum of the squares of the minors of this size of the Jacobian of F_{X^*} is real-analytic on the connected open set $\mathcal{O}_{X^*}$ and does not vanish at θ^*, so its zero set has measure zero, and so has the complement of $\mathcal{O}_{X^*}$ (Lemma D.19). Hence the rank on X^* is at least (13) at almost every parameter. Where E and every W^{ℓ}_{write} have rank d_{model}, it is at most (13) on every batch. So at generic parameters the functional dimension is (13). □

Remark D.20 ($\tanh$ blocks without normalization, from the theorem of Fefferman). In the case (n), Theorem 6.1(i) also follows from Theorem I.B.9 of Fefferman (1994). Without normalization, a network of $\tanh$ blocks computes

$$x^0 = x, \qquad x^{\ell} = \tanh\left(x^{\ell-1}A_{\ell}\right) \ (1 \leq \ell \leq L), \qquad y = x^L A_{L+1},$$

with $A_1 = EW^1_{\text{read}}$, $A_{\ell} = W^{\ell-1}_{\text{write}}W^{\ell}_{\text{read}}$ for $2 \leq \ell \leq L$, and $A_{L+1} = W^L_{\text{write}}D$. This is a $\tanh$ network of the kind of that paper, with one weight matrix between two layers and with all thresholds equal to zero. That paper takes $\sigma(z) = \tanh(z/2)$ and applies σ to the output as well. As σ is injective, two of our networks have the same output exactly when the nets of that paper with the hidden layers $x^{\ell} = \sigma(2x^{\ell-1}A_{\ell})$ and the output layer $\sigma(x^L A_{L+1})$ do, and then so do, for every input coordinate and every output, the nets with one input and one output that keep only these.

Theorem I.B.9 (pages 517 and 518 there) is stated for nets with one input and one output. Its hypotheses are that every weight is nonzero, and that the ratio of the weights that one node sends to two nodes of the next layer, which has q nodes, is never, in absolute value, a fraction p/r with integers p and $1 \leq r \leq 100\,q^2$. It puts no condition on the thresholds: the condition on the thresholds in the introduction of that paper serves to select one net among isomorphic ones (Lemma I.A.18 there). Its conclusion is that two such nets with the same output agree up to permutations and sign flips of the nodes of each hidden layer.

At generic parameters of our network, the entries of the matrices A_{ℓ} satisfy these hypotheses, and so do those of all nearby parameters. So nearby parameters with the same function have matrices A_{ℓ} that agree up to permutations and sign flips of the hidden units, which are the identity for parameters that are close enough. A matrix of rank d_{model} has only one factorization into two factors of rank d_{model}, up to an invertible matrix between them. So the parameters with the same matrices $A_1, \ldots, A_{L+1}$ are those of Proposition 3.2, with invertible matrices $G_0, \ldots, G_L$. Hence the network is locally identifiable up to its symmetries at generic parameters, which is the equality condition (Proposition 4.6). The proof of Theorem 6.1 does not use this theorem. Under normalization, $W^{\ell-1}_{\text{write}}$ and W^{ℓ}_{read} no longer enter through their product, and that theorem does not apply.

E Inactive hidden units: where the count fails

For ReLU, ReLU^2 and ReGLU blocks, Theorem 5.1 gives the count for one block. At depth it fails on an open set of weights, because a hidden unit can be inactive at every input. We take $T = 1$, and write c_k, e_k and a_k for the read column, or gate column, the value column and the write row of hidden unit k of block 1.

Proposition E.1 (Inactive hidden units)**.** *Consider networks of $L \geq 2$ ReLU, ReLU^2 or ReGLU blocks without normalization, each with more than d_{model} hidden units. There is a nonempty open set of weights on which a hidden unit of block 2 is inactive at every input. At almost every point of this set the functional dimension is at most $P - (L+1)d^2_{\text{model}} - s - (2d_{\text{model}} - 1)$, and at most $P - (L+1)d^2_{\text{model}} - s - (3d_{\text{model}} - 2)$ for ReGLU blocks. So these networks do not satisfy the equality condition.*

Proof. Let a hidden unit of block 2 have the read column r, its gate column in a ReGLU block, and let $u = h_0$.

ReLU and ReLU^2. The hidden unit of block 2 reads the number $h_1 r = \sum_k \sigma(uc_k)\, a_k r$, with $\sigma \geq 0$. It is positive at no input when $a_k r < 0$ for all k, which holds on a nonempty open set of weights.

ReGLU. The gate of the hidden unit of block 2 reads $z(u) = \sum_k \mathrm{ReLU}(uc_k)\,(ue_k)\,a_k r$, and the sign of a term depends on the value ue_k. We choose weights at which every gate of block 1 opens where its value is positive. Let q be the number of hidden units of block 1 and $0 < \delta < 1/(q\, d_{\mathrm{model}})$. For every k choose an index $i(k) \leq d_{\mathrm{model}}$, with every index chosen by some k. Let c_k have the entry 1 at $i(k)$ and zeros elsewhere, let e_k have the entry 1 at $i(k)$ and $-\delta$ elsewhere, and let $a_k r = -1$ for all k. Then

$$z(u) = -\sum_k \mathrm{ReLU}(u_{i(k)})\Big(u_{i(k)} - \delta \sum_{j \neq i(k)} u_j\Big) \;\leq\; -(1 - \delta\, q\, d_{\mathrm{model}}) \sum_{i\,:\,u_i > 0} u_i^2,$$

so $z(u) < 0$ if u has a positive entry, and $z(u) = 0$ if it has none. Suppose that, at weights that converge to these, the gate of the hidden unit of block 2 is positive at unit rows u_n, and let u be a limit of the u_n. Then $z(u) \geq 0$, so u has no positive entry. For large n, a hidden unit k with $u_{i(k)} < 0$ has a closed gate at u_n. A hidden unit k with $u_{i(k)} = 0$ has a positive value at u_n, since its value at u is $-\delta \sum_{j \neq i(k)} u_j > 0$, so its term is not positive. Hence the gate is not positive at u_n, a contradiction. The function z is positively homogeneous in u, so at all weights close to the chosen ones the gate is positive at no input.

The rank. On this open set of weights, take a batch of inputs at which no hidden unit sits at its kink. The columns of the Jacobian that belong to the weights of the inactive hidden unit vanish: $2d_{\mathrm{model}}$ columns, or $3d_{\mathrm{model}}$ for ReGLU. The other columns form the Jacobian of the network without this hidden unit, which computes the same function. That network has $P - 2d_{\mathrm{model}}$ parameters, or $P - 3d_{\mathrm{model}}$, its blocks have at least d_{model} hidden units, and its symmetries have the dimension $(L+1)d_{\mathrm{model}}^2 + s - 1$, or $(L+1)d_{\mathrm{model}}^2 + s - 2$ for ReGLU, since the inactive hidden unit had one rescaling, or two. By Lemma B.1 they move generic parameters in as many independent directions, which lie in the kernel of the Jacobian. So at almost every point of the open set the rank on every such batch is at most

$$(P - 2d_{\mathrm{model}}) - \big((L+1)d_{\mathrm{model}}^2 + s - 1\big) \;=\; P - (L+1)d_{\mathrm{model}}^2 - s - (2d_{\mathrm{model}} - 1),$$

and at most $(P - 3d_{\mathrm{model}}) - \big((L+1)d_{\mathrm{model}}^2 + s - 2\big)$ for ReGLU. □

F Numerical ranks

This appendix gives the ranks to which Section 7 refers. Each network has an embedding, L blocks of one kind and a readout, without biases, with $d_{\mathrm{in}} = d_{\mathrm{out}} = d_{\mathrm{model}} + 2$. The weights have independent entries $\mathcal{N}(0, 1/n)$, where n is the number of rows of the matrix, and the inputs have independent $\mathcal{N}(0,1)$ entries. The Jacobian of the map from all P parameters to the outputs on a batch of more than $3P/d_{\mathrm{out}}$ inputs is formed by automatic differentiation, in double precision. The rank is read at the largest gap of the spectrum, the largest ratio of two consecutive singular values. It is a lower bound on the functional dimension, and where it equals the bound of Theorem 4.3, it is the functional dimension.

For each kind of MLP block and each of the three cases of normalization, we computed the rank of 8 networks: $(d_{\mathrm{model}}, L, d_{\mathrm{hidden}}) = (3,2,3)$, $(3,3,6)$, $(4,2,4)$ and $(4,3,8)$, with two draws of weights and inputs each. For attention blocks, on sequences of $T = 3$ tokens with $d_{\mathrm{model}} = 4$ and without normalization, we computed it for $L = 1$ with $H = 1$ and $H = 2$ heads, for $L = 2$ with $H = 2$ and $H = 4$, and for $L = 3$ with $H = 2$, again with two draws each. In all 106 networks the rank is the bound $P - (L+1)\dim \mathcal{G} - s$ (Table 2).

Blocks	Normalization	At the bound	Smallest gap	The count is
tanh	none	8 of 8	7×10^8	proved, Theorem 6.1(i)
tanh	RMSNorm, $\varepsilon = 0.1$	8 of 8	5×10^7	proved, Theorem 6.1(i)
tanh	RMSNorm, $\varepsilon = 0$	8 of 8	3×10^7	proved, Theorem 6.1(i)
GELU	none	8 of 8	4×10^9	Conjecture 1
GELU	RMSNorm, $\varepsilon = 0.1$	8 of 8	3×10^8	proved, Theorem 6.1(ii)
GELU	RMSNorm, $\varepsilon = 0$	8 of 8	7×10^7	proved, Theorem 6.1(ii)
SwiGLU	none	8 of 8	4×10^6	proved, Theorem 6.1(iii)
SwiGLU	RMSNorm, $\varepsilon = 0.1$	8 of 8	2×10^9	proved, Theorem 6.1(iii)
SwiGLU	RMSNorm, $\varepsilon = 0$	8 of 8	3×10^9	open
GeGLU	none	8 of 8	3×10^6	proved, Theorem 6.1(iii)
GeGLU	RMSNorm, $\varepsilon = 0.1$	8 of 8	9×10^8	proved, Theorem 6.1(iii)
GeGLU	RMSNorm, $\varepsilon = 0$	8 of 8	6×10^8	open
attention, $L = 1$	none	4 of 4	6×10^{12}	proved, Theorem 5.1
attention, $L = 2$	none	4 of 4	2×10^{11}	open
attention, $L = 3$	none	2 of 2	8×10^5	open

Table 2: Networks whose numerical rank is the bound $P-(L+1)\dim \mathcal{G}-s$, by kind of block and normalization, with the smallest gap in the spectrum at the rank, and what this paper shows for the count.

G With skip connections: where the reduction stops

The networks of this paper have no skip connections. Most networks in use have them: a block adds its input to its output. This appendix shows where the reduction of Section 3 stops in such a network, and what follows for two statements of Wang and Wang (2025) and for a conjecture of Henry (2026). Networks have no normalization here, unless it is named.

A block with its skip connection, in the form of a block. A block with a learned linear skip, $h \mapsto hA + f(hW_{\text{read}})\,W_{\text{write}}$ with a trainable $d_{\text{model}} \times d_{\text{model}}$ matrix A, still has the form (1), with d_{model} more hidden units that apply the identity (Lin and Jegelka, 2018, Section 4). Write $A = A^{\text{read}}A^{\text{write}}$, and let $\tilde{f}\big([\,u \mid z\,]\big) = \big[\,u \mid f(z)\,\big]$, the identity on the first d_{model} columns and f on the others. Then

$$hA + f(hW_{\text{read}})\,W_{\text{write}} \;=\; \tilde{f}\Big(h\,\big[\,A^{\text{read}} \mid W_{\text{read}}\,\big]\Big)\begin{bmatrix}A^{\text{write}}\\ W_{\text{write}}\end{bmatrix}.$$

The changes of basis of Definition 3.1 act on these reads and writes as on any others. At the hidden state h_ℓ, which block ℓ writes and block $\ell+1$ reads,

$$A_\ell^{\text{write}} \mapsto A_\ell^{\text{write}}\,G_\ell, \qquad A_{\ell+1}^{\text{read}} \mapsto G_\ell^{-1}A_{\ell+1}^{\text{read}}, \qquad \text{so that} \qquad A_\ell \mapsto G_{\ell-1}^{-1}\,A_\ell\,G_\ell.$$

Two facts stop the reduction of Proposition 3.3.

The symmetry between the two skip matrices is trivial. Applied to the submatrix $A_{\ell+1}^{\text{read}}$, the reduction turns it into the identity and moves it into the write before it, $A_\ell^{\text{write}} \mapsto A_\ell^{\text{write}}A_{\ell+1}^{\text{read}}$. Likewise, inside one block only the product $A^{\text{read}}A^{\text{write}}$ enters the function. These steps remove the parameters that writing one skip as two matrices had added, and none of W_{read} and W_{write}.

A skip connection ties two hidden states. The skip of block ℓ reads $h_{\ell-1}$ and writes into h_ℓ. A change of basis that reduces a submatrix of $W_{\text{read}}^{\ell+1}$ also changes the skips on both sides of h_ℓ. A learned skip absorbs the change. An identity skip cannot: $A_\ell = I$ is fixed, and $G_{\ell-1}^{-1}\,I\,G_\ell = I$ only if $G_{\ell-1} = G_\ell$. So with identity skips the changes of basis of Definition 3.1 preserve the form of the network only when they are the same at all hidden states: for every block ℓ,

$$E \mapsto EG, \qquad W_{\text{read}}^{\ell} \mapsto G^{-1}W_{\text{read}}^{\ell}, \qquad W_{\text{write}}^{\ell} \mapsto W_{\text{write}}^{\ell}G, \qquad D \mapsto G^{-1}D. \tag{52}$$

The reduction then removes d_{model}^2 parameters in the whole network, where Proposition 3.3 removes $(L+1)d_{\text{model}}^2$. Earlier work uses this one change of basis, in its orthogonal form, to compress and to merge transformers (Ashkboos et al., 2024; Theus et al., 2025).

Two statements of Wang and Wang (2025). Wang and Wang (2025) state that the symmetry group of a transformer is the product of the groups of its layers (their Theorem 7), and that the group of a feed-forward layer whose activation is not a polynomial consists of the permutations of its hidden units (their Lemma 6, with their Assumption A9, which names GELU and ReLU). We note a counterexample to each, and for this reason we do not engage further with that paper.

The product. That paper takes a sequence of token embeddings as the input of the model. In a transformer the embedding and the readout are parameters as well, and the map (52) then preserves the function for every invertible G. It moves the embedding, which no symmetry of a single layer does, so it is not a product of symmetries of the layers. That paper allows LayerNorm before each sub-layer, with gains and biases. There the same holds for every orthogonal Q with $\mathbf{1}^\top Q = \mathbf{1}^\top$, where $\mathbf{1}$ is the column of ones: the embedding, the writes and their biases are multiplied by Q on the right, as in (52), and a read W with the bias b that follows a LayerNorm with the gain γ and the bias β becomes

$$W' = \operatorname{diag}(\gamma)^{-1} Q^\top \operatorname{diag}(\gamma)\, W, \qquad b' = b + \beta\,(W - W').$$

In a transformer of three layers of this kind, with GELU feed-forward layers and a trainable embedding and readout, this map with a Q far from the identity changes the outputs by 10^{-14}, at outputs of size 15, and by 19 when one layer is left unchanged. In the networks of this paper, which have no skip connections, a change of basis at every hidden state couples two consecutive layers (Proposition 3.2), so a product over the layers does not describe their symmetries either.

The feed-forward layer. ReLU is not a polynomial, and a ReLU hidden unit with the read column c and the write row a has the rescaling $(c, a) \mapsto (\lambda c, a/\lambda)$, $\lambda > 0$. The value of a hidden unit of a SwiGLU layer, the feed-forward layer of current language models, can be rescaled in the same way (Appendix B). These continuous symmetries are not permutations, and the paper does not mention them. For a coordinate-wise activation, the matrices that pass through it had been determined by Godfrey et al. (2022, Theorem E.14): permutations times diagonal matrices, with the positive diagonal matrices for ReLU and the permutations alone for GELU.

A conjecture of Henry (2026). For an attention-only transformer of L layers with skip connections and without normalization, Henry (2026, Conjecture 2.4) conjectures that the parameters with the same function form, generically, a set of L times the dimension that they have for one layer. The symmetries inside the layers account for this dimension. When the embedding and the readout are parameters, the map (52) adds d_{model}^2 dimensions to the set, since it moves the embedding and the symmetries inside the layers do not. For the layers alone, with the embedding and the readout fixed, the map (52) is not available, and the count of the conjecture is not affected.

What we do not determine. Whether a network with skip connections has further continuous symmetries, beyond the symmetries inside its blocks and the one change of basis (52), is a question on its functional dimension, and we do not determine it here.